

MASTER'S THESIS

ISO-MISSION-TIME RANDOM-NOISE ERROR BUDGETS FOR LIFE WITH ARBITRARY NULLING AND NOISE MODELS

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Abstract

The LIFE space mission aims to detect and characterize nearby terrestrial exoplanets in the mid-infrared. Its scientific return depends on how much additional instrumental random noise can be tolerated without making the required observing program too long. This thesis develops an Agnostic Mission Simulator (AMS) and a trade-space explorer that convert a wavelength-dependent aggregate random-noise allowance into a 90th-percentile mission time, then invert it to locate the largest tolerated allowance. They establish a previously unstated property of the LIFE requirements studied here: they are ill-conditioned. Located where the mission-completion curve is steep, such a requirement amplifies rather than transmits uncertainty in the model behind it. Five distinct perturbation routes show this across several scales: a 27% error in one background term moves the derived allowance by factors of 1.5 to 11; a model difference of 1.2×10^{-4} in mission time moves individual endpoints by up to 11.9%; correcting an off-by-one completion rule in the inherited scheduler moves every endpoint by 3.5 to 95%; resampling the synthetic universe ensemble alone moves the reported amplitude by half its magnitude at the tightest operating point; and the occurrence-model configuration alone moves the feasibility boundary by 5 to 15%. A single dimensionless margin governs the sampling effect—the target's distance from its own feasibility boundary in sampling standard deviations—and orders all twenty-five operating points measured here and yields a quotability criterion on the comparable flat allowance: fixed on eight points, it classifies the seventeen held out correctly, and holds on eighty independent population realizations under a pre-registered protocol, with $z \geq 3$ a conservative sufficient margin.

The background error above is not hypothetical. Reformulating the source calculation analytically exposed a normalization defect in the inherited local-zodiacal term. At the published LIFE yield-study configuration, the chain reproduces the published yields with the defect reinstated and lowers them by about 7% corrected—an external consistency check and a quantified correction in one measurement.

Final calculations compare flat and wavelength-weighted noise budgets between realizable architectures for the two standard LIFE target catalogs, each response derived from aperture positions and beam combiner. For the second-order reference array, noise concentrated at short wavelengths is least damaging; a fourth-order null suppresses geometric stellar leakage by four orders of magnitude and reverses that preference toward long wavelengths, in both catalogs and at every target. Such a null requires not a minimum number of apertures but a combiner with two dark outputs that pair into a chopped fourth-order null: a four-design survey exhibits this at four, five, and six apertures, and finds that designs succeed or fail by their response's shape against the mission's achievable baselines—the four-aperture collinear design is fastest of all on the least collecting area. The allowances are aggregate amplitudes at a defined reference plane: an impact study, not an engineering design.

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Introduction

Why LIFE matters and how a science goal becomes an instrument requirement

Human curiosity extends far beyond Earth. We seek to understand how life began and whether Earth is its only home in the Galaxy. The LIFE space mission turns that broad question into a concrete observational program: detecting nearby temperate terrestrial exoplanets in the mid-infrared, measuring their thermal spectra, and using those measurements to constrain planet populations, atmospheric composition, and possible habitability [1, 3].

This scientific promise makes the practical definition of LIFE especially consequential. A mission can deliver those measurements only if its instrument can be built and operated within credible performance limits. This project therefore asks how much aggregate instrumental random noise LIFE can tolerate, and where in wavelength that noise is least damaging, before the required observing program becomes too long. Answering it helps define instrument boundaries, test feasibility, and increase confidence in the mission concept and the conclusions drawn from its measurements.

The reference LIFE studies establish the astrophysical source model, signal extraction, and expected detection and characterization performance for a double-Bracewell interferometer [1, 2, 7]. An aggregate instrumental random-noise term is not itself new, and Section 1.2 records what this work inherits. What the reference studies do not provide is its inversion: the wavelength-dependent *iso-mission-time* boundary, that is, the set of noise-budget amplitudes that all lead to the same required mission time, for a mechanism-independent aggregate random-noise term. This thesis fills that narrower gap with two linked tools. The *Agnostic Mission Simulator* (AMS) maps a specified noise budget and

operating point to mission time; the *Trade Space Explorer* (TSE) inverts that mapping to find the largest tolerated budget amplitude. A controlled change from a second- to a fourth-order transmission proxy then moves the spectral region that tolerates the most added noise from short to long wavelengths. The direction of that change follows from the known suppression of geometric stellar leakage at a deeper null; what the completed exploration establishes is that the reversal survives the full nonlinear scheduling and percentile chain, and that it holds under equal integrated added noise rather than only under equal ramp endpoints.

This work takes the double Bracewell as its reference geometry but is not restricted to it. It is mechanism-agnostic for the aggregate random-noise allowance, in that individual independent random terms are represented by their common equivalent rate, and architecture-agnostic for the instrument, in that the transmission is derived from an arbitrary combiner rather than written in closed form for one array. The result is therefore a bridge quantity. It translates a science-level completion time into an allowed single-output random count rate at a defined *pre-efficiency reference plane*. That plane is the point in the signal chain at which photon rates have been multiplied by the collecting area but not yet by the optical throughput or the detector quantum efficiency, so a rate quoted there is independent of how those two factors are eventually apportioned; Section 2.6 fixes it precisely. A later engineering allocation can map detector, optical, thermal, and control-noise terms to that plane and test their combined variance against the allowance. This thesis does not perform that subsystem allocation and cannot decide which hardware implementation is easiest. It does, however, evaluate the allowance for more than one realizable array, so the part of the answer that belongs to the mission can be separated from the part that belongs to the instrument.

1.1 Research Questions

This thesis addresses the following central question.

Using an efficient architecture-agnostic simulator for arbitrary physics-conforming nulling and noise models, can instrument-agnostic requirements be determined on the aggregate random-noise error budget of the LIFE space mission such that it completes Experiment 1—the inherited detection-and-characterization program for 50 qualifying catalog planets—within a reasonable mission time, under pessimistic assumptions on the physical limits?

The question names a means as well as an end. The end is a requirement: one aggregate allowance on independent random noise, referred to a defined plane and expressed so that a later subsystem allocation can be tested against it. The means is a simulator that

is agnostic in two distinct senses, one about the noise and one about the instrument. It must represent the noise without committing to a physical mechanism, and it must represent the interferometer without committing to a particular beam combiner, since a requirement that holds only for one architecture is not a requirement on the mission. That the simulator must also be efficient is not a convenience: the inversion evaluates the same mission many times, and a requirement that cannot be recomputed cannot be revised.

The question is therefore structured into five sub-questions, labelled here and closed one by one in Section 7:

- Q1** *What is the dimension of the simulator's trade space, and can it be reduced without compromising fidelity?*
- Q2** *Which source-model computations can be reused when only global filters or budget amplitudes change, and what does that reuse buy?*
- Q3** *Can the response of an arbitrary, physically realizable nulling architecture be evaluated at the cost of the closed-form reference, rather than by a per-target spatial grid?*
- Q4** *Can the maximum tolerated amplitude be located for flat and linear-ramp random-noise budgets at a fixed, deliberately pessimistic operating point?*
- Q5** *Does the resulting requirement depend on the nulling architecture, and if so, in what way?*

1.2 Relation to Prior Work

The tools and the source model used here are inherited, so it is worth stating precisely what is taken over and what is added. LIFE paper II supplies the double-Bracewell response, the signal-extraction formalism, and the LIFEsim source model on which the simulator in Section 3.6 is built, and its appendix constrains instrumental perturbations mechanism by mechanism [2].

The direct predecessor is Dannert's doctoral thesis [7]. It already introduces an aggregate instrumental photon-noise term that collects all additional photon-noise contributions, defines a fundamental-noise-limited criterion comparing that term against the astrophysical floor, and studies the resulting effect on detection yield. The aggregate random-noise quantity used here is therefore not a new object, and the present work does not claim it as one. Nor is requirement derivation itself new to that work. Its final chapter sweeps aperture diameter against perturbation level, extracts point designs that characterize thirty

habitable-zone planets within a five-year schedule, and reads instrument requirements off them—0.59 nm optical path difference and 0.04 % relative amplitude in its balanced case—benchmarked against the stability demonstrated by comparable missions.

Four things separate the present work from it. The allowance here is *mechanism-agnostic*: it aggregates any independent random term referable to a common plane rather than being stated for named perturbations, so it does not presuppose which mechanism dominates. It is *located by a root-find* to a stated mission-time tolerance rather than read from a swept grid. It is *resolved in wavelength* across a tested family of budget shapes, which is what allows the spectral question to be asked at all. And it is *architecture-agnostic*, so the part of the answer that belongs to the mission can be separated from the part that belongs to the array. The two sets of requirements are therefore complementary rather than competing: one gives numbers for specific mechanisms in a specific array, the other an envelope that any mechanism must fit inside, for any array.

Two earlier master's theses occupy neighbouring ground, one from this group and one from a different institution. Norbruis [15], at Delft University of Technology, built a LIFEsim-derived performance model, used P-Pop populations as a test set, and ran grid and multi-objective optimisation over array geometry, aperture area, and instrument temperature to produce Pareto fronts against a detection figure of merit. That work optimises the design for yield; the present work holds the design fixed and solves for a tolerated noise allowance at fixed mission time. Binkert [16] addresses signal extraction and molecular-feature detection during the search phase, a different layer of the same mission and without overlap with the question asked here.

The question itself is older than LIFE. For the Darwin and TPF-I studies, Lay derived how systematic instrumental errors couple into a nulling measurement and where the resulting requirements bind [9], and later extended the analysis to instability noise and its suppression [19]. Chazelas et al. translated the same concern into instrumental stability requirements for Darwin-class missions [20], Kaltenegger et al. into requirements on the achievable rejection ratio [21], and the TPF-I technology plan [22] and the Darwin mission summary [23] record how those studies allocated noise among subsystems at fixed integration time. What that literature does not contain is the inversion performed here—holding the mission's completion time fixed and solving for the largest tolerable aggregate random-noise amplitude—nor any statement about the conditioning of the number such an inversion returns.

Against that background, the central claim of this thesis is a property of the requirement itself rather than any particular value of it.

The iso-mission-time requirements studied here are ill-conditioned. Such a

requirement is located where the completion curve is steep, because that is where a mission-time target worth adopting lies, and it therefore does not merely transmit uncertainty in the model it is derived from but amplifies it. Five distinct perturbation routes establish this, spanning several scales and mechanisms—from a tens-of-percent physical error through a 10^{-4} implementation difference to a rule change, pure resampling, and the population model itself (the routes share simulator, scheduler, and scaffold: distinct probes, not statistically independent evidence):

- A 27.3 % error in one background term, the inherited local-zodiacal normalization, moves the derived allowance by up to a factor of eleven (Section 4.2, Table 4.1).
- A model difference of 1.2×10^{-4} in mission time, between two formulations of the same reference array, moves three of twelve endpoints by 4.3, 6.8 and 11.9 % (Appendix B).
- Correcting the inherited scheduler's completion rule—an off-by-one against the stated experiment, a fraction of a percent in the quantity it thresholds—moves every endpoint amplitude by 3.5 to 95 % (Section 4.3).
- Resampling the synthetic universe ensemble the mission-time percentile is taken over, changing nothing physical at all, moves the reported amplitude by half its own magnitude at the tightest operating point, and six percent of three hundred resamples there leave no usable positive allowance at all (Section 5.4). How badly a given amplitude is affected is governed by one dimensionless quantity—the margin between its mission-time target and its own feasibility boundary in sampling standard deviations—which orders all twenty-five operating points measured here: fitted on eight, it holds on seventeen held-out points across four architectures at rank correlation -0.94 (flat-family spread).
- Regenerating the synthetic populations themselves with the public generation tool, with and without the binary-occurrence suppression, moves the feasibility boundary by 5–15 %—an order of magnitude beyond its sampling spread—while the margin criterion continues to classify every cell correctly as a conservative bound across eighty independent realizations (Section 5.4).

The consequence is practical and, as far as the LIFE literature goes, unstated: a requirement of this kind must be quoted with the background model that produced it, and a subsequent correction to that model invalidates it rather than perturbing it. It also means the common practice of adopting the tightest feasible mission-time target places the resulting requirement in precisely the least determined part of the space.

Establishing that claim is what the rest of the work is for; each supporting piece states exactly what in it is new and what is inherited. The iso-mission-time inversion supplies the requirement to be conditioned. The closed-form treatment of the circularly symmetric backgrounds removes image size as a production setting and makes thousands of inversions affordable (Section 4.2). The transmission model is not itself new—writing the response in terms of aperture positions and a beam-combiner matrix dates to Lay [9] and Guyon et al. [13]—but its cost is: the rotation average of any single-baseline geometry collapses to a one-dimensional function of a single similarity variable, so an arbitrary architecture is tabulated once and thereafter evaluated for the price of the closed-form reference (Section 3.3). That shows the conditioning is a property of the requirement, not of one array. And the local-zodiacal normalization defect, exposed by the analytic reformulation in inherited code, is the worked example: correcting it lowers the yield of a fixed search campaign by about seven percent—consequential, not merely a vehicle for the argument.

One further result is independent of the conditioning claim: the spectral preference between budget shapes inverts with null order, survives the full scheduling and percentile chain rather than following from leakage scaling alone, and survives substituting a realizable combiner for an idealized proxy—the one quantity robust to every perturbation applied in this work.

Four terms recur in fixed senses. A *realizable response* is computed from aperture positions and a lossless beam-combiner matrix—optics that could in principle be built; an *idealized proxy* is a $\sin^{2p}$ response written for its null order alone, no array behind it. What follows is an *impact study*: it measures what a stated mission-time requirement can support under retained assumptions. It is not an *engineering design*, which would add optical-loss, instability, and control analyses.

The work behind these claims is stated once, because the document is the only record its reader has. This thesis contributes four thousand lines of simulation and analysis code over the inherited LIFEsim core; a closed-form reformulation of the three circularly symmetric background terms, validated at seven boundaries; the discovery and quantification of two inherited-code defects—the local-zodiacal normalization and the scheduler's completion rule—both since confirmed upstream; an external consistency check against published LIFE yields; a trade space of several thousand scheduler inversions; and the bootstrap, survey, and margin analyses of Chapters 5 and 6. What is inherited is named above; the rest is new.

The remainder follows the causal chain: the experiment, targets, and error budget (Section 2); the simulator and transmission model (Section 3); the trade-space reduction and analytic response (Section 4); the completed exploration (Section 5); and what the allowances do and do not imply (closing chapters).

Background and Procedures

From a synthetic planetary population to a measurable mission signal

This project is conducted within the scope of the LIFE space mission. To connect its scientific objective to an engineering requirement, we first need to establish what the experiment asks the instrument to observe and how those observations become a mission-time calculation. LIFE is a proposed space-based mid-infrared nulling interferometer for detecting and characterizing nearby temperate terrestrial planets [1, 2]. Its current concept separates light collection from beam combination: four free-flying collector spacecraft feed a central beam-combiner spacecraft while the formation rotates about the line of sight. The long effective baseline supplies angular resolution that a single space telescope of comparable total collecting area could not provide.

The scientific reason for observing in the mid-infrared is twofold. A temperate planet emits much of its thermal radiation in this range, reducing the star-planet contrast relative to visible reflected light, and several atmospheric molecules have diagnostically useful absorption bands there. Nulling interferometry, proposed for this purpose by Bracewell [11] and extended to imaging arrays by Angel and Woolf [12], does not form a conventional resolved image of every system. Instead, it suppresses the on-axis star, modulates off-axis emission through rotation, and measures a wavelength-dependent differential signal. Detecting that signal constrains planet populations; accumulating a spectrum can constrain atmospheric composition. The latter remains an inference problem rather than an automatic identification of life [1, 3].

This thesis retains the four-aperture double-Bracewell response used by LIFEsim [2]. Two aperture pairs first produce destructive and constructive outputs. The destructive outputs are recombined with a relative phase shift to produce two dark outputs whose difference changes sign for an off-axis source. The reference response has a second-order intensity null: close to the optical axis, transmitted stellar intensity grows quadratically with angular offset. Figure 2.1 shows this optical logic.

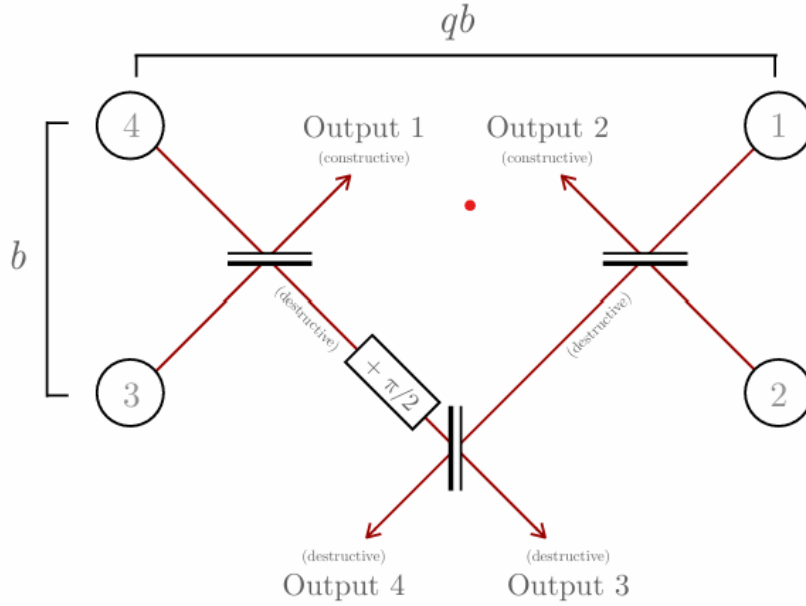


Figure 2.1: Double-Bracewell interferometer layout reproduced from Dannert et al. [2]. Light from apertures 4 and 3, and from 1 and 2, is combined to form destructive and constructive outputs. The destructive outputs are recombined with a relative phase shift of $\pi/2$ to form two differential dark outputs with a second-order null.

2.1 The Experiment and Assumptions

This work adopts the legacy LIFESim/AMS configuration called *Experiment 1*: the mission must detect and characterize 50 catalog planets with radii between 0.5 and 1.5 Earth radii, host-star temperatures between 4370 and 7310 K, and a positive catalog habitable-zone flag. The thresholds and instrument settings used throughout the completed data collection are summarized in Table 2.1. They define this study's controlled comparison; they should not be read as a claim that every value is the latest top-level LIFE mission requirement.

Quantity	Adopted value	Role in the calculation
Qualifying sample	50 planets	Experiment-completion target
Planet radius	0.5–1.5 $R_{\oplus}$	Catalog eligibility filter
Host temperature	4370–7310 K	Catalog eligibility filter
Habitable zone	Catalog flag required	Catalog eligibility filter
Detection threshold	SNR 7	Common detection campaign
Characterization threshold	SNR 44.1	Broadband characterization proxy
Orbit epochs	5	Detection plus four follow-up visits
Collector diameter	4 m each	Photon collecting area
Optical throughput	0.15	Pre-detector transmission
Quantum efficiency	0.60	Photon-to-count conversion
Photon conversion efficiency (PCE)	0.09	Throughput $\times$ quantum efficiency
Spectral band and resolution	4–18.5 μm , $R = 20$	31 finite wavelength bins
Baseline ratio and bounds	$r = 6$, 10–100 m	Imaging/nulling geometry and clipping

Table 2.1: Fixed LIFESim/AMS settings used for all final simulations. SNR 7 follows the LIFE yield-study detection convention [4]. The characterization value 44.1 is the integrated broadband equivalent of a legacy per-channel requirement of SNR 10 at 11.2 μm and $R = 50$ [7].

Both thresholds are conventions inherited deliberately, and their origins bound what they mean. The detection value of SNR 7 is the LIFE yield-study convention [1, 4]: for a photon-noise-limited measurement with near-Gaussian statistics it corresponds to a per-trial false-alarm probability of order 10^{-12} , small enough that a survey of thousands of stars with many independent spectral and positional trials still expects no false detection, with margin left for the non-Gaussian residuals that real signal extraction leaves [2]. The characterization value of 44.1, integrated over the band, is the broadband equivalent of a legacy per-channel requirement of SNR 10 at 11.2 μm and $R = 50$, the signal quality at which atmospheric-retrieval studies recover the main molecular features of a temperate

terrestrial atmosphere [3]. Adopting both unchanged keeps the mission times of this work directly comparable with the published yield studies. The characterization threshold deserves particular care. It is not a universal statement that every molecular feature is measurable at bulk SNR 44.1. It is a retained broadband proxy that permits a consistent mission-time comparison with the inherited simulation framework [7]. Similarly, the fixed 4 m collectors and 9% photon conversion efficiency set photon rates but are not optimized in this thesis. The output is conditional on all these assumptions.

These assumptions specify the required scientific outcome, but not the individual planets that must supply it. We therefore next define the simulated target populations on which the mission-time calculation is based.

2.2 Target Catalog Selection

The LIFE target catalogs are created from a common scaffold of 4505 nearby F, G, and K stars out to roughly 50 pc (Fig. 2.2); its composition, verified directly from the star columns of the catalog files, is 854 F, 1319 G, and 2332 K hosts. This scaffold was inherited together with the catalogs, and its provenance was verified at the identifier level for this work: all 4505 stars—4415 carrying TESS Input Catalog and 90 Gaia DR3 identifiers—are members of the Habitable Worlds Observatory Preliminary Input Catalog [34], of which the scaffold is a selected subset of roughly a third. The star count and distance limit quoted here were verified directly from the catalog files, and the identifier cross-match is archived with the reproducibility products. The LIFE Science Team's own record confirms and completes this account (LIFE Science Team, private communication, 2026): the mission's reference star catalog, compiled in October 2025, holds 14,571 F, G, K, and M stars—the 12,944 stars of HPIC, built from the TESS Input Catalog and Gaia DR3 with a 50 pc distance limit and magnitude cuts $T \leq 12$ and $G \leq 12$, plus 1,627 stars from the LIFE Stellar Catalog v4 [35], predominantly late M dwarfs excluded from HPIC by its magnitude limits. That all 4505 scaffold stars cross-match to HPIC is consistent with that structure: the scaffold selection drew from the reference catalog's HPIC portion, and none of the fainter catalog-v4 additions survive it.

It is deliberately broader than the stellar sample of LIFE paper VI, which is restricted to single and wide binary main-sequence stars within 20 pc [4]. The $\eta_{\oplus}$ values adopted in Table 2.2 were averaged over that smaller sample, so the occurrence normalization and the stellar scaffold used here are not drawn from the same stellar population. This is acceptable for the present purpose, because the comparison throughout is between operating points and budget shapes evaluated on one fixed scaffold, but it means the absolute planet counts in Table 2.3 should not be read against the yields reported in

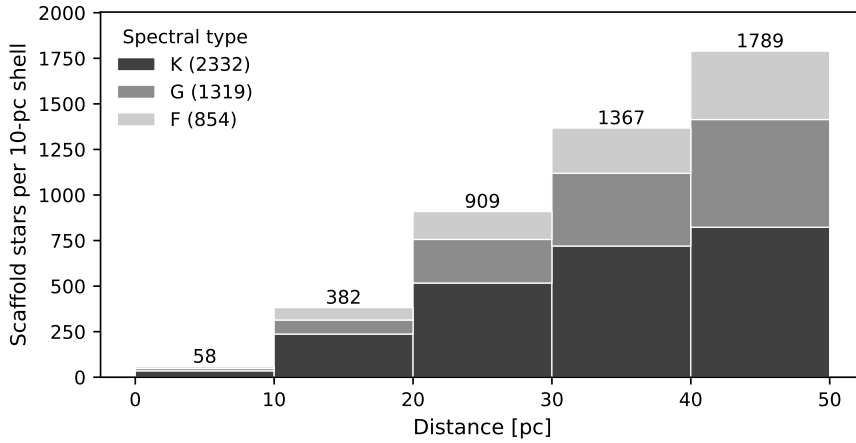


Figure 2.2: Distance distribution of the 4505-star scaffold in 10-pc shells, stacked by spectral type. The population grows roughly with shell volume—58, 382, 909, 1367, and 1789 stars in successive shells—so most targets sit in the outer half of the 50 pc sphere, where they are least resolved and hardest to detect; K stars dominate every shell. The scaffold holds no M dwarfs: they fail the Experiment 1 host-temperature window before any catalog is drawn.

LIFE paper VI. Since planets are not known around most of them, the population must be synthesized. P-Pop draws occurrence and orbital properties constrained by Kepler statistics and places a hypothetical planetary system around every stellar target [5]. One complete draw over the stellar scaffold is called a *universe*. The 100 universes in a catalog are therefore alternative population realizations, not 100 independent observations of the real Galaxy.

We use two *target catalogs*, *Hab2Min* and *Hab2Max*, with 100 universes each. The terminology is kept strict throughout this work: the stellar side—the 4505-star scaffold selected from the mission’s reference *star catalog*—is always called the scaffold, while *catalog*, unqualified, always means one of these two target catalogs, the scaffold plus its synthesized planet populations. They implement the lower and upper “hab2 stars” extrapolation scenarios of the Bryson et al. occurrence model as adopted in LIFE paper VI [6, 4]. The quantity $\eta_{\oplus}$ is the average number of exo-Earth candidates per Sun-like star, not the fraction of stars hosting at least one such planet. *Hab2Min* represents the less favorable and *Hab2Max* the more favorable occurrence scenario; their difference is driven chiefly by the extrapolation beyond 500 d, where Kepler completeness no longer directly constrains the occurrence surface. Both retain the same stars, so comparing them changes the hypothetical planets rather than the accessible stellar neighborhood. M dwarfs, though abundant in the upstream reference catalog, fail the Experiment 1 host-temperature cut and are therefore absent from the scaffold altogether.

The catalog ensemble has two roles. It exposes the scheduler to different possible planet populations, and it supplies the distribution from which the 90th-percentile mission time

Catalog	$\eta_{\oplus}$
Hab2Min	$0.19^{+0.19}_{-0.10}$
Hab2Max	$0.34^{+0.38}_{-0.19}$

Table 2.2: Mean numbers of exo-Earth candidates per Sun-like star used to define the two occurrence scenarios [6, 4]. The asymmetric bounds are the reported 1σ occurrence-model uncertainties and are dominated by occurrence-rate and long-period extrapolation uncertainty.

Catalog	Universes	Planet rows	Stars	Eligible planet rows	Eligible planets/universe
Hab2Min	100	486 244	4505	205 715	1958–2154
Hab2Max	100	729 925	4505	318 654	3049–3331

Table 2.3: Inventory measured from the exact input files used in this work after loading them through the same P-Pop interface as the AMS. “Eligible” applies only the Experiment 1 radius, host-temperature, and habitable-zone filters; visibility, SNR, and scheduling cuts are applied later. The median eligible counts are 2058.5 for Hab2Min and 3194 for Hab2Max. LIFE paper VI uses a narrower exo-Earth-candidate subset, $0.8a_p^{-0.5}R_{\oplus} \leq R_p \leq 1.4R_{\oplus}$, for its occurrence analysis; that definition does not replace the broader Experiment 1 radius filter used here [4].

is taken. Catalog eligibility alone does not imply detectability: transmission, noise, magnitude, field of regard, and the finite observing campaign all act later. This distinction explains why target selection precedes signal collection in the narrative.

The catalog fixes the systems and planets to be observed; the next step is to determine how light from each hypothetical planet is separated from its much brighter surroundings.

2.3 Signal Collection: Transmission Map, Output Subtraction, and Modulation

Before quantifying the SNR, we must establish how the interferometer separates an off-axis planet from bright backgrounds. A transmission map $T(\alpha, \beta)$ gives the fraction of incident flux reaching one output at sky offset (α, β) . Following the notation of LIFE paper II, the nulling baseline b sets the broad null envelope, while the imaging baseline rb sets the fine differential and modulation fringes; the baseline ratio is $r = 6$ in the nominal settings. For each star, b is selected using the habitable-zone-centre prescription and clipped to the instrument bounds. A common field-of-view taper attenuates off-axis light. The star lies on the central null, whereas an off-axis planet samples the surrounding fringes [2].

The two destructive outputs are differential by subtraction. Their symmetric mean back-

grounds cancel in the difference, but their independent photon-counting variances remain and add. A planet generally contributes to both outputs, but unequally, and therefore survives the subtraction (Fig. 2.3a, right).

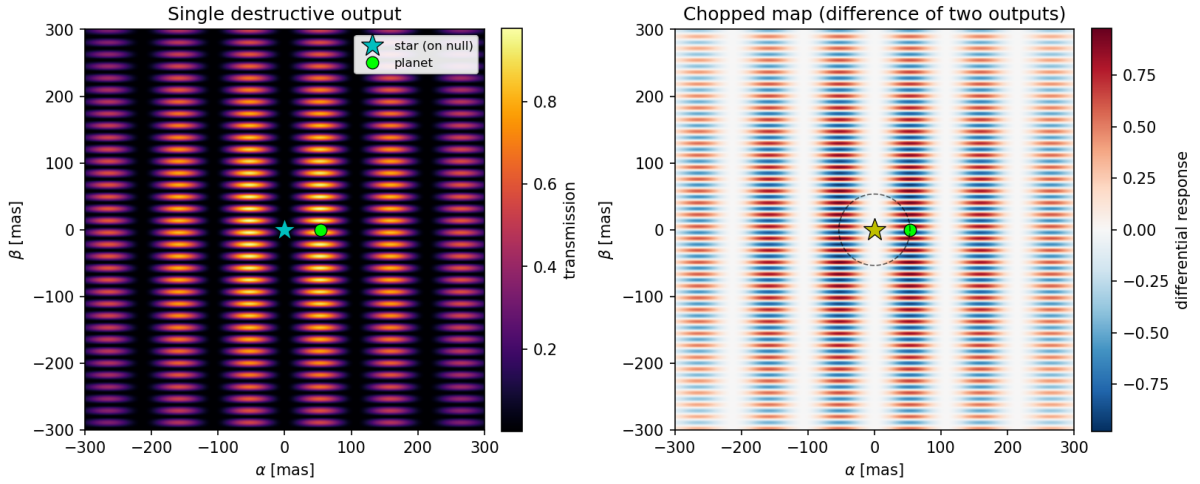
The final ingredient is *modulation* through array rotation. Over an observation the entire array is slowly rotated about the line of sight. A planet at a fixed angular separation is thereby swept through the alternating positive and negative lobes of the differential map, so that its contribution to the differential output oscillates in time, tracing out a characteristic *transmission curve* as a function of rotation angle (Fig. 2.3b). The symmetric backgrounds, being nearly axisymmetric, stay approximately constant under this rotation: Lay reports that the stellar and local-zodiacal geometric leakage rates are nominally independent of the array rotation angle and therefore appear at zero frequency, and Dannert et al. state correspondingly that for rotationally symmetric sources the detected signal does not depend on the rotation angle of the array [9, 2]. The planetary signal is therefore encoded as a time-modulated component riding on an essentially static background, which allows it to be isolated in subsequent analysis [10, 2]. This is the origin of the temporal average $\langle \cdot \rangle$ over one full array rotation that appears throughout the SNR derivation below.

The calculation does not simulate a time-series estimator sample by sample. It uses the root-mean-square response over a complete, uniformly sampled rotation. For a differential planet transmission $m(\theta) = T_3(\theta) - T_4(\theta)$, the effective signal coefficient is therefore

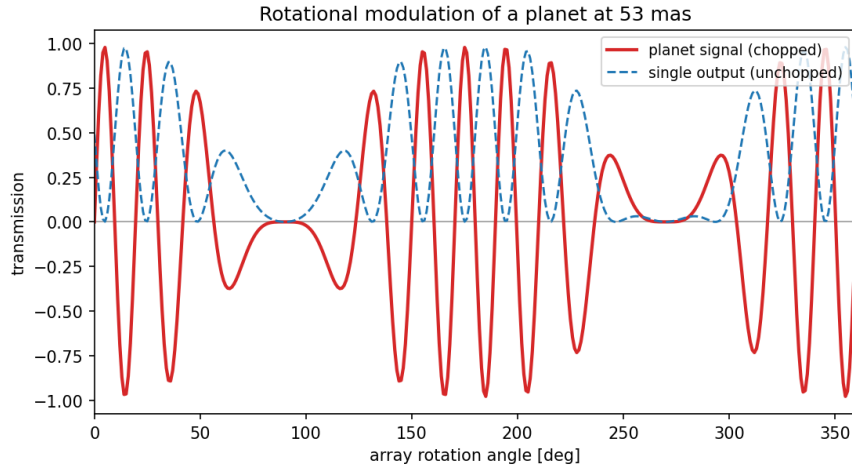
$$s = \left[\frac{1}{2\pi} \int_0^{2\pi} m^2(\theta) d\theta \right]^{1/2}. \quad (2.1)$$

This is the matched-modulation amplitude for white, rotation-independent variance under the adopted normalization. A constant symmetric brightness cancels from the difference, but it still sends photons to each output and therefore remains in the variance. Figure 2.3 shows the undifferenced and differential maps over which these coefficients are averaged.

Two consequences of this scheme carry directly into the SNR. First, the planetary *signal* is collected through the differential map, whereas the planetary *photon noise*, which is insensitive to the sign of the modulation, is collected through a single, undifferenced output; the two therefore enter with different transmission weights. Second, because the map is integrated over a full rotation, the effective response becomes radially symmetric and depends only on the angular separation of the planet and on the nulling baseline. This rotation-averaged response, further attenuated by a field-of-view taper that models the loss of off-axis coupling into the instrument's optical fibers, is the *transmission efficiency* introduced in Section 2.6. Having established what reaches the detector and how it is weighted, we can now quantify whether the planet can be distinguished from noise.



(a) Transmission maps of the double Bracewell array.



(b) Rotational modulation of the marked planet.

Figure 2.3: Signal collection in the double-Bracewell reference geometry at $10.4\,\mu\text{m}$, $b = 20\text{ m}$ and nominal baseline ratio $r = 6$, zoomed to the central $\pm 300\text{ mas}$. (a) A single destructive-output map and the difference of the two. Symmetric mean contributions cancel in the difference, whereas an off-axis planet generally has unequal transmission in the two outputs. (b) Rotation modulates the differential planet signal; the undifferenced single-output response used for its photon-noise term has a different weighting.

2.4 The Signal-To-Noise Ratio (SNR)

Everything the interferometer collects funnels into one question per planet: does its modulated signal stand out of the noise? The SNR makes that question quantitative: the expected modulated count signal divided by the standard deviation of the associated random count fluctuation. The implementation evaluates finite wavelength bins. Its reference planes are easy to confuse because planet flux is naturally specified at the observer, whereas extended backgrounds have already been integrated over collecting area. Table 2.4 makes the convention explicit.

Symbol	Meaning at the SNR interface	Units
$F_{p,i}$	Planet photon flux at the collectors, integrated over bin i	$\text{ph s}^{-1} \text{m}^{-2}$
A	Total effective collecting area in the source convention	m^2
η	Optical throughput times quantum efficiency	1
s_i	RMS differential planet-response coefficient	1
n_i	RMS single-output planet-noise coefficient	1
B_i	Single-output astrophysical rate after area, before η	ph s^{-1}
$N_{I,i}$	Added single-output rate after area, before η	ph s^{-1}
t	On-source integration time	s

Table 2.4: SNR symbols and reference-plane convention. The reported spectral budget $N_I(\lambda)$ is converted to a bin rate by $N_{I,i} = N_I(\lambda_i)\Delta\lambda_i$.

For bin i , the expected differential planet counts follow from its incident flux, collecting area, efficiency, time, and RMS modulation response. Each destructive output is a Poisson stream. If the two output counts have means μ_3 and μ_4 , then

$$\text{Var}(C_3 - C_4) = \text{Var}(C_3) + \text{Var}(C_4) = \mu_3 + \mu_4 \quad (2.2)$$

when the streams are independent. Symmetric backgrounds have equal rotation-averaged rates in both outputs, so subtracting their means removes the DC signal but doubles their single-output variance. With $s_i = \sqrt{\langle(T_3 - T_4)^2\rangle}$ and $n_i = \sqrt{\langle T_4^2\rangle}$, the implemented expected signal counts and variance are

$$C_{S,i} = F_{p,i}A\eta t s_i, \quad (2.3)$$

$$V_i = 2\eta t [B_i + N_{I,i} + AF_{p,i}n_i]. \quad (2.4)$$

Hence

$$\text{SNR}_i = \frac{C_{S,i}}{\sqrt{V_i}}. \quad (2.5)$$

Rates have units ph s^{-1} per bin, $C_{S,i}$ is a count, V_i a count variance, and $\sqrt{V_i}$ a count standard deviation. The subtraction cancels the symmetric mean but differences two destructive outputs, so their independent photon-counting variances add and give the factor two. Output-swap/rotation symmetry makes the rotation-averaged undifferenced planet photon-noise rates equal; the same two-output variance sum therefore gives the factor two for the planet term. Assuming independent bins, the bulk SNR is

$$\text{SNR}_{\text{tot}} = \sqrt{\sum_{\lambda_i} \text{SNR}_{\lambda_i}^2}. \quad (2.6)$$

It scales as $\sqrt{t}$; $\text{SNR}_{1\text{h}}$ uses $t = 3600 \text{ s}$. This source-model convention follows LIFE paper

II [2]. Correlated fluctuations and systematic biases are outside this photon-statistical model [9, 7]. From this point onward, SNR refers to bulk SNR unless otherwise stated.

The SNR expression identifies the quantities that limit a measurement, but the error budget can only be defined after those noise contributions have been separated by physical origin.

2.5 Non-Systematic Noise Sources

Three astrophysical light sources stand between the instrument and the planet, and the model distinguishes them by both physical origin and statistical treatment. Stellar leakage arises because a real star has finite angular radius: only its centre lies exactly on the null, while light from the resolved disk is transmitted. Local-zodiacal emission is thermal radiation from dust in the Solar System; exozodiacal emission is the analogous dust contribution around the target star. The adopted exozodiacal surface-brightness model follows the smooth, face-on disk prescription used in LIFEsim and is tied to established exozodiacal modeling [8, 2]. All three contribute photon shot noise even where output subtraction cancels their symmetric mean.

Table 2.5 collects these terms with their physical origin, the dependence retained in the model, and the boundary of that treatment.

The question is how much instrumental random noise can be allowed in addition to the astrophysical backgrounds while retaining the adopted mission target. Here the *random-noise error budget* is operationally an equivalent added single-destructive-output photon-rate spectral density $N_I(\lambda)$, in $\text{ph s}^{-1} \mu\text{m}^{-1}$, referred to the same pre-efficiency plane as the astrophysical background. It represents independent Poisson-like random terms that enter as additive variance after differencing. It neither represents correlated or systematic biases nor assigns the allowance to individual subsystems. To turn that budget into an engineering question, we next express the physical contributions as inputs that can be varied or held fixed.

2.6 Inputs and Instrument Performance

An *input* is a value or function used to describe a candidate mission calculation. A *variable* input is a target-dependent physical signal or noise channel; changing it triggers SNR recomputation (for example stellar leakage or planetary transmission). An *agnostic* input is a global visibility filter or overhead applied without recomputing source physics (for example limiting magnitude, field of regard, or slew time).

Term	Physical origin	Modeled dependence	depen- dence	Treatment and boundary
Stellar leakage	Finite stellar disk transmitted around the central null	λ , stellar radius and temperature, distance, b , null order		Uniform circular disk; photon noise only
Local zodiacal	Solar-System dust foreground	λ , target ecliptic lat- itude, field of view		Locally uniform circular foreground; photon noise only
Exozodiacal	Dust in the target system	λ , stellar luminosity and distance, disk radius		Smooth face-on axisymmetric disk; photon noise only
Planet	Desired thermal emission	λ , radius, tempera- ture, separation and rotation		Differential signal plus its own photon noise
Added budget N_I	Equivalent aggregate instrument contribu- tion	Imposed spectral family and ampli- tude		Independent Poisson-like vari- ance at the pre- efficiency single- output plane
Excluded terms	Calibration bias, drift, correlated instability, confusion	Architecture and data-reduction de- pendent		Not represented by N_I

Table 2.5: Noise taxonomy and model boundary [2, 8]. “Photon noise only” means that a source’s mean differential signal may cancel while the associated shot-noise variance remains. Excluded systematic and instability terms require a separate architecture-dependent treatment [9, 7].

The word *agnostic* is used in two distinct senses in this thesis, and they should not be conflated. In the title and in Section 1 it is *mechanism*-agnostic: the instrumental allowance $N_I(\lambda)$ does not commit to a physical origin for the random terms it aggregates. In the sense just defined, and throughout the description of the simulator, it is a *computational* classification: an agnostic input is one whose variation does not require the source physics to be recomputed. The first sense is a modelling choice about what the result means; the second is an implementation property that makes the repeated evaluations affordable. An input can be agnostic in the second sense while being entirely architecture-specific, as slew time is.

The ideal $N_I = 0$ case is the physical lower bound: it has maximum SNR and minimum mission time, but it is the *hardest* engineering requirement because it demands a perfectly noiseless added contribution. It is a mathematical reference origin, not an easy design. Negative budgets are outside this model. Requirements are relaxed upward from this origin until the iso-mission-time boundary is reached. If, at some wavelength, the added

budget is lower than the total astrophysical noise, the calculation is *astrophysical-noise-limited* there; if it exceeds it, it is *instrumental-noise-limited*.

As an example, consider again the stellar leakage. Typically, a star emits more strongly in the shorter wavelengths, so we might add some margin to the shorter wavelengths and observe the impact on mission time. This motivates the short-weighted ramp tested in Section 5; the opposite ramp tests the zodiacal-dominated long wavelengths. Neither shape redistributes or subtracts noise elsewhere.

Planetary emission enters through the rotation-averaged transmission efficiency and a common field-of-view taper. As defined in Section 2.3, b sets the null envelope and rb the fine fringes in nominal settings; b is selected per star from the habitable-zone centre and clipped to the allowed bounds. The field-of-view taper represents common off-axis coupling loss without assuming a particular taper formula.

We can now collect the complete set of variable inputs and the physical quantities each of them depends on. Every noise contribution reaches the instrument through the transmission map, which is why each additionally carries the wavelength dependence characteristic of its blackbody source. The variable inputs are as follows:

- Stellar leakage. Depends on the wavelength, the stellar temperature, the stellar radius, and the distance to the star.
- Local-zodiacal noise. Depends on the wavelength and the ecliptic latitude of the target.
- Exo-zodiacal noise. Depends on the wavelength, the stellar luminosity, and the distance to the star.
- Planetary signal. Depends on the wavelength and, through the transmission efficiency, on the nulling baseline and the field of view taper.

Counted naively, the search space is enormous: each of the four physical channels may carry its own multiplicative and additive modifier—eight free functions of the listed variables—so the unrestricted problem is infinite-dimensional before anything is fixed. The calculations actually reported are far smaller. Once catalog, null order, operating point, and spectral family are fixed, the search runs over a single number, the amplitude A ; letting magnitude, field of regard, and slew time vary adds three more scalars; freeing the two ramp endpoints gives five in total. Catalog, null order, spectral family, and target mission time remain discrete scenario choices. This hierarchy motivates the reductions of Section 4.1.

In addition to the variable ones, there are inputs termed *agnostic*: global filters or over-heads scanned without recomputing source physics, not quantities literally independent of every architecture. They are evaluated on simple grids. The agnostic inputs are as follows:

- Limiting magnitude. Describes the maximum K-band magnitude object that the instrument can detect.
- Field of regard. The implemented filter marks a target visible when its absolute ecliptic latitude is at most the FoR. Thus FoR 65° retains the band from -65° to $+65^\circ$ and excludes the polar caps; FoR $\pi/2$ retains the full sky.
- Slew time. Defines the average time it takes for the instrument to center on a new target.

Knowing the starting points and limits, we can now go back to the performance of the instrument. We employ a practical science iso-performance approach. Specifically, we declare an instrument better-performing if it can complete the scientific goals of the mission in less time. The variable and agnostic inputs together span a large space of possible choices, which we call the *trade space*. To estimate an error budget, the *Agnostic Mission Simulator* (AMS), described in Section 3.6, maps inputs to mission time. Here a set budget means a specified wavelength-dependent additive shape together with a selected operating point.

2.7 The Trade Space

These inputs define a forward map from an instrument description to mission time, but the engineering question asks for its inverse: the largest added budget compatible with a chosen time. The raw space cannot be sampled directly because it contains free functions. Even a finite ten-dimensional surrogate, sampled at only ten values per dimension, would require 10^{10} AMS evaluations. At an optimistic one second per evaluation, that coarse grid would take approximately 317 years. This estimate is illustrative rather than a claim that the raw functional space has exactly ten dimensions. It shows why the workflow must first restrict the budget to interpretable spectral families, then exploit monotonic searches and reusable computations instead of applying a uniform grid.

The completed TSE runs invert this mapping for null orders two and four. They therefore show both how much aggregate random-noise amplitude is tolerated within each tested shape and how the modeled null order changes the spectral ordering of those allowances.

Methods

Building a simulator that is agnostic to both the noise mechanism and the array geometry

The background has established the quantities mapped from a description of the array to mission time. We now construct that mapping, with a mechanism-agnostic aggregate random-noise allowance and a transmission model that accepts any physically realizable nulling architecture rather than one written in closed form.

3.1 Variable Input Treatment

A free interpolation grid would preserve maximum flexibility but create a poorly constrained functional search. Instead, the AMS modifies each modeled photon-flux density F —stellar leakage, for example— through a multiplicative function M and an additive function A :

$$V(\alpha_1, \dots, \alpha_n) = M(\alpha_1, \dots, \alpha_n) \times F + A(\alpha_1, \dots, \alpha_n), \quad (3.1)$$

where M is a multiplicative modifier and A an additive modifier. For an *extra-noise* channel, $M \geq 1$ and $A \geq 0$ preserve the astrophysical reference, with the zero-budget case $M = 1$, $A = 0$. These inequalities do not describe signal degradation: that requires a separate convention, such as $0 \leq M \leq 1$ or a negative additive term constrained to keep the signal non-negative. The condensed study reported here varies additive background noise only.

3.2 The Reference Double-Bracewell Response

The simulator remains mechanism-agnostic with respect to the aggregate random-noise allowance throughout. Its treatment of the instrument, by contrast, begins from the reference array and is then generalized. For the reference double Bracewell, the first pairwise combinations produce a $\sin^2$ nulling envelope, while the second combination produces complementary imaging fringes. Let α and β be angular sky offsets along the nulling and imaging-baseline directions. With the LIFE notation b for nulling baseline and r for the imaging-to-nulling baseline ratio [2], the phases are $\phi = \pi b \alpha / \lambda$ and $\psi = \pi r b \beta / \lambda$. The two reference destructive outputs are

$$T_{3,2} = \sin^2(\phi) \cos^2(\psi - \pi/4), \quad (3.2)$$

$$T_{4,2} = \sin^2(\phi) \cos^2(\psi + \pi/4). \quad (3.3)$$

Their difference is $T_{3,2} - T_{4,2} = \sin^2(\phi) \sin(2\psi)$, up to the chosen output-sign convention. Close to the optical axis, $\sin \phi \simeq \phi$, so the nulling envelope grows as ϕ^2 and has intensity null order two.

Written this way the response is exact, but it is exact for one array. A requirement derived through it is a statement about the double Bracewell rather than about the mission, and the central question asks for a simulator that accepts arbitrary physics-conforming nulling models. The remainder of this section replaces the closed form with a description that any nulling interferometer satisfies.

3.3 Transmission of an Arbitrary Nulling Architecture

A nulling interferometer is specified completely by where its collectors sit and how their beams are combined. Let the apertures occupy positions $\mathbf{r}_k$ in the array plane and let the beam combiner be described by a matrix U with one row per output and one column per aperture. A source at sky offset $\boldsymbol{\theta}$ imposes a phase $2\pi \mathbf{r}_k \cdot \boldsymbol{\theta} / \lambda$ on aperture k , so output j receives

$$T_j(\boldsymbol{\theta}) = \frac{1}{N} \left| \sum_{k=1}^N U_{jk} \exp\left(\frac{2\pi i \mathbf{r}_k \cdot \boldsymbol{\theta}}{\lambda}\right) \right|^2, \quad (3.4)$$

with N the number of apertures. Everything the simulator needs follows from Eq. 3.4: the null order, the behaviour under subtraction, the rotation-averaged background response, and the throughput. Nothing is assumed about the architecture beyond the array being lossless, which makes the rows of U orthonormal.

The factor $1/N$ fixes a convention that matters as soon as designs with different aperture counts are compared. With unit amplitude at each aperture the array collects N units of intensity and a lossless combiner conserves it, so dividing by N makes the outputs sum to unity and a response is then directly the fraction of collected light reaching that output. Throughput therefore ceases to be a separate quantity requiring its own bookkeeping: it is the number of outputs used for science divided by N . This is worth stating explicitly, because holding throughput fixed while changing the response is exactly the error an idealized shape model invites.

Equation 3.4 reproduces the closed form above when U is the double Bracewell. Expressed as four apertures on a rectangle with the nulling baseline along α and the imaging baseline along β , and with the two Bracewell nulls recombined at relative phase $\pi/2$, it agrees with $T_{3,2}$ and $T_{4,2}$ to a relative 3×10^{-14} , and an independent power-law fit to its near-axis response returns a null order of 2.0000 without being told what to expect.

3.4 Architectures Considered

Five designs are used in this work, summarized in Table 3.1 and sketched in Fig. 3.1; the pentagon appears twice because its two output pairs are two instruments on one array. The first is LIFE's reference array. The others achieve a fourth-order null in structurally different ways, and comparing them establishes a constraint that no idealized response can express—and corrects one this work itself initially got wrong.

Architecture	N	Order	Dark outputs	Subtraction	Pair light
Double Bracewell (reference)	4	2	2	yes	50 %
Angel Cross	4	4	1	no	25 %
Double triple nuller	6	4	2	yes	33 %
Kernel-5 pentagon, deep pair	5	4	2	yes	40 %
Kernel-5 pentagon, shallow pair	5	2	2	yes	40 %
Collinear kernel	4	4	2	yes	50 %

Table 3.1: Architectures evaluated in this work. Dark outputs—the nulled outputs at the design's quoted order that serve as science outputs—and the fraction of collected light follow from the row structure of an orthonormal combiner matrix rather than from any fitted quantity. The reference's fourth output, the difference of its two pair-bright rows, is also dark on axis but is not a nulled science output and is not counted; the 50 % and 25 % figures reproduce those quoted for Darwin and the Angel Cross by Guyon et al. [13].

The Angel Cross [12] places two crossed Bracewell pairs [11] on a square. Its four orthogonal outputs comprise one bright output, two second-order outputs, and a single output whose amplitude vanishes to second order and whose intensity therefore vanishes

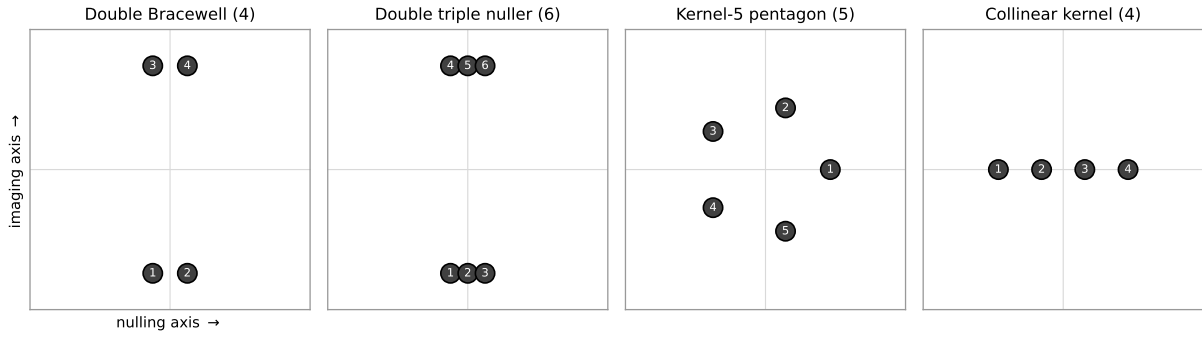


Figure 3.1: Aperture geometries of the surveyed designs, drawn from the same constructors the results use, at unit nulling baseline and the production imaging ratio of six (pentagon and collinear designs have a single characteristic scale). The reference and the double triple nuller null along the horizontal axis and image along the vertical one; the collinear kernel places all four apertures on the nulling axis.

to fourth. What output subtraction requires is stated precisely in the next paragraph, and the Angel Cross fails it for a structural reason: among its rows exactly one has an amplitude vanishing beyond first order, and one such row is not enough. **What a fourth-order null with output subtraction requires is not a minimum aperture count but a combiner with two rows whose amplitudes vanish to at least second order:** two such rows r_1, r_2 remix losslessly into a conjugate pair of fourth-order outputs whose intensity difference modulates with source azimuth. The remix is the unitary pair $(r_1 \pm i r_2)/\sqrt{2}$, so losslessness is automatic; each new output's intensity, $\frac{1}{2}|a_1 \pm i a_2|^2$, is built from amplitudes that are individually second order in the offset and is therefore fourth order; and the chopped difference, $2\text{Im}(a_1 a_2^*)$, modulates with azimuth provided the two row responses are not everywhere in phase—the one degenerate case, which two distinct rows of the surveyed designs never realize. Section 5.8 exhibits this at four apertures on a line and at five on a pentagon; an earlier draft of this work claimed six as the minimum, and both constructions refute it.

The double triple nuller realizes the condition in LIFE's own template; that name is used throughout, abbreviated to *triple nuller* in tables, and the design is also referred to descriptively as the six-aperture fourth-order design when its aperture count or null order is the point. It is the direct analogue of the reference architecture with each two-aperture Bracewell pair replaced by a three-aperture nuller of amplitudes $1 : -2 : 1$. A single such triple has amplitude proportional to $2(\cos \psi - 1)$, second order in the offset, so its intensity vanishes to fourth order; this is the one-dimensional $2(N - 1)$ rule for $N = 3$ [13]. Two triples separated along the imaging baseline and recombined at relative phase $\pi/2$, exactly as the two Bracewell pairs are in the reference design, yield two fourth-order outputs of opposite modulation sign, and output subtraction is available again. Six apertures are thus the smallest realization within the two-dimensional, mirror-symmetric family LIFE's design belongs to—a statement about the template, not about aperture

count.

The subtraction pair has a further property worth naming. In both the reference design and the double triple nuller the two dark outputs are complex conjugates of one another row by row, and the difference of their intensities is a kernel observable in the sense of Martinache & Ireland [18]: its response to small per-aperture piston errors cancels at second order, and its response to relative amplitude errors cancels identically, since a real perturbation produces the same intensity in conjugate rows. Both properties hold numerically on the matrices of Table 3.1: over random piston draws the mean leakage of either dark output grows as σ^2 while the spread of the subtracted observable grows as σ^3 , and pairing a chopped output with the non-conjugate dark output of the reference design instead recovers the σ^2 scaling. The output subtraction LIFE relies on is therefore a kernel observable, and the requirement that two dark outputs exist is, in this language, the requirement that the combiner admit a conjugate pair of fourth-order rows.

The *maximum* achievable null order rises with aperture number [13], so orders beyond a given template's ceiling are bought with collectors; within a fixed aperture count, however, the combiner decides how much of that ceiling is realized with chopping—the four-aperture collinear kernel of Section 5.8 reaches a chopped fourth-order null at the same N as the reference. Kernel nullers formulated for an arbitrary number of apertures [14] are the natural family in which further designs would be expressed, and Eq. 3.4 accepts them unchanged.

The transmission model of Section 3.3 accepts any combiner, which makes the space of architectures as unbounded as the space of noise functions was before the first reduction. This study evaluates four designs spanning that space along its most informative axes: the reference double Bracewell, the six-aperture double triple nuller of Section 3.4, and the two kernel designs of Section 5.8—a five-aperture pentagon and a four-aperture collinear array, both carrying fourth-order conjugate pairs. The Angel Cross is excluded by the condition of Section 3.4: with a single deep-capable row it admits no conjugate pair. Each architecture is evaluated at its own optimal baseline rather than the reference's, so the comparison is between designs each doing the best it can. Four designs spanning aperture counts four to six and both null orders are a survey of the accessible kernel families, not of the full design space.

Collector size, not total area, is held fixed: every design is built from the 4 m collectors of Table 2.1, so the six-aperture array has 75.4 m² of collecting area against the reference's 50.3. This convention is forced by the simulator's own physics: collecting area and single-aperture field of view both follow from the collector diameter, as $N\pi(D/2)^2$ and $\lambda/2D$, so fixing total area instead would mean 3.27 m collectors whose beams subtend 1.5 times the solid angle and admit correspondingly more local-zodiacal foreground; holding area

fixed while keeping the reference's field of view would describe no instrument at all.

The comparison is therefore not like-for-like: part of the six-aperture design's gain is bought with that extra area rather than the deeper null. The opposite convention is equally self-consistent and points the other way. Evaluated at equal total collecting area, with 3.27 m collectors, the same design reaches a Hab2Max zero-budget time of 6.18 yr against the reference's 5.32: 16 % *slower* rather than faster. The architecture comparison of Section 5.8 is therefore bracketed by the two conventions; which is the right systems-level question this thesis does not settle. What survives both is the ordering of the budget families.

3.5 Why an Arbitrary Architecture Remains Affordable

Evaluating Eq. 3.4 on a spatial grid for every target and every wavelength would be far more expensive than the closed form it replaces, and would forfeit the reductions of Section 4. It is not necessary. The array rotates during an observation, and for a circularly symmetric source that is equivalent to averaging the response over the azimuth of the source. If the geometry scales with a single baseline, so that $\mathbf{r}_k = b \mathbf{u}_k$ for fixed $\mathbf{u}_k$, the phase in Eq. 3.4 depends on the target only through

$$x = \frac{\pi b \theta}{\lambda}, \quad (3.5)$$

the same dimensionless separation the reference response depends on. The rotation-averaged response of any such architecture is therefore a one-dimensional function of x , and can be tabulated once per design and interpolated thereafter. Throughout, x denotes this variable evaluated at a general sky angle. Where a resolved stellar disk of angular radius $R_\star$ is averaged over, the natural argument is instead $x_\star \equiv 2\pi b R_\star / \lambda$, which is twice x evaluated at $\theta = R_\star$; the factor of two is carried by the Bessel identity used there and is kept explicit to avoid confusing the two.

The consequence is that the cost of an arbitrary combiner is the cost of the reference. Measured on the same catalog and hardware, a full evaluation takes 14.3 s with a general combiner against 14.7 s with the closed form, and the tabulated background response agrees with the analytic Bessel reduction of Appendix A to a relative 6×10^{-12} . The architecture-agnostic model is thus not a slower fallback used where the closed form is unavailable; it is the same calculation, obtained numerically rather than by hand.

3.6 The Agnostic Mission Simulator (AMS)

The pieces now assemble into the tool this thesis is built around. The AMS maps one set of inputs to mission time in two phases. The *simulation phase* computes or reuses an unfiltered one-hour SNR for every catalog planet, then applies the global visibility filters. The *distribution phase* schedules the filtered targets and returns the required percentile mission time.

3.6.1 The Simulation Phase

Figure 3.2 separates the expensive *SNR flow* from the inexpensive *filter flow*. The SNR flow fills a persistent array with one-hour SNRs for every target and reruns only when a physics-governing quantity changes. The filter flow always returns a copy in which targets failing the magnitude or field-of-regard cut have SNR zero; it never overwrites the unfiltered cache.

When the SNR flow runs, it iterates over the complete catalog. Within that recomputation, stellar leakage, local-zodiacal noise, and the unit-exozodi response are reused among planets sharing one host; planet transmission remains target-specific. Magnitude, field of regard, and slew-time changes reuse the unfiltered SNR array and rerun only filtering and time distribution. By contrast, any change to budget amplitude or spectral shape changes the bin variance, invalidates the SNR array, and requires catalog-wide SNR recomputation.

3.6.2 The Distribution Phase

We now have a full set of filtered SNRs for every target in the input planet catalog. The only remaining step is to use LIFEsim to distribute the available observation time across the targets and then find the required mission time.

The distribution rests on photon-limited SNR accumulation as the square root of integration time. The SNR flow yields $\text{SNR}_{1\text{h}}$ for each target, so

$$t_{\text{int}}(q) = \left(\frac{q}{\text{SNR}_{1\text{h}}} \right)^2 \quad [\text{h}] \quad (3.6)$$

for a fresh integration to requested bulk SNR q . The implemented follow-up calculation does not subtract SNR already accumulated during detection. For a selected planet, with

slew time t_{slew} , it charges

$$t_{\text{orbit}} = (n_{\text{orbits}} - 1) [t_{\text{int}}(7) + t_{\text{slew}}], \quad (3.7)$$

$$t_{\text{char}} = t_{\text{int}}(44.1) + t_{\text{slew}}. \quad (3.8)$$

This convention treats the four follow-up orbit epochs and the characterization as separate observations. It is intentionally reported as implemented; alternative reuse of detection photons would reduce the follow-up cost.

The complete scheduler is:

1. Optimize one common detection sequence over all 100 synthetic universes.
2. Stop when the completion check reports 50 qualifying detections in 90% of universes. The inherited implementation tested both thresholds strictly—51 planets in 91 universes with 100 universes—which Section 4.3 corrects to the stated rule used for all headline results.
3. Assign the resulting shared detection-campaign time to every represented universe.
4. In each universe, retain eligible planets that were detected, rank them by their one-hour SNR at maximum separation, and select the best 50.
5. Add four orbit follow-ups and one characterization observation per selected planet using the equations above.
6. Return the 90th percentile of the resulting per-universe total times.

The AMS is consequently a perfect-information ensemble yield model, not an adaptive discovery scheduler. In experiment-limited mode, the configured 0.8 observing efficiency neither caps nor rescales the returned time. The inherited implementation tested “completion of Experiment 1” strictly—at least 51 detections in at least 91 universes, with empty universes omitted from the time sheet. Section 4.3 audits both choices, corrects the completion rule for all headline results, and measures the empty-universe omission to have no effect on the production ensemble.

Two consequences follow for how every time reported in this thesis should be read. First, the returned quantity is the time the scheduler charges: the sum of on-source integration and slew, accumulated over the shared detection campaign, the four orbit follow-ups, and the characterization observation. It is not a calendar duration, because no duty cycle, downlink, calibration, or safe-mode overhead is applied to it. Second, the observing efficiency of 0.8 configured in Table 2.1 would, if applied as a duty cycle, map a charged time T to an elapsed duration $T/0.8$; the 5.5-year target used throughout would then

correspond to roughly 6.9 years of elapsed mission. Every target, contour, and budget amplitude reported here is stated in charged time, so comparisons among them are internally consistent. They should not be read directly against a mission calendar, and the allowances in Section 5 are correspondingly optimistic if the reader interprets the target as elapsed time.

The returned percentile is fixed at 90%. If 100 universe times are retained, a distribution-free binomial calculation gives approximately 95.6% coverage for the population 90th percentile between the 84th and 96th order statistics. Fewer retained universes require different ranks. The original production campaign retained contour endpoints but not every per-universe time vector; the bootstrap and survey campaigns of Chapter 5 retain complete per-universe component times for every replicate (Appendix B) and state intervals accordingly.

3.6.3 Validation of the SNR Flow

The physics and numerical rewrite were checked at several distinct boundaries; Table 3.2 states what each check can and cannot establish.

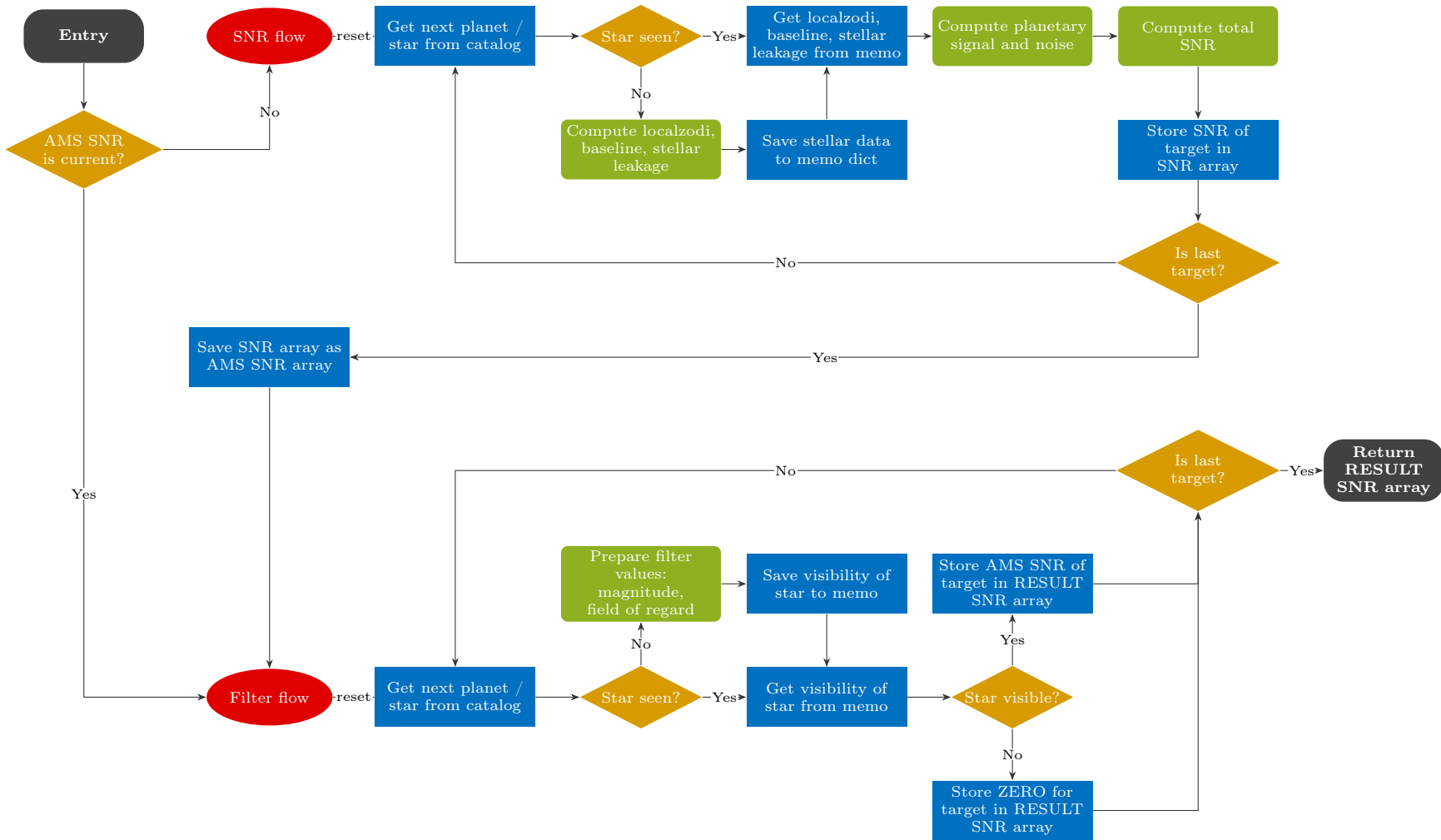


Figure 3.2: AMS simulation phase. Dark rounded tiles mark entry or exit, red ellipses denote flows, amber diamonds decisions, green boxes calculations, and blue rectangles memory operations. Physics or budget changes trigger the SNR flow and replace the unfiltered cache. The filter flow always copies that cache and sets targets outside the magnitude or field-of-regard limits to zero; it never destroys unfiltered values.

Comparison	Result	Interpretation
Shared-route SNR comparison (two thousand sampled targets)	Maximum relative difference 4.2×10^{-16}	Confirms wiring and vectorization against shared source modules
8-point versus 64-point inverse-wavelength quadrature, $R = 5, 10, 20, 50$	Approximately machine precision for sampled stellar and exozodiacal terms	Establishes convergence of within-bin integration for the retained smooth spectra
Analytic azimuthal response versus 2×10^6 angle samples, orders 2, 4, 6	Relative agreement 10^{-16} for $x \gtrsim 1$, rising to 10^{-12} at $x = 0.2$ and order six	Checks the general even-order Bessel reduction; the larger residual is cancellation in the coefficient sum where the response is itself of order 10^{-5}
Uniform-disk average versus dense two-dimensional grid	Agreement at approximately 10^{-5}	Residual is consistent with reference-grid discretization
Order-four stellar and local-zodiacal terms versus image sampling, 15 stars	Below 0.7% at image size 400	End-to-end check against an independent spatial discretization
Exozodiacal quadrature self-convergence, radial orders 2, 4, 6	Below 10^{-5}	Checks the one-dimensional extended-disk integration
Order-four and order-six exozodiacal terms versus image sampling	Up to about 10% on a few stars	Reference-grid aliasing near the Kennedy inner cutoff, not an order-four error: the same stars and grid deviate by 4.5–5.4% at order two
Exozodiacal radial sampling, 25-star check	Worst case 1–3% after the log-spaced substitution, against up to 35% before it	A first uniform-in-radius implementation undersampled the steep profile near the inner cutoff; the defect was found by this check and fixed
Invalid odd null order	Explicit error	Prevents applying the even-order derivation outside its domain

Table 3.2: Validation matrix for the AMS and analytic source calculation. Agreement with shared code establishes numerical consistency, not independent empirical validation of the astrophysical assumptions: the strongest equality, AMS against the standard source routine, is deliberately narrow since both routes share physical modules, and none of the checks validate the face-on exozodiacal assumption or the scheduler—the external yield validation of Section 4.4 closes part of that gap. The first, third and fourth rows were recomputed for this document from the committed source, and their scripts are listed in Appendix B; the remaining entries are carried over from the development record and were not re-derived here.

Trade Space Reduction

Reducing an intractable trade space without discarding the relevant physics

The raw functional trade space cannot be explored directly. This section therefore reduces it in two steps: first by combining admissible independent random terms at one reference plane, and then by replacing spatial pixel grids with analytic integrals where source symmetry permits. Each reduction retains a stated validation boundary.

4.1 Condensation of Astrophysical Noise

The eight unrestricted modifier functions are too conditional to produce a readable engineering bridge quantity. The first reduction therefore asks which terms can be represented by one equivalent independent random-noise contribution without changing their treatment in the SNR variance.

We reduce the trade space by exploiting the additive background terms in the bin variance. We define a wavelength-dependent instrumental term $N_I(\lambda)$; in bin i , $N_{I,i} = N_I(\lambda_i) \Delta\lambda_i$ is added alongside the corresponding B_i term.

$$N_B(\lambda) = \sum_{i=1}^n N_i(\lambda) + N_I(\lambda).$$

As is evident from the blackbody behavior of the background noise sources, we must keep the wavelength dependence to maintain physical relevance. Figure 4.1 illustrates the construction: an imposed $N_I(\lambda)$ is added on top of the astrophysical background, and the shaded area between the two curves is the allowance being solved for.

The reduction is valid when terms map to a defined independent random allowance at the same reference plane; it excludes correlated/systematic terms and is not an allocation to subsystems. This study tests flat and two linear-ramp forms of N_I , not arbitrary functions. Increasing the amplitude within one specified family increases mission time; the iso-mission-time boundary then defines its tolerated amplitude.

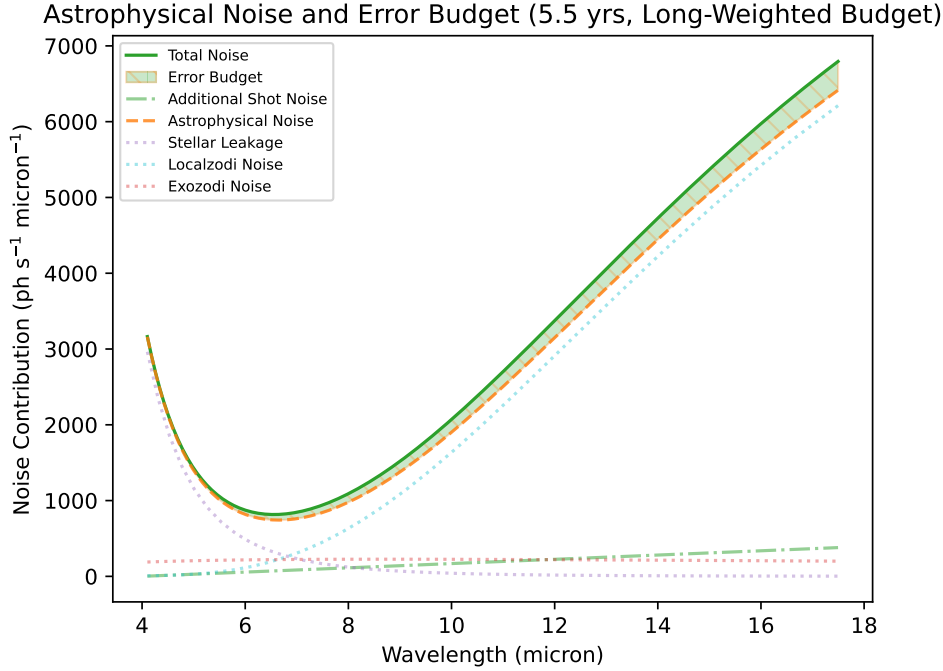


Figure 4.1: Construction of an imposed random-noise budget on top of the astrophysical background. The dashed green line is the imposed $N_I(\lambda)$, here the long-weighted linear ramp of Section 5.2; the solid green line is the resulting total background, and the shaded area between them is the allowance itself. All curves are photon-rate spectral densities at the pre-efficiency single-output plane, not variances. The configuration is Hab2Max at null order two and the common operating point, with the amplitude at the largest value compatible with the 5.5-year target; the same panel recurs as Fig. 5.1(c). The background curves belong to the simulator's single plotted target, which is hotter and more distant than the eligible median, so the panel illustrates the construction rather than an average background; the ensemble-median background an allowance should be compared against is reported in Section 6.1 (artifact `background_context.tsv`, Appendix B.2).

4.2 Analytic Background Evaluation and Spectral Resolution

The analytic rewrite of the background evaluation removes the science dependence on image size. Circularly symmetric sources permit an analytic azimuthal and radial reduction of the background integrals: the stellar disk is treated as uniform and circular, local zodiacal light as locally uniform, and exozodiacal light as smooth and face-on. Inclined or asymmetric exozodiacal disks require an angle-resolved treatment and are outside this

reduction. Image size now affects visualization and independent cross-checks only, not production science SNR.

The reduction follows from expanding the even power $\sin^{2p} \phi$ into cosines. Azimuthal averaging then uses the standard Bessel integral representation [17],

$$\frac{1}{2\pi} \int_0^{2\pi} \cos(z \cos \varphi) d\varphi = J_0(z). \quad (4.1)$$

For null order $n = 2p$, azimuthally averaging one destructive output at angular radius ϑ gives

$$\langle T_{2p} \rangle_{\varphi}(\vartheta) = \frac{1}{2} \frac{\binom{2p}{p}}{4^p} + \sum_{j=1}^p (-1)^j \frac{\binom{2p}{p-j}}{4^p} J_0\left(\frac{2\pi j b \vartheta}{\lambda}\right). \quad (4.2)$$

For a uniform disk, radial area averaging replaces each $J_0(jx_*)$ term by $2J_1(jx_*)/(jx_*)$, following the standard Bessel derivative identity [17], where $x_* = 2\pi b R_*/\lambda$ and R_* is the angular disk radius, as introduced in Section 3.5. The limiting value is one as $jx_* \rightarrow 0$, so the expression is evaluated with its analytic limit near the origin. The Bessel power series then gives the small-star limits

$$\langle T_2 \rangle_{\text{disk}} = \frac{x_*^2}{32} + \mathcal{O}(x_*^4), \quad (4.3)$$

$$\langle T_4 \rangle_{\text{disk}} = \frac{x_*^4}{256} + \mathcal{O}(x_*^6). \quad (4.4)$$

Taking the ratio of Eqs. 4.3 and 4.4, the fourth-to-second-order stellar-leakage ratio therefore approaches $x_*^2/8$. The same general average is used inside one-dimensional radial quadrature for a tapered source or exozodiacal profile. Rotation does not “average away” a circular background; rather, circular symmetry makes its integrated response independent of rotation angle [9, 2]. Its mean may cancel in the difference, but its photons still set the shot-noise variance.

The local-zodiacal correction arose from a geometric inconsistency in the removed pixel-grid path. That path is inherited LIFESim code and predates this work; the analytic reformulation did not introduce the defect, it exposed it. The physical field of view is circular, with solid angle proportional to πR_{FoV}^2 , but one normalization divided the integrated circular flux by the area $4R_{\text{FoV}}^2$ of its enclosing square. It therefore suppressed the uniform foreground by $\pi/4$.

The decisive evidence that this is a defect rather than a modelling choice is internal to the inherited code: the untapered branch of the same routine normalized by a circular mask, while the default tapered branch normalized by the square pixel sum, so the two

branches disagreed with each other on the same physical configuration. Restoring the circular normalization multiplies the local-zodiacal rate by

$$\frac{4}{\pi} = 1.2732 \dots \quad (4.5)$$

an increase of 27.3% in the limit of an untapered field of view, which is the case Appendix A derives. The production configuration applies a Gaussian field-of-view taper, and there the factor is not exactly $4/\pi$: both paths carry the same solid-angle prefactor, so the correction is the ratio of two averages of the transmission, one over a rotationally symmetric disk and one over the square evaluation grid whose corners break that symmetry. Measured against the old grid and extrapolated in resolution, that ratio converges to approximately 1.267 rather than to 1.273, and the difference is the taper rather than discretization noise. The correction is therefore an increase of about 27%, with $4/\pi$ as its untapered limit rather than its exact value. This is a normalization correction, not a new dust model.

Because local-zodiacal shot noise is present for every target, the change propagates nonlinearly through SNR, target ranking, and the 90th-percentile mission time. Comparing the corrected and uncorrected paths over every qualifying target, with the code otherwise identical, lowers the one-hour signal-to-noise ratio by a median of 9.2% at null order two and 9.8% at order four for Hab2Max, and by 9.1% and 9.8% for Hab2Min. The distribution is skewed rather than tight: the sixteenth-to-eighty-fourth percentile range runs from about -11% to -2.7% , the small shifts belonging to targets whose noise budget the local-zodiacal term does not dominate. The scope of that figure should be read carefully: it applies to results computed through the default tapered local-zodiacal path.

A shift in per-target signal-to-noise ratio is not yet a statement about anything the LIFE literature publishes, so it is worth converting it into one. Yield studies report detections within a fixed search campaign rather than the time needed to complete one, so the optimizer is run in its fixed-time mode, at the standard 2.5-year search and with nothing changed but the local-zodiacal normalization. The reference array then detects, per synthetic universe,

$$\begin{aligned} \text{Hab2Max: } & 63.0 \rightarrow 58.9 \text{ planets } (-6.5 \%), \\ \text{Hab2Min: } & 49.2 \rightarrow 45.9 \text{ planets } (-6.8 \%), \end{aligned}$$

where the first value is computed with the inherited normalization and the second with the corrected one. A yield quoted through the defective path is therefore optimistic by about seven percent in detection count, which is several times the run-to-run scatter of the scheduler and comparable to the differences between mission configurations that yield

studies are designed to distinguish.

Two qualifications belong with that number. It is computed with this thesis's configuration rather than by rerunning any specific published study, so it states the size of the effect rather than a corrected value for any particular table in the literature; reproducing a given result would require that study's own instrument settings and catalog. And the effect is not uniform across targets, since the local-zodiacal term dominates the noise budget for some targets and not others, so a yield is shifted by more than the median per-target degradation would suggest. What the figure does establish is that the defect is worth correcting before the next yield study rather than after it. The per-category yields are the archived artifact `localzodi_yield_impact.tsv` (Appendix B.2).

Within this study the effect can be measured directly, and it is far larger than the median shift in signal-to-noise ratio suggests. Restoring the analytic $\pi/4$ factor reproduces the pre-correction path to within the small margin by which that factor differs from the realized ratio, so the two configurations differ in the defect and in nothing else of consequence. Table 4.1 reports both quantities the thesis is built on. Under-counting the local-zodiacal foreground shortens the mission time required with no added budget by 0.50 to 0.84 yr, between 10 and 16%. The tolerated flat allowance, however, changes by factors of 1.5 to 11.

Catalog	Null order	Zero-budget time [yr]		Flat allowance [$\text{ph s}^{-1} \mu\text{m}^{-1}$]		
		Pre-correction	Corrected	Pre-correction	Corrected	Ratio
Hab2Max	2	4.82	5.32	752	142	5.3
Hab2Max	4	3.86	4.47	994	641	1.6
Hab2Min	2	6.57	7.41	686	62	11.0
Hab2Min	4	5.20	6.04	897	597	1.5

Table 4.1: Effect of the inherited local-zodiacal normalization defect on the two quantities this study reports, at the common operating point and each catalog's primary mission-time target. The pre-correction column is obtained by restoring the suppression factor, so the two configurations differ only in the defect. Both arms were measured under the strict completion rule then in force; the corrected rule of Section 4.3 shifts the absolute values but not the internal comparison, which is what the table reports. Amplitudes are the values returned by the search rather than the rounded entries of Table 5.1; Appendix B.3 records the difference between the two.

The disproportion is the important part. A 27.3% error in a single background term produced up to an order-of-magnitude error in the derived requirement, because the amplitude is defined where the completion curve is steep: suppressing part of the background shortens the zero-budget time and leaves far more headroom before the target is reached. The Hab2Min second-order case is the extreme, where a corrected zero-budget time of 7.41 yr against a 7.5 yr target leaves almost no room, and the allowance collapses from

690 to 60. Had the defect gone unnoticed, the requirements reported here would have been up to an order of magnitude too permissive.

Transmission and source spectra are integrated within every wavelength bin using an eight-point Gauss–Legendre rule in inverse wavelength $u = 1/\lambda$. This coordinate is natural because interferometric phases are proportional to b/λ and therefore vary nearly linearly with u . Across spectral resolutions 5, 10, 20, and 50 and sampled catalog stars, the stellar-leakage and exozodiacal terms agree with a 64-point reference to approximately machine precision. The adopted resolving power remains $R = \lambda/\Delta\lambda = 20$, which produces 31 bins over 4–18.5 μm . That choice is a compute-accuracy trade rather than a physical one, and Figure 4.2 shows both sides of it. The compute cost of a full catalog pass grows steadily with R , since the number of bins sets the width of every per-target source evaluation. The accuracy side is regime-dependent. Referred to an $R = 100$ evaluation, the SNR computed at $R = 20$ is converged to a few parts in 10^4 when the astrophysical background sets the bin variance, but departs by orders of magnitude more once the modeled per-detector and thermal instrument-noise terms dominate it. The production configuration used here is the former case: those instrument-noise modules are deliberately absent, since representing them by an aggregate allowance is precisely the reduction of Section 4.1, so the reported results sit in the converged regime. The figure does not establish how an imposed N_I of comparable magnitude would behave, because its spectral shape is prescribed rather than thermal; that remains a boundary of the retained binning rather than a demonstrated error.

Thus spatial image size is no longer a science setting, while wavelength-bin integration is explicitly converged for the retained binning. Appendix A records the derivation and numerical conventions in greater detail.

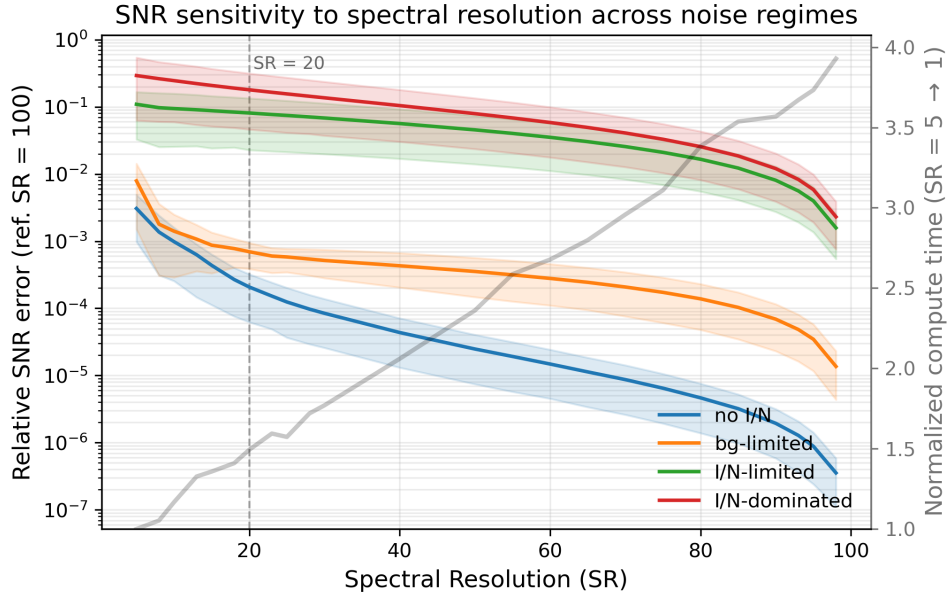


Figure 4.2: Sensitivity of the computed SNR to spectral resolving power, referred to an $R = 100$ evaluation, for four noise regimes: no instrumental term, background-limited, instrument-limited, and instrument-dominated. Solid lines are medians over sampled catalog targets; shaded bands show their spread. The grey curve, read against the right axis, is the compute time of a full pass normalized to $R = 5$. The dashed line marks the adopted $R = 20$.

4.3 Correction of the Scheduler's Completion Rule

The analytic reformulation exposed one defect in the inherited source model; auditing the inherited scheduler against the stated experiment exposed a second. The experiment of Section 2.3 completes when fifty qualifying detections are reached in at least ninety of one hundred universes. The inherited implementation tests both thresholds strictly—a universe counts only beyond fifty detections, the experiment only beyond ninety universes—so it in fact requires fifty-one detections in ninety-one universes. It also drops universes with no surviving detections from the mission-time percentile. The second choice turns out to have no effect on the production ensemble, which contains no empty universes; the first biases every mission time upward, by 0.06 to 0.32 yr at the operating points of Chapter 5—up to six times the endpoint search tolerance.

Correcting the rule to the stated experiment moves every endpoint amplitude upward, by 3.5 to 95 % across the twenty-four cells, with the largest shifts at the tightest targets: near the feasibility boundary, a fraction-of-a-percent change in the completion rule is amplified into a factor-of-two change in the derived allowance, the same mechanism as Section 6.4 in yet another guise. The family ordering of Chapter 5 is unchanged in all twenty-four cells. All headline results in this work use the corrected rule; Appendix B retains the strict-rule values beside them, and the comparison is archived as the `scheduler_boundary` record

(Appendix B.2).

4.4 External Validation Against Published Yields

Every validation to this point is internal: consistency between this work's own computational paths, or reproduction of the inherited simulator by code derived from it. One check against a number produced outside this work closes that gap. LIFE paper VI [4] publishes predicted detection yields for a 2.5-yr search with four 2m collectors at 5 % total throughput over Sun-like stars within 20 pc, for the same two occurrence scenarios the catalogs of Section 2.2 implement. Configuring this work's chain to those mission parameters, restricting the stellar scaffold to the same Sun-like definition, and running the fixed-campaign yield twice—once with the local-zodiacal defect deliberately reinstated, once corrected—tests two predictions at once, since the published yields were computed with the defect-era simulator.

Both predictions hold. With the defect reinstated, the chain lands on the published medians: 11.8 detected rocky planets in the optimistic habitable zone against the published 11 for the pessimistic occurrence scenario, and 14.7 against 14 for the optimistic one, with every planet category inside the published intervals in both scenarios. With the correction applied, the yields drop by 7.0 to 7.3 %. The chain therefore reproduces published LIFE science at its own configuration, and the normalization correction of Section 4.2 quantifies, at that configuration, by how much yields computed through the inherited path are optimistic. Three caveats bound the comparison: the stellar scaffold is not the published target list, the observing-sequence optimization differs in detail, and the planet-category windows use solar-normalized insolation bounds; the agreement is between medians well inside published uncertainty intervals, not a bit-level reproduction, and the comparison checks this work's chain against published results rather than re-evaluating those results. The per-category yields are archived in the reproducibility record (Appendix B.2).

Trade Space Exploration

From the reduced trade space to an iso-mission-time
random-noise allowance

With the trade space reduced, we now invert the AMS mapping. All results use the analytic background model of Section 4.2 and a deliberately conservative common operating point unless a plotted axis varies it. The working LIFE field-of-regard reference is 90° ; 65° is substantially more restrictive but still permits acceptable mission times. Current project estimates place the K-band limiting magnitude near 8; magnitude 7 is a pessimistic lower bound, and the completed scans show little remaining sensitivity near that value. A 10 h slew is the inherited baseline [7]; 12 h adds an intentionally simple 20% margin, while longer slews rapidly remove feasible budget space. The choices are therefore conservative comparison points, not optimized requirements. Hab2Max is evaluated at study targets of 5.5 and 6 yr; Hab2Min uses 7.5 and 8 yr. These targets are chosen, not inherited. The LIFE concept is postulated to run for roughly five years, so a shorter target is the more demanding and therefore the more informative requirement; for each catalog the lower target is the smallest half-year step at which a non-trivial positive allowance still exists at the nominal null order. Below it the zero-budget calculation already exceeds the requested time and no allowance can be reported. The second target adds half a year to expose how steeply the allowance responds to a relaxation of the science schedule. Because the two catalogs reach that feasibility threshold at different times, they are not compared at a common mission time; differences between Hab2Max and Hab2Min in Table 5.1 therefore combine the occurrence scenario with the different target, and should not be read as an occurrence effect alone. Appendix B records this

operating point together with the data provenance, the reproduction of the reported amplitudes, and the statistical interpretation of the sampled percentile.

5.1 The Iso-Mission-Time Budget

With the simulator built and the trade space reduced, the question of Chapter 1 can finally be asked in numbers. The *iso-mission-time budget* is the tolerated scalar amplitude of one imposed $N_I(\lambda)$ shape at a stated operating point. It is expressed in $\text{ph s}^{-1} \mu\text{m}^{-1}$ for the flat case or as a ramp endpoint. “Below” applies only to the same imposed shape and reference calculation.

The search starts with an amplitude bracket from 0 to $50,000 \text{ ph s}^{-1} \mu\text{m}^{-1}$. At every iteration the lower bracket endpoint is admissible and the upper endpoint exceeds the target mission time. An exponential fit through the endpoint evaluations proposes the next amplitude; it accelerates the search but does not define the answer. The new evaluation replaces one endpoint according to its mission time, so the bracket invariant is retained. The search stops when mission time lies within 0.05 yr of the target or the amplitude bracket is at most $10 \text{ ph s}^{-1} \mu\text{m}^{-1}$ wide. Reported amplitudes are therefore search point estimates; Section B.3 quantifies the resulting uncertainty by rerunning the complete search and recording the mission time each reported amplitude actually delivers. The two alternative stopping rules do not support a universal ± 5 amplitude uncertainty: a time-tolerance stop may leave a wider amplitude bracket, whereas a bracket-width stop only bounds the final numerical search interval.

Within a fixed target list, added independent variance can only lower individual SNRs. The full mission-time map also includes discrete rankings and a sampled percentile, so global monotonicity is an empirical property rather than a formal theorem. The completed scans showed nested contours and no folded iso-time branches at the sampled resolution. If the zero-budget calculation already exceeds the requested time, no non-negative allowance exists; the plotted dark region represents this physically infeasible operating point, not a failed numerical search.

5.2 The Shape of the Error Budget

The first question is how the affordable budget depends on its *distribution* over wavelength. We compare three single-amplitude families:

$$N_{\text{flat}}(\lambda) = A, \quad (5.1)$$

$$N_{\text{short}}(\lambda) = A \frac{18.5 \mu\text{m} - \lambda}{14.5 \mu\text{m}}, \quad (5.2)$$

$$N_{\text{long}}(\lambda) = A \frac{\lambda - 4 \mu\text{m}}{14.5 \mu\text{m}}. \quad (5.3)$$

Here A is the reported amplitude. The ramp endpoints are imposed by the choice of shape, not independently inferred wavelength limits. Figures 5.1 and 5.2 show the Hab2Max 5.5-year solutions.

Table 5.1 collects the final amplitudes for both architectures. Targets follow the rule used throughout: the lowest half year above that configuration's own zero-budget time—5.30 and 7.31 yr for the reference, 3.63 and 4.98 yr for the deeper null under the corrected completion rule—and the half year after it. Comparing the two at a common target would place the fourth-order design far from its own feasibility boundary, where the search saturates and the endpoint ceases to be informative. Both designs are built from 4 m collectors, so the six-aperture array collects 75.4 m^2 against the reference's 50.3 ; Section 3.4 states that convention and its consequences. Both halves come from the same general-combiner path, so the comparison shares a provenance, and the rounding convention is recorded in Appendix B.

The ordering is consistent across both catalogs and both mission-time targets. For the reference array the short-weighted endpoint is largest; for the deeper null the long-weighted endpoint is largest. There are no exceptions in the eight rows.

Relaxing the target by half a year increases the allowance sharply for either architecture, but not by the same factor. Each configuration is compared against its own primary target, so both start the same distance above their own feasibility boundary. The Hab2Max flat budget grows from 200 to 820 for the reference, a factor of 4.1, and from 350 to 890 for the triple nuller, a factor of 2.6; for Hab2Min it rises from 120 to 510, a factor of 4.2, against 270 to 520, a factor of 2.0. The reference therefore converts additional schedule into tolerated noise appreciably more readily, which is the counterpart of the result in Section 5.8: the deeper null buys its margin in schedule rather than in noise tolerance, so relaxing the schedule returns less to it than to the shallower design.

A ramp endpoint and a flat amplitude are not directly comparable quantities, so it is worth checking that the reversal is not an artifact of the chosen reporting convention. A

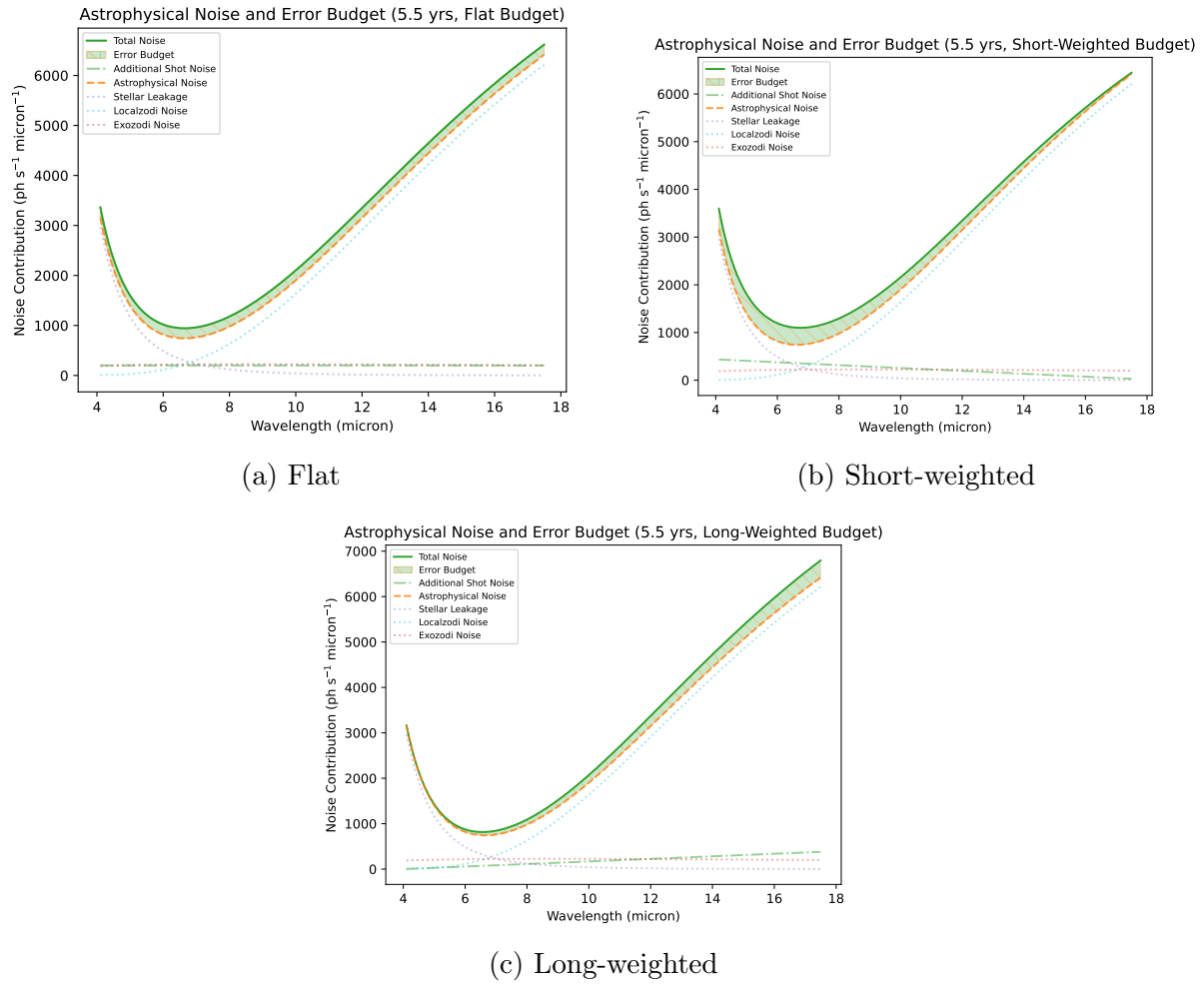


Figure 5.1: Final budgets for the reference double-Bracewell array, Hab2Max at the common operating point and 5.5-year target. Stellar leakage dominates the short-wave astrophysical background, so the short-weighted ramp admits the largest endpoint. As in Fig. 4.1, the amplitudes are ensemble quantities but the background curves beneath them are those of the single plotted target described there, at the 83rd percentile in distance and the 90th in effective temperature; they are not an ensemble average.

ramp rising linearly from zero to A across the band has mean $A/2$, so a flat budget of $A/2$ adds the same band-integrated noise; the integral-neutral ramp endpoint is therefore twice the flat amplitude in the same row. Measured against that reference, the preferred family clears the neutral value in every row of Table 5.1: the short-weighted endpoint at order two and the long-weighted endpoint at order four each exceed twice their row's flat amplitude in all four rows, so the reversal holds under equal integrated added noise and not only under equal endpoints. Under the historical strict completion rule this test failed in exactly one cell, and that cell was also where two formulations of the same array agreeing to 1.2×10^{-4} in mission time disagreed by 12 % in amplitude—the amplification of Appendix B surfacing in a place where it flipped a stated conclusion. The corrected rule removes the marginal case; the episode remains instructive as a measure of how close to the boundary such verdicts can sit.

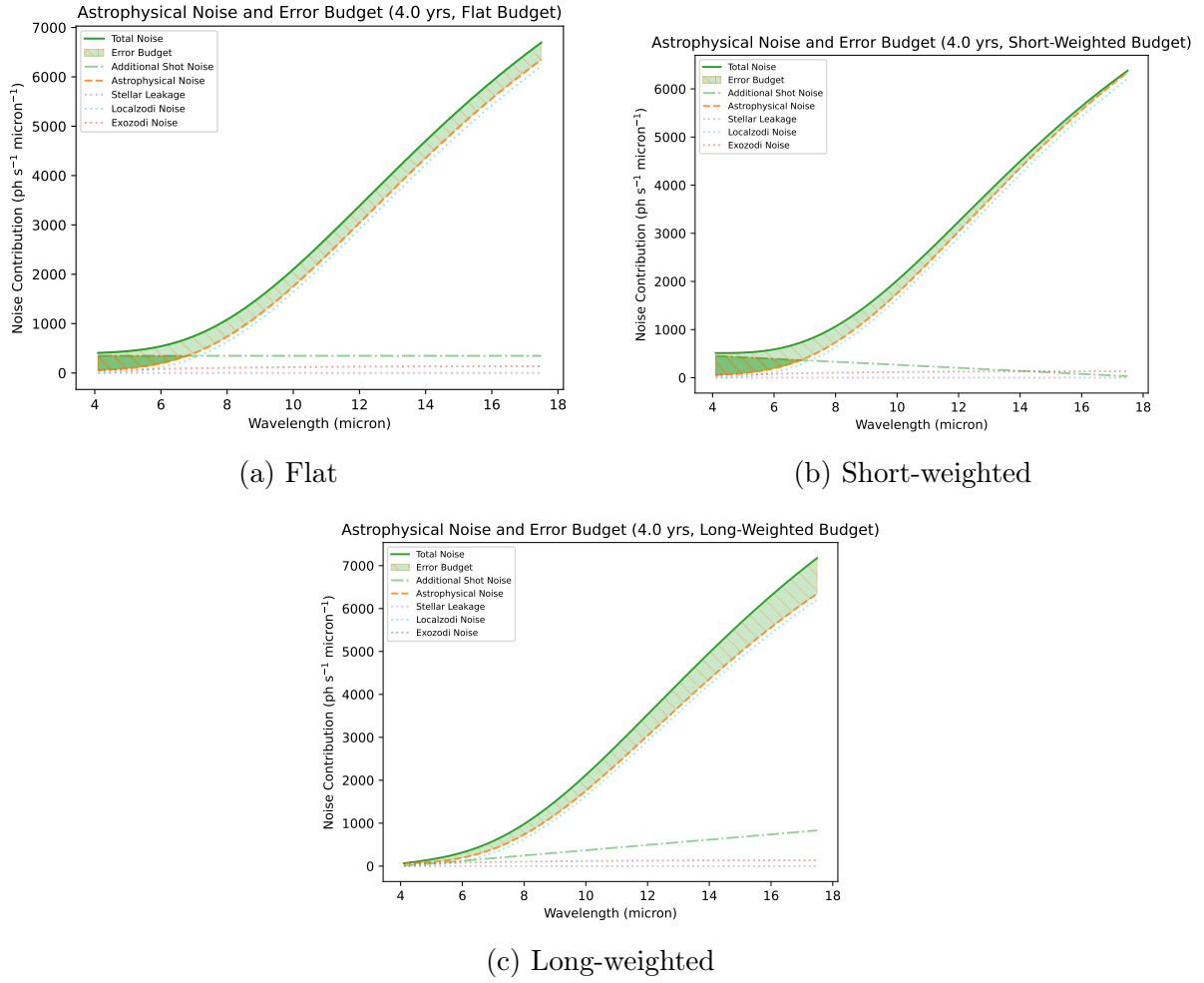


Figure 5.2: Final budgets for the six-aperture fourth-order design, at the same Hab2Max operating point as Fig. 5.1 but at this design's own 4.0-year primary target, and built from the same 4 m collectors, so it carries half again the collecting area. Geometric stellar leakage lies on the floor across the whole band, so the short-wavelength background no longer conceals added noise while the rising local-zodiacal term at long wavelengths still does; the long-weighted ramp consequently admits the largest endpoint. As in Figs. 4.1 and 5.1, the background curves are those of the single plotted host star rather than an ensemble average—the same star as in Fig. 5.1, so the architecture comparison between the two figures is unaffected by that choice.

The strong response to a half-year relaxation is not evidence of linear sensitivity. These target times lie near a steep part of the completion curve, where a small SNR loss can move marginal planets below the selected set and where the 90th-percentile universe can change identity. The budget endpoints should therefore be interpolated only through a fresh AMS inversion, not by scaling the values in Table 5.1.

That steepness has a second consequence, and it determines how the rows of Table 5.1 should be read. Section 5.4 measures how well each amplitude is determined, across three hundred bootstrap replicates per cell and twenty-five operating points spanning four architectures. The margin z in the table—the distance of each target above its configuration's

Catalog	Architecture	Target [yr]	Flat Amplitude	Short-wt. [ph s ⁻¹ μm ⁻¹]	Long-wt. [ph s ⁻¹ μm ⁻¹]	z	Rel. s.d. [%]	None
Hab2Max	Double Bracewell	5.5	200	440	410	2.13	48–52	18/300
Hab2Max	Double Bracewell	6.0	820	1800	1400	5.69	19–22	0/900
Hab2Max	Triple nuller	4.0	350	460	890	4.52	22–26	0/300
Hab2Max	Triple nuller	4.5	890	1600	2100	9.58	13–16	0/900
Hab2Min	Double Bracewell	7.5	120	280	230	3.02	39–42	3/300
Hab2Min	Double Bracewell	8.0	510	1100	1000	5.82	20–22	0/900
Hab2Min	Triple nuller	5.5	270	360	780	5.49	23–27	0/300
Hab2Min	Triple nuller	6.0	520	960	1400	9.54	14–16	0/900

Table 5.1: Final amplitudes under the corrected completion rule (Section 4.3), for the second-order reference array and the six-aperture fourth-order design of Section 3.4, each sized to its own response peak and each evaluated at *its own* mission-time targets. The last three columns record how well each row is pinned down: the margin z , the relative sampling spread over 300 or 900 bootstrap replicates, and the number of replicates returning no positive allowance. **Only rows with $z \gtrsim 3$ should be read as requirements**, as Section 5.4 establishes; the primary Hab2Max target sits below that margin. Amplitudes are rounded to two significant figures, the precision the search supports (Appendix B); raw search values are retained there. The largest endpoint in each row is set in bold; a ramp value is its non-zero imposed endpoint.

own feasibility boundary, in sampling standard deviations of that boundary—orders the outcome monotonically: at $z = 2.1$ the relative spread is about 50 % and six percent of replicates admit no positive allowance, at $z = 3$ the spread passes through 40 %, and by $z \gtrsim 5$ it settles near 20 %. Rows below $z \approx 3$ are reported for completeness, not as requirements.

The rows at $z \gtrsim 3$ are the ones to read as requirements, and the remainder of this work treats them as such: the flat allowances 820 and 510 ph s⁻¹ μm⁻¹ for the reference array on the two catalogs, and 890 and 520 for the six-aperture design, each determined to about 20 %. The primary Hab2Max row at $z = 2.1$ is retained because it is informative in a different way. It shows what an iso-mission-time requirement looks like when its target is set as tightly as the mission permits, which is the natural choice and the one this work made before measuring the consequence. Read that way it is a worked example rather than a result, and Section 5.4 states the criterion that separates the two cases.

The two-dimensional scans in Fig. 5.4 confirm that the inversion is not an artifact of comparing only the axis endpoints. At order two, equal-mission-time contours extend farther along the short-wavelength-budget axis. At order four, they extend farther along the long-wavelength-budget axis.

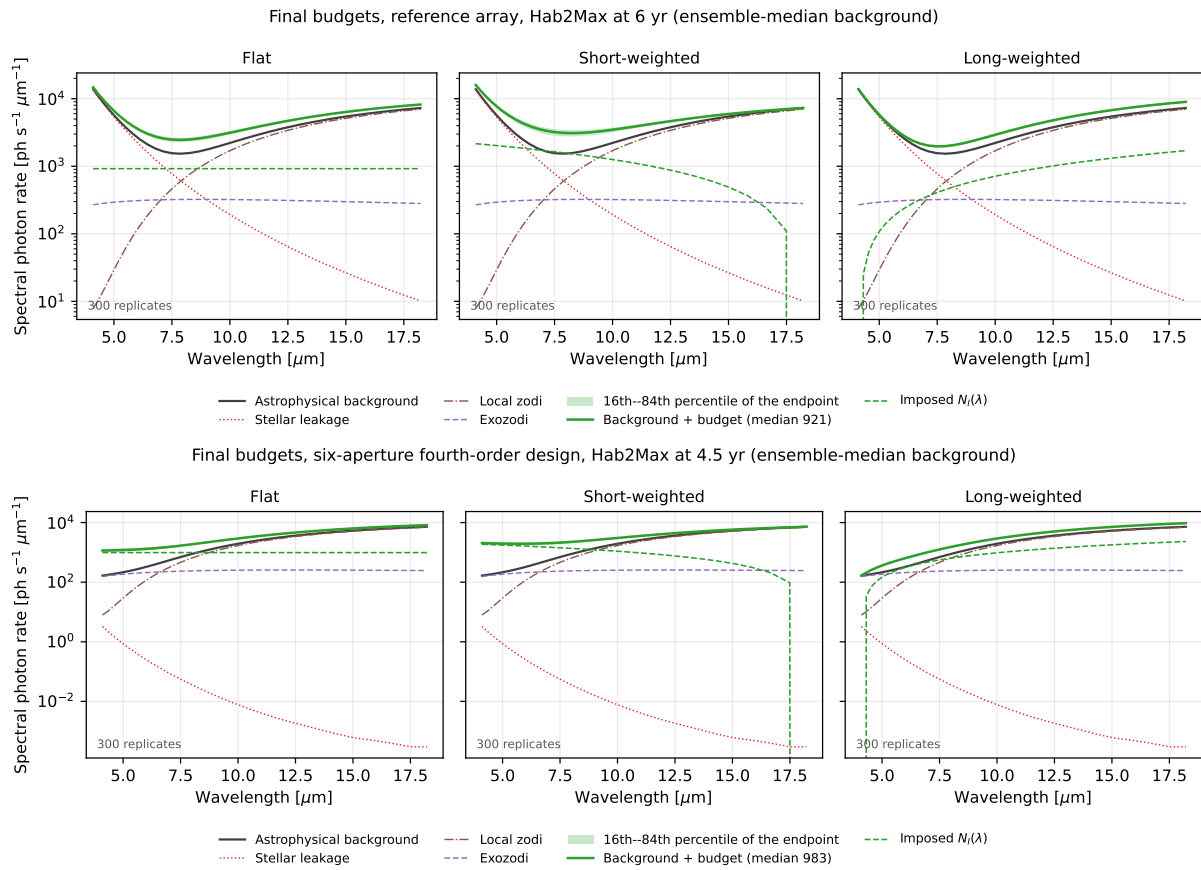


Figure 5.3: Final budgets at the two well-determined Hab2Max operating points, for the reference array (top) and the six-aperture fourth-order design (bottom). The background here is the *ensemble median* over catalog stars rather than the single plotted target of Figs. 5.1 and 5.2, so amplitude and background describe the same population; the shaded band is the 16th to 84th percentile of the endpoint over bootstrap replicates. The contrast between the panels is the mechanism behind the spectral reversal: stellar leakage dominates the reference’s short-wavelength background, whereas the fourth-order null drops it below every other term, leaving a background that rises monotonically toward long wavelengths.

5.3 The Random-Noise Error Budget

Table 5.1 is the numerical answer to the central research question for the tested families and operating points. For any chosen row and shape, the reported value is the amplitude at which the search terminated—within the 0.05 yr time tolerance of the target or, in a minority of rows, at the bracket-width stop, with achieved times recorded in Appendix B. That achieved time may lie slightly above or slightly below the target, as Table B.2 records, so a reported amplitude should be read as the boundary located to within the search tolerance rather than as a guaranteed admissible bound. The flat family is the most directly comparable scalar allowance. The ramps test where added random variance is least damaging, but their unequal spectral integrals mean that their endpoint amplitudes

must not be interpreted as a global cost ranking.

The result also does not provide independent limits at individual wavelengths. For example, the zero at one end of a ramp is imposed by its definition, not inferred from the experiment. Converting these aggregate allowances into an instrument requirement requires a separate allocation model that maps subsystem noise terms to the same reference plane and combines their independent variances.

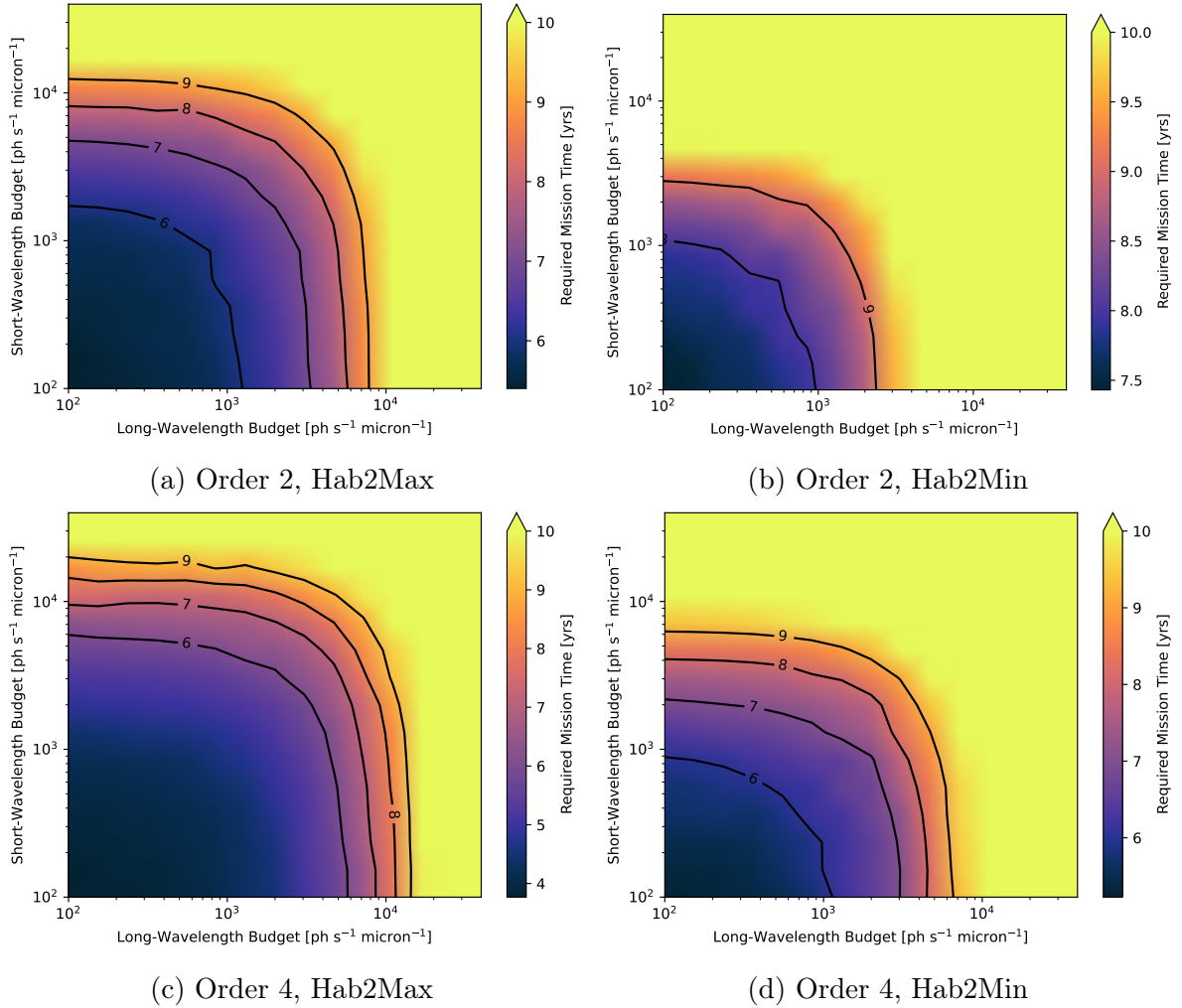


Figure 5.4: Required mission time over independent short- and long-wavelength ramp endpoints at magnitude 7, FoR 65°, and slew 12 h. Black curves mark integer mission-time contours. The preferred axis reverses between null orders in both catalogs.

5.4 How Well Determined Is a Reported Amplitude?

Every value in Table 5.1 is a point estimate, and the mission time it is solved against is the 90th percentile over 100 synthetic universes. Each reported amplitude is therefore a statistic of a finite sample, and a different draw of universes would have returned a

different value. This section measures how different, for all eight operating points, and identifies what controls it.

The measurement is a nonparametric bootstrap over the universe ensemble [24]. Each replicate draws 100 universes with replacement from the 100 available, relabels them, and relocates the endpoint from scratch, with the same catalog, the same stars, the same instrument and the same search tolerance throughout. Nothing physical changes between replicates; only the composition of the ensemble does. Three hundred replicates were run for each cell at the requirement targets and one hundred at the primary targets, individually seeded so the campaign shards reproducibly; a further seventeen operating points—intermediate targets on the same configurations and both targets of the two survey architectures of Section 5.8—were run at fifty replicates each and reserved for out-of-sample validation.

The amplitude at the tightest target is only marginally determined

At the tightest operating point the spread is half the value itself. For the reference array on Hab2Max at 5.5 yr the relative standard deviation is 48–52 % across the three families, and eighteen of the three hundred family-replicates return no usable positive allowance; in every such case the resampled mission reaches the target only within the search tolerance of its own zero-budget time, so no meaningful positive budget exists. A reported amplitude at this operating point is the centre of a wide distribution rather than a bound the mission is known to satisfy.

What controls it

The controlling quantity is how far the mission-time target sits above the configuration's own zero-budget time, measured in units of the sampling spread of that time. Running the same resamples but evaluating only the zero-budget mission time gives that spread directly: across all eight configurations measured here—four architectures on two catalogs—it is 1.9 to 3.1 % of the zero-budget time. Writing

$$z = \frac{T^* - \overline{T_0}}{\sigma(T_0)}, \quad (5.4)$$

where T^* is the mission-time target and $\overline{T_0}$ and $\sigma(T_0)$ are the mean and standard deviation of the zero-budget mission time over the resamples, orders the operating points by how well their amplitudes are determined. Throughout this section and wherever the criterion is applied, the classified quantity is the *flat-family* relative spread: flat is the directly comparable scalar allowance (Section 5.3), and it is the family the pre-registered

independent-population campaign evaluates; the other two families are reported alongside as a range. On the eight operating points of Table 5.1 the rank correlation between z and the flat-family relative spread is -0.98 , with a bootstrap 95 % interval of $[-0.95, -0.52]$ over points; on the seventeen held-out operating points, which include the two survey architectures the relation was never fitted on, it is -0.94 .

The margin must be taken against the resample mean rather than against the reported zero-budget time. Bootstrapping a high quantile is biased—the resampled order statistic is discrete and over-dispersed relative to the true sampling distribution [26, 27]—and for the tightest configurations the two differ by an appreciable fraction of a standard deviation. Because $\sigma(T_0)$ is two to three percent of T_0 in every configuration measured, a margin expressed as a percentage of the zero-budget time is a serviceable proxy, with $z \approx 3$ corresponding to roughly 8 %—but it is a proxy, and it degrades exactly where the bias is largest.

Below $z \approx 2$ the spread approaches the value itself and replicates begin to return no allowance. At $z \gtrsim 3$ the spread passes through 40 % and declines steadily thereafter, reaching 20 % by $z \approx 5$ and 10 % by $z \approx 12$. **The practical consequence is that a mission-time target should be set at least three sampling standard deviations above the configuration's zero-budget time if the resulting requirement is to be quoted as a number.** The 40 % level itself is this study's operational quotability threshold—the point at which the spread stops being comparable to the value—adopted as a working boundary rather than derived from an external engineering criterion. That statement was fixed on the eight operating points of Table 5.1 and then tested on seventeen operating points it had never seen, including both survey architectures of Section 5.8: all fifteen held-out points at $z \geq 3$ have flat-family spreads below 40 %, and both held-out points below $z = 3$ have spreads above it (Fig. 5.5). The boundary is soft rather than sharp, and the family range shows it: at the held-out $z = 3.46$ point the short-weighted family reaches 42 %, the one family-range excursion above the threshold among the fifteen, while the flat family stays at 38 %.

The criterion was then tested beyond the shared catalog ensembles, on eighty genuinely independent population realizations generated with the public P-Pop tool on the same stellar scaffold, under a pre-registered protocol with fixed seeds, thresholds, and analysis (Appendix B). Forty realizations omit and forty apply the binary-occurrence suppression, bracketing the catalogs' own configuration. Between independent realizations the picture repeats: spreads decline monotonically with the between-realization margin, infeasible draws appear only below $z \approx 2$, and none of the six tested cells at $z \geq 3$ —in either configuration—exceeds 40 % spread or loses a draw (the full thirteen-cell matrix, with bootstrap intervals on every spread and margin, is Table B.4). The hard thresh-

old itself is soft: two cells just below $z = 3$ came in at 30–38 %, better determined than a step at $z = 3$ predicts, so the transition is a band at $z \approx 2$ –2.5 and $z \geq 3$ is a *conservative* sufficient margin. Between-realization $\sigma(T_0)$ is comparable to or smaller than the within-catalog bootstrap values, so the bootstrap intervals reported above are themselves conservative—a statement demonstrated for the two occurrence-model configurations tested, not in general. Under the tightest-half-year rule used throughout this work, the primary Hab2Max target sits at $z = 2.1$ in the marginal regime, while every other reported target clears the criterion. Section 5.3 reports on that basis.

Architecture	Catalog	Target [yr]	z	Relative s.d.	No allowance
Collinear kernel (h)	Hab2Max	3.5	1.04	83–86 %	26 %
Double Bracewell	Hab2Max	5.5	2.13	48–52 %	6 %
Kernel-5 (deep pair) (h)	Hab2Max	5.5	2.36	52–56 %	5 %
Double Bracewell	Hab2Min	7.5	3.02	39–42 %	1 %
Kernel-5 (deep pair) (h)	Hab2Min	8.0	3.02	34–37 %	0 %
Collinear kernel (h)	Hab2Min	5.0	3.17	32–39 %	0 %
Double triple nuller (h)	Hab2Min	5.25	3.46	36–42 %	0 %
Double Bracewell (h)	Hab2Max	5.75	3.91	25–29 %	0 %
Double Bracewell (h)	Hab2Min	7.75	4.42	28–30 %	0 %
Double triple nuller	Hab2Max	4.0	4.52	22–26 %	0 %
Kernel-5 (deep pair) (h)	Hab2Min	8.5	5.29	19–24 %	0 %
Double triple nuller	Hab2Min	5.5	5.49	23–27 %	0 %
Double Bracewell	Hab2Max	6.0	5.69	19–22 %	0 %
Collinear kernel (h)	Hab2Max	4.0	5.74	18–25 %	0 %
Double Bracewell	Hab2Min	8.0	5.82	20–22 %	0 %
Double triple nuller (h)	Hab2Max	4.25	7.05	15–17 %	0 %
Double Bracewell (h)	Hab2Min	8.25	7.22	19–20 %	0 %
Kernel-5 (deep pair) (h)	Hab2Max	6.0	7.24	21–24 %	0 %
Collinear kernel (h)	Hab2Min	5.5	7.37	18–22 %	0 %
Double Bracewell (h)	Hab2Max	6.25	7.47	16–18 %	0 %
Double triple nuller (h)	Hab2Min	5.75	7.52	18–21 %	0 %
Double triple nuller	Hab2Min	6.0	9.54	14–16 %	0 %
Double triple nuller	Hab2Max	4.5	9.58	13–16 %	0 %
Double triple nuller (h)	Hab2Min	6.25	11.57	13–14 %	0 %
Double triple nuller (h)	Hab2Max	4.75	12.11	10–11 %	0 %

Table 5.2: How well each reported amplitude is determined, against the margin z between its mission-time target and its own feasibility boundary in units of the sampling spread of that boundary, for all twenty-five operating points measured in this work. Rows marked (h) were held out of the relation’s original statement: intermediate targets and both survey architectures of Section 5.8. The ordering holds across four architectures, two catalogs, aperture counts of four to six, null orders two and four, and absolute targets from 3.5 to 8.5 yr, none of which appears in z . Relative spread is quoted as a range over the three budget families; the criterion itself classifies the flat-family spread (Section 5.4). The last column is the fraction of replicates returning no usable positive budget.

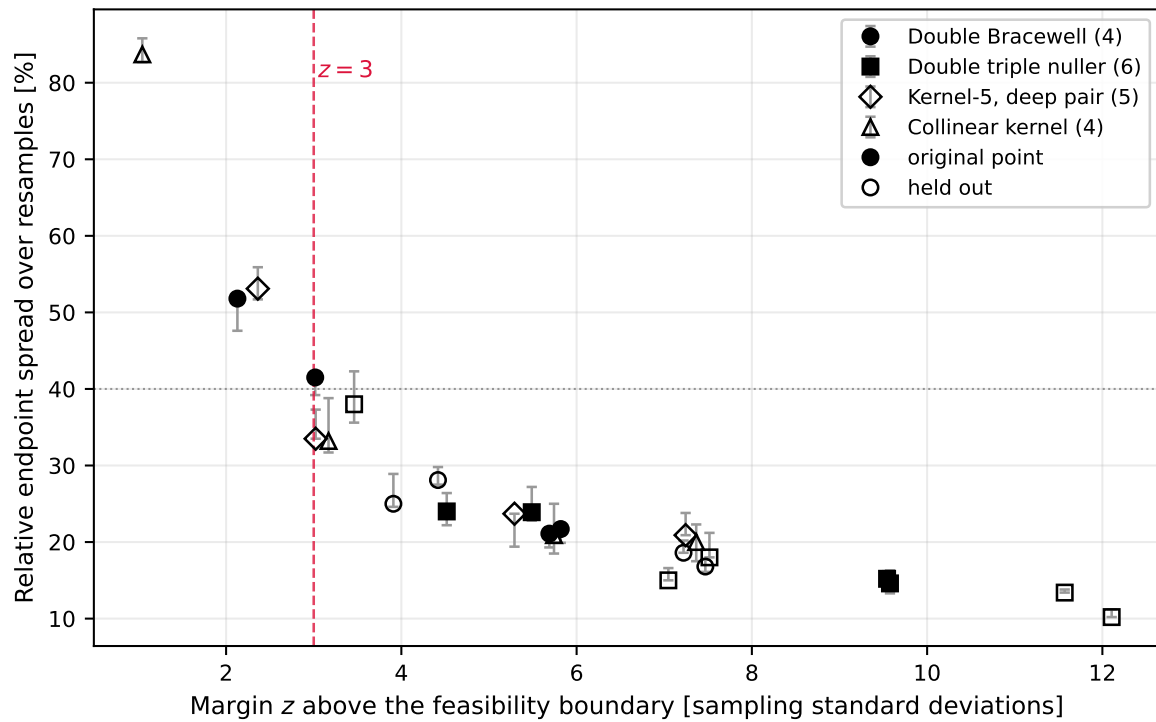


Figure 5.5: The z -relation across every operating point measured in this work. Filled markers are the eight points of Table 5.1 on which the relation was stated; open markers are the seventeen held-out points, including both survey architectures, which entered only after the relation was fixed. Markers show the flat-family spread, the variable the criterion classifies; error bars span the three budget families. The dashed line marks $z = 3$; the dotted line the 40 % spread level it corresponds to. Every held-out point falls on the same declining relation, so the criterion is a property of the requirement rather than of any one array.

What survives

The *ordering* of the budget families does. Comparing families within each replicate rather than across replicates, the preferred family's endpoint exceeds the flat amplitude in every one of the ninety-four feasible replicates of the Hab2Max 5.5 yr cell, and in every feasible replicate of every other cell measured. The full three-way ordering holds in 79 % of replicates at that tightest point—the exceptions all have the two ramps within a few tens of percent of each other—rising to 96–99 % at the requirement targets. The same resampling that moves the amplitudes by half their value leaves the ranking essentially untouched.

This is the sharpest instance of the amplification described in Section 6.4, because no modelling error is involved: the variability is intrinsic to estimating a requirement from a finite ensemble near a feasibility boundary. It also separates what this study establishes from what it merely reports. The conclusions about spectral *shape* are supported by every

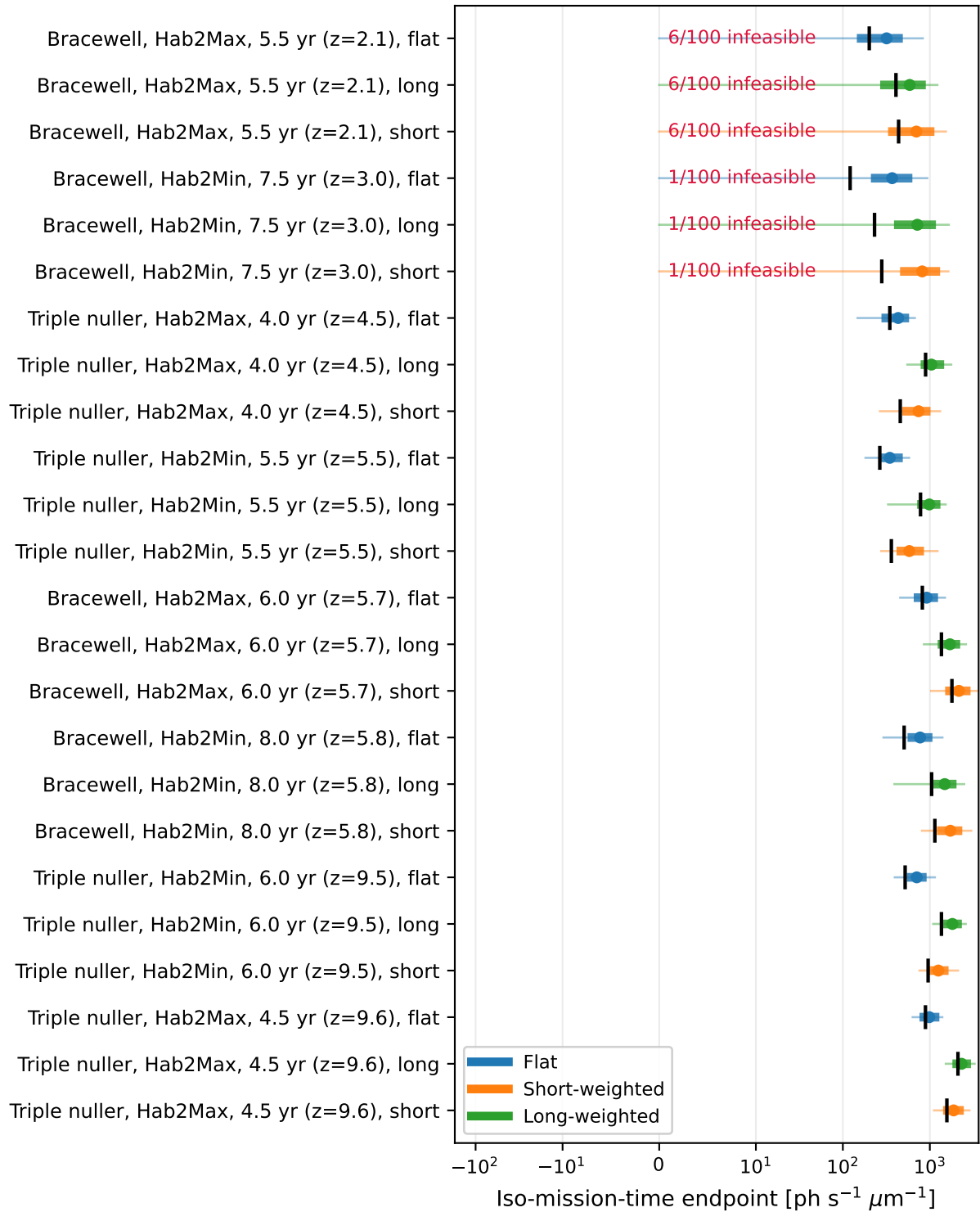


Figure 5.6: Bootstrap distribution of every reported amplitude, one row per cell of Table 5.1, ordered by the margin z . The thick bar is the 16th to 84th percentile over 300 or 900 replicates, the thin bar the full range, the dot the median, and the black tick the reported value. Rows for which some replicates returned no usable allowance are annotated with the count. The ordering of the axis is the finding: rows near the top, whose targets sit closest to the feasibility boundary, carry intervals of half their own values, while rows near the bottom are pinned to better than twenty percent.

measurement made here; the conclusions about *magnitude* are supported only where z is large enough.

Search tolerance is not among the limiting uncertainties: removing the solver's achieved-time jitter from the replicate endpoints changes no cell's spread by more than 0.7 percentage points and no classification (Appendix B). Three limitations attach to the interval itself. A bootstrap of a high quantile is coarse, since the resampled order statistic is discrete and over-dispersed relative to the true sampling distribution [26], so these spreads are upper bounds on the standard error rather than unbiased estimates. Three hundred replicates resolve the central interval and the 2.5th–97.5th percentile range reported alongside it, a count consistent with standard guidance for percentile intervals [28, 25]; the extreme tails remain unquantified. And the absolute location of the feasibility boundary is conditional on the occurrence-model configuration: the independent-realization campaign shows the binary-suppression choice alone moves T_0 by 5–15%, an order of magnitude beyond sampling uncertainty, and the thesis catalogs lie between the two tested configurations. The z -calibration transfers across configurations; the boundary itself does not, and the catalogs' exact generation configuration is bracketed rather than reproduced (Section 6.4). Every per-replicate endpoint, per-universe component time, and search bracket is retained in the `uncertainty` archive (Appendix B.2).

5.5 Dependence on the Agnostic Inputs

A requirement quoted at one operating point immediately invites the question of what happens away from it. The remaining scans answer it for the flat budget, varying two agnostic inputs at a time. They therefore show operating-point dependence of the flat allowance and cannot automatically be transferred to either ramp family. The scans were computed under the strict completion rule then in force; the corrected rule of Section 4.3 shifts every absolute level, but the feasibility boundaries and the contour topology, which are what this section reads, shift together.

Figure 5.7 varies limiting magnitude and field of regard at fixed 12 h slew time. In all four panels, widening the field of regard has the larger effect because it admits more potential targets. The magnitude dependence largely saturates near the adopted value of 7: once most high-SNR qualifying stars pass the magnitude cut, admitting still fainter targets adds little to the optimized top-50 sample. Field of regard acts differently because it removes entire sky regions irrespective of target brightness.

The fourth-order grids quantify that asymmetry directly. At the adopted field of regard of 65° , sweeping the limiting magnitude across the full range from 6 to 9 changes the allowance by a factor of only 1.26 for Hab2Max and 1.23 for Hab2Min. Sweeping the

field of regard at fixed magnitude 7 changes it by 14.4 and 3.3 over the same grids. For this architecture the requirement is therefore set by pointing access far more than by target brightness, and its iso-budget contours are correspondingly close to vertical.

The two architectures share that qualitative dependence but not its extent. The fourth-order design stays feasible further into restrictive pointing, down to 58° for Hab2Max and 61° for Hab2Min against the reference's 65° ; the grid spacing is 3.6° , so those margins are one and two grid steps and are resolved rather than precisely measured. Its allowance also peaks higher, at 560 and $414 \text{ ph s}^{-1} \mu\text{m}^{-1}$, though the comparison is not like for like: each design is scanned at its own mission-time target and the six-aperture array carries half again the collecting area. Two thirds of the Hab2Max grid and three fifths of the Hab2Min grid admit a positive budget, against roughly half for the reference. The deeper, larger design therefore both reaches operating points the reference cannot and tolerates more noise where both are feasible, which is what half again the aperture buys.

Figure 5.8 varies slew time and field of regard at fixed limiting magnitude 7. Longer slews consume time without adding SNR and therefore reduce the admissible budget. The dependence is especially important because each selected planet receives multiple follow-up visits: slew overhead is charged during the shared detection campaign, four additional orbit epochs, and characterization. Wide field of regard and short slew time therefore act together: more targets are available and less overhead is charged for moving between them.

Slew time is by a wide margin the more damaging of the two agnostic inputs examined in this section. At the adopted field of regard of 65° , the fourth-order allowance falls by a factor of 12.2 for Hab2Max and 63.5 for Hab2Min across the tested slew range, against the factors of 1.26 and 1.23 that the entire magnitude range produces on the same architecture. Overhead, unlike target brightness, is charged on every visit and cannot be recovered by the scheduler.

Both architectures produce the same wedge: feasibility requires either a short slew or a wide field of regard, and fails in the upper-left corner where neither holds. The fourth-order design occupies a larger wedge, reaching down to 40° of pointing access for Hab2Max and 47° for Hab2Min at the shortest slews, where the reference is already exhausted. Even so, only 38.7% and 38.2% of the two grids admit any positive budget, so most of this plane is not a trade at any noise level but simply infeasible, and that remains true of the more capable design. Inside the wedge its allowance peaks at about 1800 and $1500 \text{ ph s}^{-1} \mu\text{m}^{-1}$. The pattern is consistent with Section 5.8: schedule overhead binds hardest exactly where the instrument is otherwise capable, and collecting area cannot buy back slew time.

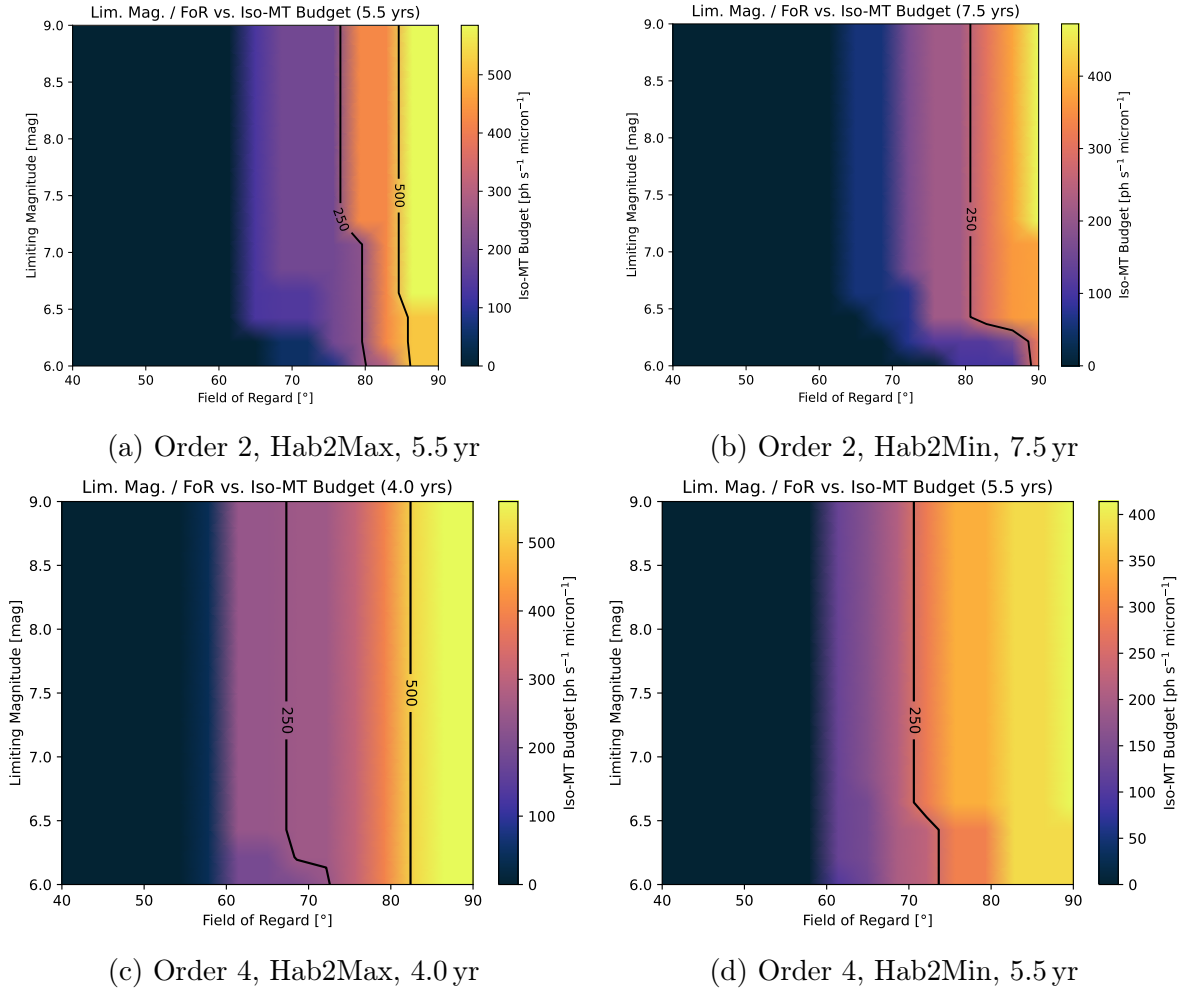


Figure 5.7: Final iso-mission-time flat-budget scans over limiting magnitude and field of regard, with slow time fixed at 12 h. Panels (a) and (b) are the second-order reference array; panels (c) and (d) are the six-aperture fourth-order design of Section 3.4. Each design is sized to its own response peak, built from the same 4 m collectors, and scanned at its own primary mission-time target (5.5 and 7.5 yr against 4.0 and 5.5 yr), so the six-aperture array carries half again the collecting area and the panels are not a like-for-like comparison. Each grid is 15×15 points (3.6° field-of-regard spacing). Dark regions have no admissible positive budget; the fourth-order design stays feasible to 58° and 61° against the reference's 65° , and admits a positive budget over 66.7% and 60.0% of its grids. **Colour scales differ between panels**, peaking at 560 and $414 \text{ ph s}^{-1} \mu\text{m}^{-1}$ for the fourth-order panels.

5.6 Catalog Dependence

The Hab2Min and Hab2Max calculations show the same qualitative shape ordering and the same null-order inversion. Hab2Min requires longer mission-time targets and generally returns smaller allowances at otherwise matched operating points, consistent with its less favorable planet population. Because the adopted mission-time targets differ between catalogs, the numerical difference cannot be attributed to $\eta_\oplus$ alone. Figure 5.9 illustrates

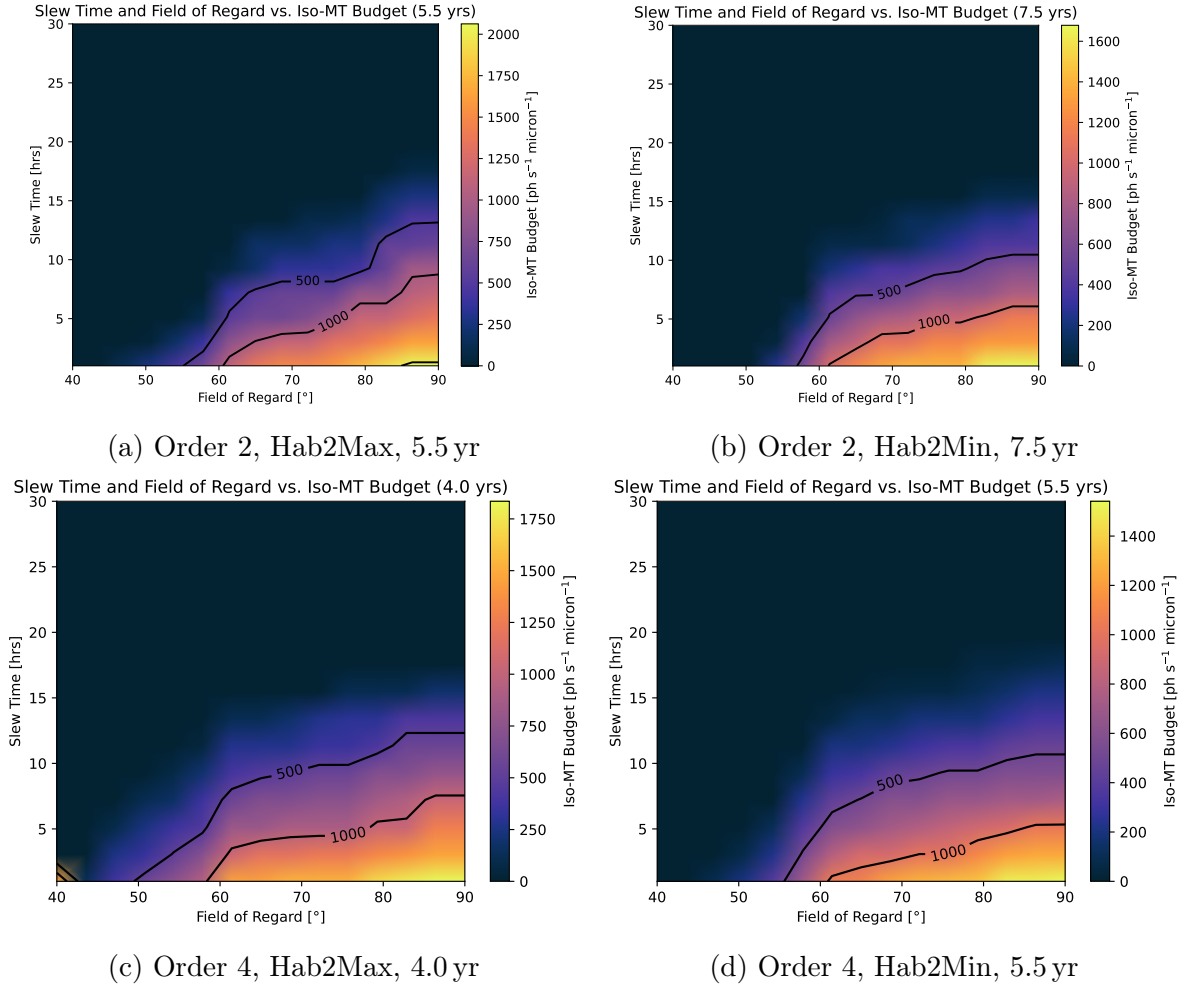


Figure 5.8: Final iso-mission-time flat-budget scans over slew time and field of regard at limiting magnitude 7. Panels (a) and (b) are the second-order reference array, panels (c) and (d) the six-aperture fourth-order design, each at its own baseline prescription and primary target, and each built from the same 4m collectors, so the six-aperture array carries half again the collecting area and the panels are not directly comparable. Dark regions have no admissible positive budget: 61.3% and 61.8% of the two fourth-order grids are infeasible at any noise level. Both architectures give the same wedge, feasible only where the slew is short or the field of regard wide; the fourth-order design reaches down to 40° and 47° of pointing at the shortest slews and peaks at about 1800 and $1500 \text{ ph s}^{-1} \mu\text{m}^{-1}$ inside it. **Colour scales differ between panels.**

the target filtering for one Hab2Max selection and is not evidence of catalog independence. It also explains why the magnitude dependence saturates while the field-of-regard dependence does not. Of the 26.7% of Experiment-1-eligible host stars excluded at the adopted cuts, the magnitude limit accounts for 20.5 percentage points and the field of regard for 7.5. The magnitude cut removes many more stars, but it removes the faint ones, which contribute little to a top-50 sample; the field-of-regard cut removes a wedge irrespective of brightness and so takes good targets with it. That the larger cut is the less consequential one is the whole content of the saturation seen in Section 5.5.

5.7 Why the Spectral Preference Reverses at a Deeper Null

Figure 5.3 shows the mechanism directly, and this section states what it rests on. For the reference array the stellar-leakage term dominates the background below about $8\text{ }\mu\text{m}$; for the six-aperture design it lies below every other term across the whole band. What remains once it is gone rises monotonically toward long wavelengths, so that is where added variance can be hidden, and the tolerant ramp changes ends.

The suppression is the one predicted analytically. Equations 4.3 and 4.4 give a fourth-to-second-order ratio of $x_\star^2/8$, and across the 4505 catalog stars $x_\star = 2\pi b R_\star/\lambda$ stays above 0.005 over the modeled band, outside the cancellation-sensitive regime of the direct Bessel sum. The median leakage ratio is about 6.3×10^{-5} at $10\text{ }\mu\text{m}$ and 1.9×10^{-5} at $18.5\text{ }\mu\text{m}$, and direct positive quadrature agrees with the analytic response over the complete catalog range.

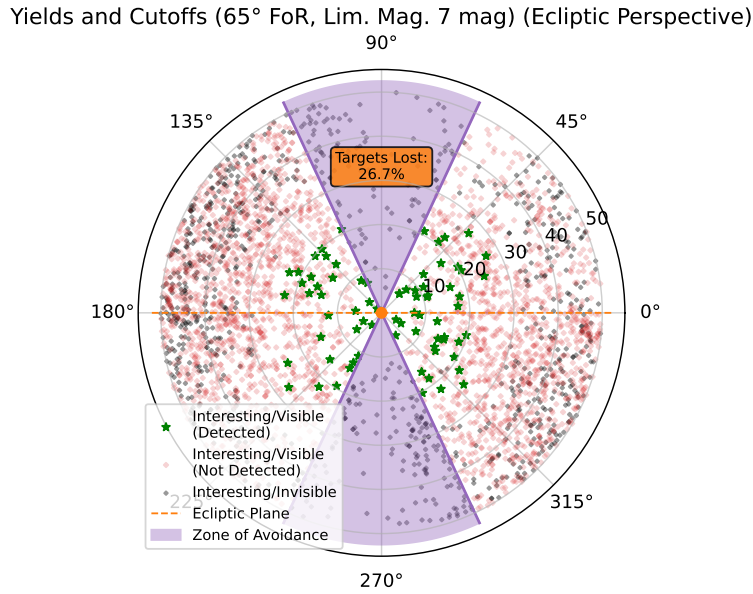


Figure 5.9: Ecliptic view of the final Hab2Max selection at limiting magnitude 7 and FoR 65°, for the second-order reference array; “interesting” denotes the 3605 Experiment-1-eligible host stars. *Reading the axes.* The radial coordinate is distance to the host star in parsec, with gridlines every 10 pc; the polar angle is ecliptic *latitude*, and stars at ecliptic longitude beyond 180° are reflected through the origin so that both hemispheres share the plane. Longitude is therefore not resolved and this is not a sky map. The dashed orange line is the ecliptic plane and the shaded wedges the zone of avoidance, $|\text{latitude}| > 65^\circ$, which is how the field of regard enters the model. The annotation counts eligible host stars given zero SNR by *either* exclusion, since one mask applies both. **The loss is dominated by the magnitude limit, not by the zone of avoidance the figure draws;** Appendix B.2 decomposes it.

One caveat limits the attribution. The collecting-area contribution separates exactly

(every photon rate is linear in aperture area, so a global rescale moves the signal-to-noise ratio by exactly the root of the area ratio), but within the remaining equal-area factor no ablation isolates leakage suppression from the extended-source and science-pair-fraction changes that accompany it, so that part of the mechanism is identified rather than proven.

Table 5.3 collects the effects that change between the two architectures alongside their consequences for the reported allowance. A deeper null does not act on the stellar term alone, and the background terms separate by how they resolve. The far-field average of the response falls from $1/4$ to $1/6$ per output—one over the aperture count, the general result for a lossless combiner—but the collecting area rises by half, and for a uniform background the two cancel exactly: any lossless design sharing a collector diameter transmits $N \cdot A_{\text{ap}} \cdot \langle R \rangle = A_{\text{ap}}$ of it, and the measured local-zodiacal ratio between the two architectures is 1.000 (Appendix A). The exozodiacal term, which the fringe pattern partially resolves, falls by a band-median factor of 0.76 (Hab2Max) and 0.74 (Hab2Min). Only for the resolved stellar disk does the fourth-order dependence remove the term almost entirely.

Effect	Double Bracewell, 4 Triple nuller, 6 apertures apertures	
Near-axis stellar leakage	Leading term $x^2/32$	Leading term $x^4/256$; median ratio 6.3×10^{-5} at $10 \mu\text{m}$
Far-field response average	$1/4$	$1/6$, i.e. one over the aperture count
Light in the output subtraction pair	50 % (two of four outputs)	33 % (two of six outputs)
Dominant short-wave background	Stellar leakage	Leakage largely removed; added short-wave variance becomes more costly
Least damaging tested ramp	Short-weighted	Long-weighted
Collecting area	50.3 m^2	75.4 m^2 (same 4 m collectors)
Zero-budget mission time, Hab2Max	5.30 yr	3.63 yr

Table 5.3: Synthesis of the physical changes between the reference array and the six-aperture fourth-order design, and their consequences for the reported allowance. The two designs share a collector diameter rather than a total collecting area, so the last two rows combine the effect of the deeper null with that of half again the light; Section 3.4 separates them.

5.8 Comparing Realizable Architectures

The preceding results describe the reference array. Because the transmission model of Section 3.3 accepts any combiner, the same inversion can be run for a genuinely realizable

fourth-order design and the two compared directly, each sized to its own response peak.

That last point is not a detail. The baseline prescription of Section 3.6 places the first peak of the differential response on the habitable-zone centre, and the constant that does so is a property of the response, not of the mission: it is 0.589 for the double Bracewell, 1.138 for the double triple nuller, 1.125 for the pentagon's deep pair, and 2.022 for the collinear kernel. Sizing the six-aperture design with the reference constant leaves it operating at 31 % of its achievable signal. A comparison made that way measures the mismatch, not the architecture.

Table 5.4 reports the headline comparison. With no added budget the six-aperture fourth-order design completes Experiment 1 substantially faster than the reference, by 31.5 % for Hab2Max and 31.9 % for Hab2Min, and it tolerates a larger flat allowance at its own operating point.

Both figures must be read with the sizing convention of Section 3.4 in view. The two designs share a collector diameter, not a total collecting area, so the six-aperture array works with 75.4 m² against 50.3: half again as much light. A large part of the gain is therefore purchased with aperture count rather than with null depth, and the comparison is deliberately not resource-neutral. The same design evaluated at equal total collecting area, with 3.27 m collectors and correspondingly wider beams, instead came out 16 % *slower* than the reference (measured under the strict completion rule; the convention dependence is geometric and carries over). The honest reading is that a deeper null does not by itself buy schedule at fixed collector size, and the null depth changes which part of the spectrum the remaining allowance should be spent on. The survey below sharpens this: a fourth-order design with *fewer* apertures than the reference can also outrun it, so the schedule gain is not a property of aperture count either, but of where each design's response puts its signal.

The gain is not uniform across the population, and the way it varies is what identifies its origin. Over the 2641 visible host stars the fourth-order design improves the one-hour signal-to-noise ratio by a median factor of 1.31 for both catalogs, with only 8 % of targets worse off and a sixteenth-to-eighty-fourth percentile range of 1.11 to 1.45. Ranking targets by their reference signal-to-noise ratio, the best fifty—the number Experiment 1 observes—improve by a median factor of 1.54, appreciably more than the population median.

The two causes separate exactly, because every photon rate in the chain—planet signal and all three background terms—is linear in the per-aperture collecting area, so a global area rescale multiplies the signal-to-noise ratio by the square root of the area ratio and dividing out $\sqrt{1.5} = 1.22$ is an ablation, not an estimate. At equal collecting area the

Catalog	Architecture	Area [m ²]	Zero-budget [yr]	Target [yr]	Flat	Long-wt.
Hab2Max	Double Bracewell	50.3	5.30	6.0	820	1400
Hab2Max	Double triple nuller	75.4	3.63	4.5	890	2100
Hab2Min	Double Bracewell	50.3	7.31	8.0	510	1000
Hab2Min	Double triple nuller	75.4	4.98	6.0	520	1400

Table 5.4: Reference and fourth-order architectures, each sized to its own response peak. Amplitudes are quoted at each design's *secondary* target rather than its primary one, since Section 5.4 shows the primary targets sit too close to the feasibility boundary for their amplitudes to be well determined; the zero-budget times are independent of the target and stand for both. Amplitudes are in $\text{ph s}^{-1} \mu\text{m}^{-1}$. **The two designs share a collector diameter of 4 m, not a total collecting area**, so the six-aperture array collects half again as much light; Section 3.4 explains why that convention is forced and what the opposite one gives. Because the targets differ, the amplitude columns are not a like-for-like comparison either: they state what each design tolerates where it actually operates.

combiner change contributes a median factor of 1.07, and 28 % of targets are worse off; over the reference's best fifty targets the equal-area factor is 1.26. That concentration is where the physics is: the best targets are the nearest and most resolved stars, so they are the ones whose noise is dominated by geometric stellar leakage, and they are the ones a fourth-order null helps most. The schedule gain therefore has two separable causes—area on every target, a concentrated combiner gain on the leakage-limited best—and only the smaller, concentrated one is attributable to the deeper null.

The mechanism is the one the null order was meant to exploit. The best targets are the nearest and brightest stars, which are also the most resolved, so they suffer most from the geometric stellar leakage that a second-order null cannot suppress. The deeper null removes it. Faint distant targets are background-limited instead, where a lower far-field transmission costs and the absent leakage was never the constraint. A mission observes the top of the distribution, not its median, and that is why a design that is worse on average completes the program sooner.

The spectral preference established in Section 5.7 survives the substitution. Across both catalogs and both mission-time targets the long-weighted family remains the most tolerant, by factors of 1.28 to 1.91, and exceeds the integral-neutral reference of twice the flat amplitude in every case. The reversal is therefore a property of a buildable architecture and not of an idealized response.

The operating-point scans of Figs. 5.7 and 5.8 carry the same message in a second form. Against the reference, the fourth-order panels remain feasible at a field of regard some four degrees more restrictive, while the allowance they admit at a generous field of regard is smaller by roughly a factor of two. Their contours are also more nearly vertical, so for

that architecture the requirement is governed by pointing access almost to the exclusion of limiting magnitude.

The survey

The condition of Section 3.4—two rows vanishing to second order, remixed into a conjugate pair—admits designs the six-aperture template does not suggest, and the survey evaluates the two smallest: the pentagon kernel nuller, whose five-point discrete-Fourier combiner carries a second-order and a fourth-order conjugate pair on the same hardware, and the collinear kernel, four apertures on a line whose order-four and order-six rows remix into a fourth-order pair carrying half the collected light. Table 5.5 summarizes all four designs; the family attribution is to Martinache & Ireland [18] and Laugier et al. [14], who note lossless five-input realizations without constructing one—the explicit combiners, the null-order identification, and the mission-level evaluation are this work's.

The outcome scrambles every simple ordering. The collinear kernel, with four apertures and 50.3 m^2 , completes the program faster than the six-aperture design with 75.4 m^2 —3.50 against 3.63 yr for Hab2Max—while the pentagon's deep pair, at five apertures and a fourth-order null, is slower than the second-order reference. The discriminator is neither aperture count nor null order but where each response puts its signal relative to the baseline envelope. The prescription clips the requested baseline to the 100 m cap for a quarter of the scaffold under the reference constant and for 87 % under the collinear one, yet the collinear design loses almost nothing to clamping: at the clamped working point it retains 61 % of its peak chopped response, against 23 % for the pentagon's shallow pair, whose narrow high-order fringes leave clamped stars nearly blind. The envelope-sensitivity axis makes the same point from the other side: doubling the cap to 200 m changes the four viable configurations' zero-budget times by at most six percent, while the shallow pair improves by a fifth and still completes no program in under fifteen years. The envelope punishes not a large sizing constant but a response without low-baseline support.

The pentagon's same-hardware comparison is the cleanest statement of that conclusion. Its two conjugate pairs share every resource—apertures, area, geometry, throughput convention—and differ only in null order and fringe structure; choosing the deep pair yields a working instrument while the shallow pair, nominally the same instrument class as the reference, does not reach any adopted target inside the envelope. A null-order comparison at fixed hardware is what Table 5.4 could not offer; it shows the response shape doing all the work.

Both survey architectures also test the conditioning claim on designs it was never fitted

Design	N	Order	Pair light	Const.	Clipped	T_0 [yr]
Double Bracewell	4	2	50 %	0.589	26 %	5.30 / 7.31
Double triple nuller	6	4	33 %	1.138	63 %	3.63 / 4.98
Kernel-5 (deep pair)	5	4	40 %	1.125	63 %	5.35 / 7.57
Collinear kernel	4	4	50 %	2.022	87 %	3.50 / 4.90

Table 5.5: The architecture survey under the corrected completion rule, every design sized to its own response peak and evaluated with the same catalogs, scheduler, and search tolerance. N is the aperture count, “Pair light” the fraction of collected light in the chopping pair, “Const.” the response-peak sizing constant, “Clipped” the fraction of the 4505-star scaffold whose prescribed baseline exceeds the 100 m cap, and T_0 the zero-budget mission time as Hab2Max / Hab2Min. The collinear kernel completes the program fastest on both catalogs with the smallest fourth-order collecting area; the pentagon’s deep pair is slower than the reference despite its deeper null. Neither follows from aperture count, null order, or clipped fraction alone—Section 5.8 locates the discriminator in the response shape at the clamped working point.

to. All eight operating points appear among the held-out entries of Table 5.2 and Fig. 5.5, and every one falls on the same z -relation as the architectures the relation was stated on. A collinear array whose zero-budget time lands a few days under a half-year gridline even supplies the lowest-margin point measured in this work, at $z = 1.0$ with a quarter of its replicates admitting no allowance—the undetermined regime, entered by construction, not misfortune.

The collinear geometry is not exotic: linear nulling arrays are the original TPF-I formation, for which the error-budget heritage of Section 1.2 was developed [9, 22]. Whether a linear formation is operationally preferable is a systems question outside this scope; the survey establishes that the design space above the conjugate-pair condition is wider than aperture counting suggests, and that the baseline envelope, not combinatorics, decides where designs succeed or fail.

Three boundaries limit the survey. No design carries an optical-loss or instability model: every comparison is between lossless combiners. The four designs sample the kernel families accessible to the conjugate-pair condition, not the full space Eq. 3.4 admits. And the collector-size convention remains in force: designs share the 4 m collector, total area varies with aperture count, and only the pentagon’s internal comparison is resource-neutral.

Discussion

Interpreting the allowances, the null-order inversion, and their boundaries

The thesis asks two connected questions: how much aggregate random noise LIFE can tolerate at a chosen mission time, and how that allowance changes when the modeled central null becomes steeper. The completed results answer both within the controlled reference calculation before the methodological and architectural boundaries are considered.

6.1 The Error Budget and Architecture Comparison

At the adopted operating point, the nominal second-order null permits flat allowances of $200 \text{ ph s}^{-1} \mu\text{m}^{-1}$ for Hab2Max at 5.5 yr—a marginal value at $z = 2.1$, reported as a worked example—and 120 for Hab2Min at 7.5 yr, which sits at the $z = 3$ criterion. Within the tested ramp families, short-weighted noise is least damaging because it is added where stellar leakage already dominates. This finding is specific to the second-order background hierarchy.

An allowance in photons per second per micron carries no sense of scale on its own, so it is worth stating what it represents against the background it is added to. Both quantities must come from the same architecture, since the background a design sees is a property of that design; the figures below are measured through the same general-combiner path as the allowances themselves. Taking the median over catalog stars at the same reference plane, the astrophysical background averages about $4300 \text{ ph s}^{-1} \mu\text{m}^{-1}$ across the band for the reference array and about 3110 for the six-aperture design.

That reduction is smaller than the null depth alone would suggest, and the reason is

instructive. Three effects act at once and in opposite directions. The deeper null lowers the far-field transmission seen by every background term, from $1/4$ to $1/6$ per deep output, a factor of $2/3$. Separately, and only for the resolved stellar disk, the $x_\star^4$ dependence removes the geometric leakage almost entirely. Working against both, the six-aperture design collects through 75.4 m^2 rather than 50.3 , so every background term is scaled up by half again. Attributing the net reduction to suppressed stellar leakage alone would therefore be wrong twice over.

The consequence for the requirement is that the two architectures are not as different as the raw allowances imply. The reference's flat allowance is roughly 4.6% of its background for Hab2Max and 1.5% for Hab2Min. The six-aperture design's are about 9.8% and 7.2% of its own, larger but of the same order. In both cases the instrument must be quiet to within several percent of an irreducible foreground, which is a demanding requirement for either design. A deeper null relaxes it by a factor of two to five, not by the orders of magnitude that a comparison against an idealized response would suggest.

The composition of that background also explains why the normalization correction of Section 4.2 propagated so strongly. Near $10\text{ }\mu\text{m}$ the local-zodiacal term supplies roughly 1760 of the $2260\text{ ph s}^{-1}\text{ }\mu\text{m}^{-1}$ present at order two, and about 1320 of 1530 at order four, so it is the dominant contribution in both cases. A 27.3% error in the largest single term of a background against which a few-percent requirement is being set is not a small perturbation. Over the same interval the stellar-leakage term falls from about 184 to $0.011\text{ ph s}^{-1}\text{ }\mu\text{m}^{-1}$ between the two orders, a suppression of roughly four orders of magnitude, consistent with the $x^2/8$ scaling of Section 5.7.

A deeper null changes that hierarchy. For the six-aperture design of Section 5.8 the long-weighted ramp becomes the least damaging family, in both catalogs and at every mission-time target. The interpretation is geometric: the fourth-order response suppresses geometric stellar leakage by roughly four orders of magnitude and removes the short-wave variance floor, so added noise at short wavelengths is no longer hidden beneath a background that was there anyway, while the rising zodiacal terms at long wavelengths still hide it.

What the architecture comparison adds is that this reordering survives the move from an idealized proxy to a realizable response, and that the schedule gain accompanying it is mostly not a null-order effect at all. The six-aperture design improves the median one-hour signal-to-noise ratio by a factor of 1.31, but $\sqrt{1.5} = 1.22$ of that is collecting area—an exact factor, since every photon rate is linear in aperture area—leaving a median equal-area combiner factor of only 1.07. What the deeper null contributes is concentrated instead: an equal-area factor of 1.26 over the best fifty targets, the nearest, most resolved stars, which are precisely the ones geometric stellar leakage limits. Because a mission is

scheduled by the top of the target distribution rather than its median, that concentrated excess is worth more to the schedule than its size suggests. The spectral reordering, by contrast, follows from null depth alone and is independent of how the apertures are sized.

This is also where an idealized comparison would have misled, and the size of the error is worth stating rather than gesturing at. Holding throughput, collecting area and the baseline prescription fixed while deepening the null describes no realizable instrument. Doing so returned a Hab2Max flat allowance of $641 \text{ ph s}^{-1} \mu\text{m}^{-1}$ at 5.5 yr where the six-aperture design run at the same target gives 138, an overstatement by a factor of 4.6, and it placed the feasibility boundary in field of regard near 47° where the realizable design reaches only 58° , some eleven degrees too far. The quantity that survives is the ordering of the budget families; the magnitudes do not.

6.2 The Agnostic Approach and the AMS

The methodological contribution is mechanism-agnostic treatment of an *aggregate* independent random-noise allowance, rather than a subsystem allocation. This common reference-plane quantity allows science completion time to constrain a later engineering allocation without prematurely choosing the physical origin of every random term.

The AMS separates its SNR and filter flows (Section 3.6). Magnitude, field-of-regard, and slew changes therefore reuse cached unfiltered SNRs, while physical or budget changes trigger recomputation. Host-star memoization further avoids repeated source calculations for planets sharing one star. These boundaries turn the repeated evaluations required by the TSE from an impractical uniform grid into targeted inversions.

6.3 What the Reductions Buy

The first reduction combines only independent random terms that can be referred to the same single-output plane as $N_I(\lambda)$ (Section 4.1). The second replaces spatial-grid evaluation of symmetric backgrounds with exact azimuthal averages and one-dimensional radial quadrature. Together they reduce a formally infinite-dimensional problem to explicit scalar searches, remove image size as a production-science setting, expose the local-zodiacal normalization error, and provide the analytic null-order scaling used to interpret the result.

They also make the inversion possible at all, and the size of that gain is not a single number. The removed grid cost scales as the square of the image size, while the closed-form background does not depend on image size at all, so the two do not differ by a fixed

factor: the advantage grows with the resolution the grid would have needed.

Measured against the pre-rewrite source model on identical input, the same eight workers and the same instrument settings, the cost of a full catalog pass is as follows. Both implementations memoize the expensive work per host star, so the comparison is made per star and is symmetric; this was verified by holding the star set fixed and increasing the planets per star from 2.6 to 218, which left the runtime unchanged in both. At an image size of 100 the grid requires about 1.2 minutes and the closed form about 0.4; at the image size of 512 used for current yield runs the grid requires about 25 minutes while the closed form is unchanged, since it never forms a grid. The reduction is therefore worth a factor of roughly three at the smaller resolution and sixty at the larger one. Adding the vectorized, host-memoized evaluation of the simulator brings a complete pass to some fifteen seconds, so the two reductions together stand at about two orders of magnitude at the resolution production actually uses.

The distinction between those contributions is worth preserving. Replacing the grid by a closed form is a change of algorithm and accounts for the resolution-dependent factor; vectorizing the evaluation is a change of implementation and accounts for a further factor of under two; and evaluating an arbitrary combiner at the same cost is neither, being a capability the grid formulation did not have at any resolution. Reporting a single figure would merge three different claims.

This is what converts the central question from a description into a procedure. A single endpoint search evaluates the mission six to eight times, a two-dimensional operating-point scan evaluates it a few hundred times, and the results reported here required several thousand evaluations in total. At the pre-rewrite cost and the production image size, the trade space of Section 5 would have taken months of computer time; the requirement could have been stated but not located, and certainly not revised when the background model changed. The efficiency the central question asks for is therefore a precondition rather than a convenience, and it is also what allows an arbitrary architecture to be evaluated at the cost of the closed-form reference rather than at the cost of a grid per target and per wavelength.

6.4 The Conditioning of an Iso-Mission-Time Requirement

That last exposure is an instance of the central result of this work, and it is worth separating from the particular defect that revealed it. An iso-mission-time requirement is located where the completion curve is steep, because that is where a mission-time target worth adopting lies. A requirement so placed does not merely transmit uncertainty in the model it is derived from; it amplifies it. Five measurements made here establish that,

and they are independent in both mechanism and scale.

The first perturbs the physics. A 27.3% error in one background term, the inherited local-zodiacal normalization, moves the derived allowance by up to a factor of eleven (Table 4.1), and the amplification is largest exactly where the target is most demanding and the result therefore most interesting.

The second perturbs the implementation. Two formulations of the same reference array, agreeing to 1.2×10^{-4} in mission time and to 1.6×10^{-4} in median per-target signal-to-noise ratio, nonetheless return endpoints differing by 4.3, 6.8 and 11.9 % in three of twelve cells (Appendix B). A model difference of order 10^{-4} therefore produces a requirement difference of order 10^{-1} , an amplification of roughly three orders of magnitude.

The third perturbs a single rule. The inherited scheduler's strict completion test—fifty-one detections in ninety-one universes where the experiment states fifty in ninety—is a boundary discrepancy of a fraction of a percent in the quantity it thresholds, and correcting it moves every endpoint amplitude, by 3.5 to 95 % across the twenty-four cells, largest exactly at the tightest targets (Section 4.3). Like the normalization defect it is a real correction rather than a synthetic perturbation, and like it the amplification is steepest where the requirement is most interesting.

The fourth perturbs nothing physical whatsoever. Resampling the synthetic universe ensemble over which the 90th-percentile mission time is taken—the same catalog, the same stars, the same instrument—moves the reported amplitude by half its own magnitude at the tightest operating point, and six percent of three hundred resamples there leave no usable positive allowance at all (Section 5.4). This is the sharpest of the five, because there is no modelling error to blame: the variability is intrinsic to estimating a requirement from a finite ensemble at a target close to the feasibility boundary.

The fifth perturbs the population model. Regenerating the synthetic populations with the public generation tool on the identical stellar scaffold, with and without the binary-occurrence suppression, moves the zero-budget feasibility boundary itself by 5–15 %—an order of magnitude beyond its sampling spread—and thereby moves or extinguishes every requirement anchored near it: at the harder configuration, two of the four conventional operating points admit no allowance in nearly every realization (Section 5.4). A configuration choice that alters the eligible planet count by a few percent relocates the boundary by years of mission time.

It is also the one that yields a criterion. Across the twenty-five operating points measured in this work the relative spread is governed by z , the margin between the mission-time target and the configuration's own zero-budget time in units of the sampling spread of that time: the rank correlation on the flat-family spread—the classifier variable throughout—

is -0.98 on the eight points the relation was stated on and -0.94 on the seventeen held out to test it, with no residual pattern detected across the sampled architectures, catalogs, aperture counts, null orders or absolute schedules. Below $z \approx 2$ the amplitude is not determined; the spread passes through 40 % at $z \approx 3$ and declines steadily beyond. Every held-out point obeys the $z \geq 3$ criterion on the flat family—the worst non-flat family grazes 42 % at $z = 3.5$ —including all eight operating points of the two survey architectures the relation had never seen, and the pre-registered independent-realization campaign extends it beyond the catalog ensembles: across eighty independent populations in two occurrence-model configurations, spreads decline monotonically with the between-realization margin, infeasible draws appear only below $z \approx 2$, and no failure was observed among the six tested $z \geq 3$ cells—with the transition itself localized to $z \approx 2\text{--}2.5$, so the hard threshold is conservative rather than exact (Section 5.4). A requirement of this kind is therefore quotable as a number only when its target is set with deliberate margin above the feasibility boundary, and the margin that matters is measured in sampling spreads rather than in years.

Sensitivity and margin practice in systems engineering manages uncertainty in the forward direction: an allowed output uncertainty is apportioned among input contributions and margins are held against growth [32, 30, 31], and statistical tolerance regions bound where a population quantity lies at stated confidence [29]. The object here is the inverse: the model is inverted at a fixed mission-time target for the largest tolerable input amplitude. We are not aware of a statement, in that literature or in the nulling-interferometry heritage of Section 1.2, that such inversely derived requirements are ill-conditioned precisely at the tightest feasible targets a program would naturally adopt, nor of an operational margin criterion for when the resulting number is quotable at all. Within LIFE the statement and the criterion are new; beyond it, the claim is only that they were not found.

Two consequences follow, and neither is confined to LIFE. A requirement of this kind must be quoted together with the background model that produced it, and a subsequent correction to that model should be treated as invalidating the requirement rather than perturbing it. And the convention of adopting the tightest feasible mission-time target—the natural choice, and the one used throughout this work—places the resulting requirement in the least determined part of the space. Where a well-determined requirement matters more than a tight one, the target should be set with deliberate headroom above the zero-budget time, and the cost of that choice is now measurable rather than a matter of judgement.

6.5 Limitations

Table 6.1 ranks the remaining boundaries by their ability to change the numerical allowance or its engineering interpretation.

Priority	Limitation	Consequence
1	Lossless combiners and one collector convention	Every surveyed design is a lossless matrix with no optical-loss or instability model, and designs share the 4 m collector, so total area varies with aperture count; the pentagon's internal pair comparison is the only fully resource-neutral cell (Section 5.8). The equal-area counterfactual reverses the six-aperture schedule advantage.
2	Baseline envelope interaction	The habitable-zone sizing rule requests baselines beyond the 100 m cap for 26 % of scaffold stars under the reference constant and up to 87 % under the survey designs, so every comparison partly reflects the envelope. Designs with broad response shoulders tolerate this; narrow-fringed responses do not, and doubling the cap does not rescue them (Section 5.8).
3	Aggregate random-noise model	Correlated instability, calibration bias, and systematic error cannot be allocated from N_I and may impose stricter requirements; correlated instrumental noise is the subject of parallel work in this group [36].
4	Sampling uncertainty	The 100 universes resolve only a sampled 90th percentile, whose 95.6% distribution-free rank interval spans order statistics 84–96. Per-universe component times are now retained for every bootstrap replicate, but the production campaign itself remains a single 100-universe draw.
5	Symmetric exozodiacal model	Inclined, clumpy, or asymmetric disks can create rotation-dependent structure absent from the face-on radial reduction.
6	Tested spectral families	Flat and two linear ramps sample three directions in a functional trade space; endpoints are not independent wavelength limits or engineering-cost rankings.

Table 6.1: Ranked limitations of the reported error budgets. Priority reflects relevance to the thesis's central numerical and architectural claims, not ease of remediation.

The first limitation sets the strongest boundary on the central comparison. Every surveyed design is an impact study of a response under retained assumptions, not a design prediction: each is evaluated from an explicit realizable combiner, but none constitutes an engineering design, since no optical-loss model, instability-noise calculation, or control requirement accompanies any of them. Correlated instrumental noise—vibrations, line-of-sight jitter, and other temporally structured terms—is the subject of parallel work in this group [36], and no allocation of the aggregate allowance across correlated contributors is attempted here; covariance-aware error-budget allocation of that kind is established practice for high-contrast missions [33], and the allowance reported here is the envelope any

such compliant inventory must fit inside, not a subsystem budget. Likewise, the point estimates inherit both search tolerance and finite-universe rank uncertainty; neither is recoverable as a single symmetric error bar from the retained products.

6.6 Outlook

The next scientific step extends the survey rather than repeating it. Four designs establish that the requirement depends on the array through the shape of its response against the baseline envelope, not through aperture count or null order alone; the conjugate-pair condition of Section 3.4 defines the family, kernel nullers for arbitrary aperture number [14] populate it, and the collector-size convention should be relaxed so that aperture count and collector size trade against each other explicitly. Optical losses and instability terms, which no design in this work carries, belong in the same extension, connecting to the group's parallel instrumental-noise effort [36]. The independent-population campaign brackets the catalogs' generation configuration between the tested suppression variants; closing that bracket with the original configuration record would replace a bounded statement with an exact one, and extending the campaign's anchor coverage across the transition band would sharpen the interval the hard threshold conservatively summarizes. The aggregate allowance should then be allocated to concrete subsystems at the same reference plane.

The scheduler and retention corrections this work identified are already applied: production campaigns here run with completion criteria matching the stated experiment and retain every per-universe component time, and both corrections, together with the analytic background formulation, are being incorporated into the ongoing rewrite of the group's mission-simulation software, so the error budgets of future LIFE studies will rest on the corrected chain. Extending the analytic framework to inclined or asymmetric exozodiacal disks would remove the next important symmetry assumption. And the baseline envelope deserves promotion from boundary condition to design variable: the survey shows it, not combinatorics, is where candidate architectures succeed or fail, and the formation's maximum stretch is an engineering parameter with a now-measurable science return.

Conclusion

What the completed simulations establish—and what remains architecture-dependent

The completed AMS and TSE answer the central research question for every tested budget family and operating point, and in doing so establish something about the answer's status. The iso-mission-time requirements studied here are ill-conditioned: located where the completion curve is steep, it amplifies rather than transmits uncertainty in the model behind it, by three orders of magnitude in the sharpest case measured here, and at the tightest conventionally adopted operating point the finite universe ensemble alone leaves it determined only to half its own magnitude (Section 6.4). The numbers below should be read in that light: quotable where the margin criterion holds, and best-available estimates rather than tight bounds everywhere else.

Under the analytic background model of Section 4.2, the corrected completion rule of Section 4.3, and at the operating points satisfying the margin criterion, the second-order null permits flat allowances of $820 \text{ ph s}^{-1} \mu\text{m}^{-1}$ for Hab2Max at 6.0 yr and 510 for Hab2Min at 8.0 yr, each determined to about 20 %. The realizable six-aperture fourth-order design, built from the same 4 m collectors and therefore carrying half again the collecting area, permits 890 and 520 at its corresponding targets of 4.5 and 6.0 yr, reaching them some thirty percent sooner than the reference reaches its own; and the least damaging tested ramp reverses from short-weighted to long-weighted. The tightest conventional target of 5.5 yr sits at a margin of $z = 2.1$, below the criterion, and its nominal allowance of 200 is reported as a worked example rather than a requirement.

The five sub-questions of Section 1.1 close as follows.

Q1 The unrestricted AMS trade space contains up to eight free modifier functions and

three scalar agnostic inputs and is therefore formally infinite-dimensional. The reported family searches reduce it to one amplitude dimension, four dimensions when all agnostic inputs vary, or five when two ramp endpoints vary independently. Whether that reduction costs fidelity must be judged separately for its two halves, and only one of them is validated here. The *numerical* reduction, analytic integration of the circularly symmetric backgrounds in place of a spatial pixel grid, removes the grid dimension and reproduces the retained source calculation to the precision stated in Table 3.2. The *modelling* reduction, representing all admissible independent random terms by a single aggregate $N_I(\lambda)$ at a common reference plane, removes the unbounded functional dimension. For terms that are genuinely independent and referable to that plane it is exact rather than approximate, because independent variances add; no fidelity is lost in the aggregation itself, and none can be recovered by splitting it. What the reduction does not establish is how large a fraction of a realistic instrumental inventory meets those two conditions. Settling that requires a covariance model of the subsystem terms and a decomposition of the signal chain finer than this work assumes, so the allowances reported here should be read as conditional on that fraction being high. Q1 is therefore answered for the numerical reduction and bounded, rather than answered, for the modelling one.

- Q2** Magnitude, field-of-regard, and slew changes reuse unfiltered SNRs and repeat only filtering and scheduling. Budget, source-physics, or null-order changes require catalog-wide SNR recomputation, although host-shared source terms remain memoized within that pass.
- Q3** The response of an arbitrary nulling architecture can be evaluated at the cost of the closed-form reference. Writing the transmission in terms of aperture positions and a combiner matrix, and exploiting the fact that the rotation average of any single-baseline geometry is a one-dimensional function of $\pi b\theta/\lambda$, gives a full evaluation in 14.3s against 14.7s for the hand-derived response, agreeing with it to a relative 6×10^{-12} . The simulator is therefore architecture-agnostic in the sense the central question requires, and not at the price of the efficiency the same question demands.
- Q4** The TSE locates maximum flat and linear-ramp amplitudes as iso-mission-time point estimates; Table 5.1 reports the complete set, including both catalogs and both mission-time targets.
- Q5** The requirement does depend on the architecture, and not only in magnitude. A fourth-order central null suppresses geometric stellar leakage by about four orders of magnitude and reverses the spectral preference toward long wavelengths, in both catalogs, at every mission-time target, and for a realizable six-aperture combiner as well as for the idealized proxy; the ordering is thus a property of the null depth

rather than of the modelling convention. Built from the reference's own 4 m collectors, the six-aperture fourth-order design also completes Experiment 1 about thirty percent sooner and tolerates a larger flat allowance at its own operating point. That gain is not free: the design carries half again the collecting area, and at equal total collecting area it is instead about sixteen percent slower than the reference. The architecture result is therefore bracketed by the two sizing conventions, and what survives both is the spectral reordering rather than the schedule (Sections 3.4 and 5.8). One part of the dependence follows from the combiner structure alone rather than from any simulation: output subtraction at fourth order requires a conjugate pair of dark outputs, which a combiner admits exactly when two of its rows vanish to second order or better. The survey of Section 5.8 realizes that condition at four apertures on a line and five on a pentagon as well as six in LIFE's template—refuting this work's own earlier six-aperture-minimum claim—and locates the practical boundary not in aperture count but in the shape of each design's response against the mission's baseline envelope, where the four-aperture collinear kernel completes the program fastest of all surveyed designs on the smallest collecting area.

One further result falls outside the sub-questions and is reported on its own account. The inherited local-zodiacal normalization is defective, and the defect is worth more than a percent-level curiosity. Dividing a circular field of view's integrated flux by the area of its enclosing square suppressed the uniform foreground by $\pi/4$; correcting it lowers the one-hour signal-to-noise ratio by a median of about 9 % and, propagated through a fixed-length search campaign, lowers the detection yield by 6.5 % for Hab2Max and 6.8 % for Hab2Min. Yields computed through that path are optimistic by roughly that margin, which is comparable to the differences between mission configurations such studies exist to distinguish (Section 4.2).

These allowances do not yet constitute subsystem requirements. They define a validated bridge from scientific yield to an explicit engineering trade space and show which spectral regions most strongly control mission time. That bridge matters because LIFE's promise—new knowledge of nearby terrestrial planets and the possibility of life beyond Earth—depends not only on asking an ambitious question, but on demonstrating that an instrument can answer it within credible limits.

Analytic Transmission Derivation

Derivation of the circularly symmetric transmission and background response

This appendix records the reduction used by the production source model. It is included to make the fourth-order result independently inspectable without requiring the reader to reconstruct the mathematics from code.

A.1 Even-Power Reduction

For a positive even null order $n = 2p$, the nulling envelope can be written as a finite cosine series:

$$\sin^{2p} x = \frac{1}{4^p} \left[\binom{2p}{p} + 2 \sum_{j=1}^p (-1)^j \binom{2p}{p-j} \cos(2jx) \right]. \quad (\text{A.1})$$

One destructive output contains this envelope multiplied by an imaging factor $\cos^2(\psi \pm \pi/4)$. For an axisymmetric source, the complementary imaging phase contributes one half after full azimuthal integration. Let a sky point at angular radius ϑ have projected nulling coordinate $\alpha = \vartheta \cos \varphi$. Substituting $x = \pi b \vartheta \cos \varphi / \lambda$ into Eq. A.1 and using the NIST Digital Library of Mathematical Functions integral representation [17],

$$\frac{1}{2\pi} \int_0^{2\pi} \cos(z \cos \varphi) d\varphi = J_0(z) \quad (\text{A.2})$$

gives Eq. 4.2. The reduction is exact for this response and any circularly symmetric surface-brightness profile. It is not an approximation based on rapid rotation. The same one-dimensional structure holds for an arbitrary combiner, where the rotation average is obtained numerically rather than in closed form (Section 3.5); the closed form given here is the special case that admits a Bessel series.

For the two orders used in the thesis,

$$\langle T_2 \rangle_\varphi = \frac{1}{4} \left[1 - J_0 \left(\frac{2\pi b \vartheta}{\lambda} \right) \right], \quad (\text{A.3})$$

$$\langle T_4 \rangle_\varphi = \frac{1}{16} \left[3 - 4J_0 \left(\frac{2\pi b \vartheta}{\lambda} \right) + J_0 \left(\frac{4\pi b \vartheta}{\lambda} \right) \right]. \quad (\text{A.4})$$

At large phase, the oscillatory Bessel terms approach zero. The corresponding far-field averages are therefore $1/4$ and $3/16$. This is the origin of the throughput caveat in the main text. These two values belong to the idealized $\sin^{2p}$ proxies derived here, not to any particular hardware: the realizable six-aperture design of Section 3.4 has a far-field average of $1/6$ per dark output, not $3/16$, and Section 5.7 quotes the realizable values. The distinction matters because the ratio between orders differs, $2/3$ against $3/4$; neither ratio describes a uniform background's absolute rate, for which the aperture-count normalization cancels the response average—any lossless combiner transmits $N \cdot A_{\text{ap}} \cdot \langle R \rangle = A_{\text{ap}}$ of a uniform source, which is why the measured local-zodiacal rate is identical between the two architectures (Section 5.7).

A.2 Uniform-Disk Average and Small-Star Limit

For a uniform disk of angular radius R , area averaging a radial Bessel term gives, using the Bessel derivative identity [17],

$$\frac{2}{R^2} \int_0^R \vartheta J_0(a\vartheta) \, \mathrm{d}\vartheta = \frac{2J_1(aR)}{aR}. \quad (\text{A.5})$$

The right-hand side is evaluated as one when $aR = 0$. Applying this to a stellar disk, so that $R = R_\star$, and writing $x_\star = 2\pi b R_\star / \lambda$ as defined in Section 3.5, the disk-averaged responses become

$$\langle T_2 \rangle_{\text{disk}} = \frac{1}{4} \left[1 - \frac{2J_1(x_\star)}{x_\star} \right], \quad (\text{A.6})$$

$$\langle T_4 \rangle_{\text{disk}} = \frac{1}{16} \left[3 - \frac{8J_1(x_\star)}{x_\star} + \frac{2J_1(2x_\star)}{2x_\star} \right]. \quad (\text{A.7})$$

The second-order result is not new: Eq. A.5 applied to the double-Bracewell response reproduces the uniform-disk stellar leakage given by Lay [10]. What the derivation above adds is the extension to arbitrary even null order through the general coefficient sum of Eq. 4.2, and its reuse inside the one-dimensional radial quadrature for tapered and extended sources in Section 4.2, neither of which follows from the second-order case alone. Using the standard Bessel power series [17], $J_1(x_\star) = x_\star/2 - x_\star^3/16 + x_\star^5/384 + \mathcal{O}(x_\star^7)$, which yields

$$\langle T_2 \rangle_{\text{disk}} = \frac{x_\star^2}{32} - \frac{x_\star^4}{768} + \mathcal{O}(x_\star^6), \quad (\text{A.8})$$

$$\langle T_4 \rangle_{\text{disk}} = \frac{x_\star^4}{256} + \mathcal{O}(x_\star^6). \quad (\text{A.9})$$

Consequently,

$$\frac{\langle T_4 \rangle_{\text{disk}}}{\langle T_2 \rangle_{\text{disk}}} = \frac{x_\star^2}{8} + \mathcal{O}(x_\star^4). \quad (\text{A.10})$$

This scaling predicts the near-disappearance of geometric stellar leakage before any mission scheduling is performed. Direct positive quadrature is retained as a numerical safeguard where subtraction of nearly equal Bessel terms could lose precision.

A.3 Radial Profiles and Wavelength Bins

For a circular source with surface brightness $I_\lambda(\vartheta)$, the transmitted spectral photon rate is proportional to

$$2\pi \int_0^{R_{\text{max}}} I_\lambda(\vartheta) \langle T_n \rangle_\varphi(\vartheta) \vartheta \, d\vartheta. \quad (\text{A.11})$$

This single radial integral replaces a two-dimensional pixel sum. The stellar disk terminates at its angular radius; the exozodiacal calculation uses the adopted smooth radial brightness profile and field-of-view taper. A source lacking circular symmetry cannot be reduced in this way.

For wavelength-bin edges λ_- and λ_+ , the calculation changes variables to $u = 1/\lambda$:

$$\int_{\lambda_-}^{\lambda_+} f(\lambda) \, d\lambda = \int_{1/\lambda_+}^{1/\lambda_-} \frac{f(1/u)}{u^2} \, du. \quad (\text{A.12})$$

Eight-point Gauss–Legendre quadrature is then applied over the finite u interval. Because the interferometric phase is proportional to u , this samples chromatic fringes more uniformly than the same number of points placed directly in wavelength.

A.4 Local-Zodiacal Solid Angle

For a locally uniform foreground with spectral radiance I_λ , the photon rate is proportional to the accepted solid angle. In the small-angle approximation, a circular field with angular radius R_{FoV} has

$$\Omega_{\text{circ}} = \pi R_{\text{FoV}}^2. \quad (\text{A.13})$$

In its tapered configuration the removed grid implementation averaged the transmission over the whole square evaluation grid, of area $\Omega_{\text{square}} = 4R_{\text{FoV}}^2$, and then multiplied that average by the circular solid angle above. Its untapered configuration instead averaged over a circular mask, and so paired the two consistently; the disagreement between the two branches on the same physical configuration is what identifies the tapered one as defective. For a foreground and a transmission that are both uniform over the domain, the mismatch is the ratio of the domains,

$$\frac{\Omega_{\text{circ}}}{\Omega_{\text{square}}} = \frac{\pi}{4}, \quad (\text{A.14})$$

so the corrected rate is the former rate multiplied by $4/\pi$. The transmission actually used is neither uniform nor confined to the circle, and the corner flux admitted by the square domain makes the realized ratio slightly smaller; measured against the removed grid and extrapolated in resolution it converges to about 1.267. The value $4/\pi$ is therefore the uniform-domain limit of the correction rather than its exact magnitude. The current analytic calculation uses the circular solid angle directly, consistent with the continuous field-of-view convention in the LIFE source model [2].

Reproducibility and Statistical Interpretation

Operating-point records and interpretation of the sampled mission-time percentile

B.1 Operating Point and Search Products

Unless an axis is explicitly varied, every reported contour uses limiting K-band magnitude 7, implemented field of regard $|\beta_{\text{ ecl}}| \leq 65^\circ$, and 12 h slew time. The flat, short-weighted, and long-weighted families are evaluated independently. A ramp amplitude always denotes its non-zero endpoint, and its other endpoint is constrained to zero by definition.

The completed products are sufficient to reproduce the reported contour endpoints and figures from the saved AMS outputs. They are not sufficient to recompute a different population quantile or a confidence interval in years because individual universe times were not retained for every evaluated amplitude. Reproducibility of future studies would improve by storing, for every search evaluation:

- the complete per-universe detection, orbit, characterization, and slew times;
- the selected target identifiers and one-hour SNRs;

- the lower and upper amplitude bracket and the stopping reason;
- the exact catalog, source-model, and scheduler configuration;
- the architecture evaluated, as aperture positions and combiner matrix, together with the baseline constant derived from its own response.

The recovered operating-point scan grids of Figs. 5.7 and 5.8 are tabulated in the archived artifact `stageb_grids.tsv`, and the ramp-endpoint grids of Fig. 5.4, recomputed under the corrected completion rule, in `linear_grids.tsv`; the bootstrap, scheduler-boundary, and external-validation products live in the sibling directories named in the corresponding sections.

B.2 Data and Source Provenance

The reformulation of the astrophysical background evaluation described in Section 4.2 was documented in a standalone technical brief covering its derivation, its validation boundaries, and the local-zodiacal normalization correction, and was reviewed and accepted by the thesis supervisor. The code state carrying that reformulation has since been placed under version control; the source hash below remains the authoritative identifier for the production campaign, because the working tree was not clean when those calculations were run.

Table B.1 records the exact local inputs and result closure used for the final thesis. SHA-256 hashes make later identity checks possible even though no immutable public archive or complete environment lock was created during the production campaign. The source hash covers the production-code closure used by the completed calculations; the result hash covers the 52 files in the complete TSE **Linear** result tree from which the thesis figures and values were taken.

The complete final archive—simulation and analysis code, thesis source and figures, and the reproducibility records of this appendix—is published in the thesis repository at github.com/nfeuchtho/LIFEsim under the tag `thesis-archive`; the SHA-256 manifest archived alongside it pins every artifact within that snapshot, and the `independent_ensembles` hash list pins the cluster-side realization catalogs that are too large to publish.

Two figure-level provenance notes belong here rather than in the captions. The field of regard enters the model as the scheduler's latitude mask (simulator source at the archived commit), which Figure 5.9 draws. Decomposing the target loss of that figure over the same 3605 eligible host stars, the field of regard alone accounts for 6.3%, the magnitude limit alone for 19.2%, and both together for 1.3%, giving marginal totals of 7.5 and

Artifact	Size or count	SHA-256
Hab2Min catalog	126 552 381 bytes	8a43af800ed5bb7540a0656506057dbb4f9dabc6be553fcf510f767ccd751aee
Hab2Max catalog	190 077 716 bytes	c372e4f665856ceb2ea7dc4c0e059b8447a119586e75e74e98001c3425da9799
Production source closure	commit 2fd5142, plus recorded working tree	00751d7dca7c095819401bd95c8ccfb64a5778ccd39469c9935eb29673b12b90
Complete final result tree	52 files; 1 895 071 bytes	61889403be458b5ac65bc7c63bde4eebf260a6251d5ed4bca3e0c2dd1d3181cc
Validation scripts	recorded script closure	040dd0bca26ba14c17a2b8cbcd2745e0fa444d0c8956c851e388e442fc1897b1

Table B.1: Identity record for the catalogs, production source, completed result tree, and validation scripts used in this work. The repository was not clean at production time; the source-tree hash, rather than the commit identifier alone, defines the recorded code state.

20.5 %; the loss is therefore dominated by the magnitude limit. The decomposition is reproduced by the archived script `decompose_target_loss.py` and is specific to this catalog, these cuts and this experiment definition.

The retained TSE **Linear** products contain rendered PDFs and text summaries, not raw per-evaluation grids, logs, or per-universe time vectors. The order-four Hab2Min 8-year ramp values were recovered from the complete vector result PDFs and cross-checked by reproducing the corresponding 7.5-year values before rounding to the precision used in Table 5.1. Consequently, the final figures and tabulated endpoints are traceable, but the production campaign cannot be rerun bit-for-bit from this record alone. A practical future release should archive the catalogs or their immutable source identifiers, the full code closure, an environment lock, the run matrix, raw grids, scheduler logs, and per-universe outputs.

B.3 Independent Reproduction of the Final Amplitudes

The amplitudes of Table 5.1 were produced during a campaign in which the working tree was not clean, and the raw per-evaluation products were not retained. To establish that they are nevertheless reproducible, the complete endpoint search was rerun from a committed code state at the identical operating point, using a single dedicated script whose output is the machine-written table below. The rerun additionally records the mission time that each reported amplitude actually delivers, which the original campaign did not retain.

That last quantity matters for interpretation. The search stops when the evaluated

mission time lies within 0.05 yr of the target *or* when the amplitude bracket has narrowed to $10 \text{ ph s}^{-1} \mu\text{m}^{-1}$, and it returns its last accepted amplitude, so a reported value belongs to the achieved time rather than to the nominal target, and the bracket-width stop can leave residuals beyond the time tolerance. Across the twenty-four strict-rule reruns of Table B.2 the residuals span -0.045 to $+0.045$ yr; across the corrected-rule record of Table B.3 they span -0.094 to $+0.040$ yr for the headline designs, and the widest survey residual is -0.125 yr. Converting a residual into an amplitude uncertainty requires the local slope of amplitude against mission time, which the secant between each catalog's two targets estimates as roughly 6×10^2 to $1.5 \times 10^3 \text{ ph s}^{-1} \mu\text{m}^{-1}$ per year, so the stopping rules correspond to an amplitude uncertainty of order 20 to $200 \text{ ph s}^{-1} \mu\text{m}^{-1}$ depending on the row. Because the response is not linear in mission time, these figures indicate the scale of the search uncertainty rather than a calibrated interval. The stored records also separate this solver jitter from the sampling spread directly: within each flat-family cell, regressing the replicate endpoints on their achieved-time residuals and removing the fitted component changes the relative spread by at most 0.7 percentage points (typically under 0.2), the amplitude-equivalent jitter is 1 to $49 \text{ ph s}^{-1} \mu\text{m}^{-1}$ across cells, and no cell moves across the 40 % level between its raw and solver-corrected values, so the spreads of Section 5.4 and every z -classification are sampling-limited, not search-limited. The reporting convention is aligned with that finding: headline tables round amplitudes to two significant figures, the appendix records store the raw search output, and no amplitude should be treated as reliable beyond its leading two digits.

For the reference array, ten of the twelve amplitudes reproduce the production campaign to the precision reported in Table 5.1, and two differ by a single rounding step, both at the Hab2Min seven-and-a-half-year target where the allowance is smallest and the fixed time tolerance therefore translates into the largest relative uncertainty. No difference exceeds one step. The Hab2Min eight-year values, which had previously been recovered from vector plots rather than from retained numerical output, are re-derived here directly from the code.

A second reproduction bounds the sensitivity of the reported amplitudes rather than their repeatability. Running the reference array through the general-combiner path instead of its closed form changes the underlying model by very little: the two agree to 0.012% in mission time and to 1.6×10^{-4} in median per-target signal-to-noise ratio, the residual being the difference between the architecture's measured response peak and the constant the source model hardcodes. The derived endpoints move considerably more. Nine of the twelve agree within one percent, but the Hab2Max six-year flat value falls by 4.3%, the Hab2Min seven-and-a-half-year flat value by 6.8%, and the Hab2Min eight-year short-weighted value by 11.9%.

That is the amplification of Section 6.4 observed with a far smaller input perturbation than the normalization correction provided, and it is the more useful demonstration of the two: a model difference of order 10^{-4} produces a requirement difference of order 10^{-1} . An iso-mission-time amplitude should therefore be quoted together with the response that generated it, and should not be transferred between calculations that differ even slightly in the underlying transmission.

The fourth-order rows carry a stronger statement, because they were computed twice on unrelated hardware. Repeating the six Hab2Max searches on a compute cluster—a different machine, operating system, worker count and major version of the numerical library—reproduced the endpoints obtained locally to between 2.6×10^{-12} and 4.1×10^{-11} in relative terms. Since the search terminates on a time tolerance rather than converging to machine precision, that agreement is not automatic, and it establishes that the reported amplitudes are properties of the model rather than of the arithmetic environment that evaluated it.

The reproduction also exposed an inconsistency in the reporting convention of the strict-rule era rather than in any calculation. The strict-era results table quoted its smallest entry as 65, which was neither the nearest ten nor two significant figures of the reproduced 62.2. Small amplitudes are where the fixed time tolerance translates into the largest relative uncertainty, so they are also where a rounding convention matters least and misleads most. The corrected headline values of Table 5.1 adopt a uniform three-significant-figure convention; within this appendix's historical strict-rule record, the values of Table B.2 are authoritative.

B.4 Rank Uncertainty of the 90th Percentile

Let $q_{0.9}$ be the population 90th percentile and let $X_{(k)}$ denote the k th ordered value in a sample of 100 independent population draws. The number of samples below $q_{0.9}$ follows a binomial distribution with $n = 100$ and success probability 0.9. Therefore the coverage probability of the interval $[X_{(84)}, X_{(96)}]$ is obtained from the corresponding binomial tail sum and is approximately 0.9557.

This interval quantifies rank uncertainty under independent sampling from the adopted population model. It does not include uncertainty in planet occurrence rates, stellar properties, dust levels, instrument settings, or scheduler design. The universes also share the same real-star scaffold. The rank statement is therefore useful but narrower than a complete physical uncertainty analysis; the campaign of the following section addresses the population-model part of that gap directly.

Catalog	Target [yr]	Flat	Short-weighted Amplitude [$\text{ph s}^{-1} \mu\text{m}^{-1}$]	Long-weighted	Achieved time [yr]
<i>Double Bracewell, four apertures</i>					
Hab2Max	5.5	142.4	298.3	210.5	5.455–5.541
Hab2Max	6.0	610.3	1303.1	1089.3	5.962–6.045
Hab2Min	7.5	62.2	197.9	156.7	7.484–7.528
Hab2Min	8.0	462.8	973.9	810.1	7.992–8.037
<i>Double triple nuller, six apertures</i>					
Hab2Max	4.0	241.4	403.1	556.9	4.011–4.045
Hab2Max	4.5	701.6	1384.9	1762.0	4.481–4.514
Hab2Min	5.5	177.3	226.1	602.2	5.481–5.534
Hab2Min	6.0	409.0	733.9	1313.7	5.991–6.036

Table B.2: **Historical strict-rule reproduction record.** Reproduction of the strict-rule production amplitudes from a committed code state at the identical operating point; the corrected-rule headline values of Table 5.1 are reproduced separately in Table B.3. Values are the raw search results, unrounded. The final column gives the range of mission times actually delivered by the three amplitudes in that row; the search stops within 0.05 yr of the target, so a reported amplitude belongs to its achieved time rather than to the nominal target. The two halves are independent in different senses: the fourth-order rows come from the same production run and check only that the record is faithful, whereas the second-order rows come from the earlier production campaign and are therefore a genuine cross-check of one calculation against another. The disagreements between them are the subject of Appendix B’s sensitivity discussion. These values were produced under the strict completion rule and are retained as the strict-rule record beside the corrected headline values of Table 5.1; the cell-by-cell comparison of the two rules is archived with the scheduler-boundary data.

B.5 Independent-Population Campaign

The independent-realization results of Sections 5.4 and 6.4 were produced under a pre-registered protocol: seeds, anchor cells, fidelity-acceptance tolerances, the classification test, the automated go/no-go gate for the corrected-configuration round, and the analysis were fixed in writing before the corresponding results existed, and the two amendments made during the campaign are recorded in the same document with their timestamps and reasons. Realizations were generated with the public P-Pop tool on this work’s own 4505-star scaffold, using the shipped Bryson occurrence models; forty realizations omit and forty apply the binary-occurrence suppression whose strength the original catalogs’ configuration lies between. All forty-plus-forty realizations passed the registered fidelity gate against the thesis catalogs’ population statistics, with zero exclusions. Table B.4 reports every tested cell. The mechanical classification verdicts are as registered—eleven of thirteen cells consistent, both exceptions in the conservative direction (spread below 40 % just under $z = 3$)—and the registered fallback, replacing the hard threshold by the measured transition band, is what Chapter 5 reports. With twenty realizations per

Catalog	Target [yr]	Flat Amplitude	Short-wt. [ph s ⁻¹ μm ⁻¹]	Long-wt.	Achieved time [yr]
<i>Double Bracewell</i>					
Hab2Max	5.5	201.2	437.6	407.9	5.467–5.540
Hab2Max	6.0	820.2	1795.1	1359.3	5.906–6.034
Hab2Min	7.5	121.3	280.1	231.4	7.453–7.467
Hab2Min	8.0	506.5	1142.3	1048.7	7.994–8.031
<i>Double triple nuller</i>					
Hab2Max	4.0	346.6	457.1	893.8	3.973–4.027
Hab2Max	4.5	890.8	1570.2	2105.3	4.488–4.517
Hab2Min	5.5	265.9	361.1	781.3	5.503–5.538
Hab2Min	6.0	520.0	955.5	1359.8	5.955–6.000
<i>Kernel-5, deep pair</i>					
Hab2Max	5.5	88.1	149.0	159.0	5.480–5.491
Hab2Max	6.0	366.4	621.9	828.8	5.964–6.037
Hab2Min	8.0	87.3	197.9	291.1	7.992–8.035
Hab2Min	8.5	310.7	523.2	771.2	8.488–8.538
<i>Collinear kernel</i>					
Hab2Max	3.5	0.0	0.0	0.0	3.499–3.499
Hab2Max	4.0	679.7	1489.8	1568.9	4.013–4.022
Hab2Min	5.0	41.0	73.6	125.0	4.971–5.000
Hab2Min	5.5	530.1	863.2	1127.5	5.375–5.488

Table B.3: Corrected-rule reproduction record for all four surveyed designs, unrounded, with the range of mission times actually delivered by the three amplitudes in each row; each design block is headed by its name. The collinear 3.5yr row is the degenerate operating point of Section 5.4: its zero-budget time sits within the search tolerance of the target, so no positive allowance exists and the searches return zero. These are the values behind Tables 5.1 and 5.5.

cell the uncertainty of an estimated spread is itself material, as the bootstrap intervals in the table show; the defensible summary is that *no failure was observed among the six tested $z \geq 3$ cells*, whose spread intervals lie entirely below 40% wherever z 's own interval clears 3, not that six successes prove universal sufficiency. The z intervals of both inconsistent cells straddle 3, which is the transition-band softness of Section 5.4 seen from the other side. The protocol, per-realization evaluations, summary statistics, and generation configuration are archived as the `independent_ensembles` record (Appendix B.2).

Config	Design	Catalog	T^* [yr]	T_0 [yr]	z [95 % CI]	Spread % [CI]	Infeas. [CI %]	Cons.	Provenance
base	Bracewell	Hab2Max	4.632	4.43±0.10	2.0 [1.5, 2.9]	39 [24, 53]	1/20 [0, 25]	yes	Amd. 1
base	Bracewell	Hab2Max	4.733	4.43±0.10	3.0 [2.4, 4.2]	30 [21, 39]	0/20 [0, 17]	no	Amd. 1
base	Bracewell	Hab2Max	5.5	4.43±0.10	10.5 [8.8, 14.7]	12 [8, 15]	0/20 [0, 17]	yes	reg.
base	Bracewell	Hab2Max	6	4.43±0.10	15.5 [13.0, 21.8]	9 [6, 10]	0/20 [0, 17]	yes	reg.
base	Bracewell	Hab2Min	6.665	6.32±0.17	2.0 [1.4, 3.8]	43 [28, 57]	1/20 [0, 25]	yes	Amd. 1
base	Bracewell	Hab2Min	6.835	6.32±0.17	3.0 [2.2, 5.6]	37 [21, 52]	0/20 [0, 17]	yes	Amd. 1
base	Bracewell	Hab2Min	7.5	6.32±0.17	6.9 [5.1, 13.0]	19 [9, 26]	0/20 [0, 17]	yes	reg.
base	Triple nuller	Hab2Max	3.148	3.03±0.06	2.0 [1.4, 3.5]	40 [20, 55]	3/20 [3, 38]	yes	Amd. 1
base	Triple nuller	Hab2Max	4.5	3.03±0.06	24.6 [19.5, 38.4]	9 [6, 11]	0/20 [0, 17]	yes	reg.
binsup	Bracewell	Hab2Max	5.5	5.65±0.14	-1.1 [-1.9, -0.6]	–	18/20 [68, 99]	yes	reg.
binsup	Bracewell	Hab2Max	6	5.65±0.14	2.5 [2.0, 3.6]	38 [27, 47]	0/20 [0, 17]	no	reg.
binsup	Bracewell	Hab2Min	7.5	7.92±0.20	-2.1 [-3.4, -1.5]	–	20/20 [83, 100]	yes	reg.
binsup	Triple nuller	Hab2Max	4.5	3.87±0.11	5.6 [4.6, 8.2]	21 [14, 26]	0/20 [0, 17]	yes	reg.

Table B.4: Every cell of the independent-population campaign: occurrence-model configuration (base / binary suppression), tested target T^* , between-realization zero-budget time T_0 (mean $\pm$ s.d., $n = 20$), margin z with a bootstrap 95 % interval over realizations, flat-family relative endpoint spread over feasible draws with its bootstrap interval, infeasible count with a Clopper–Pearson 95 % interval, whether the cell classifies consistently under the registered rule, and whether the cell comes from the original registration (reg.) or Amendment 1 (Amd. 1). Design names use the table aliases of Section 3.4. “Cons. no” rows are the two benign exceptions: spread below 40 % just under $z = 3$. Spreads are undefined (–) where nearly all draws are infeasible.

References

- [1] S. P. Quanz et al., “Large Interferometer For Exoplanets (LIFE). I. Improved exoplanet detection yield estimates for a large mid-infrared space-interferometer mission,” *Astronomy & Astrophysics*, vol. 664, A21, 2022, doi:10.1051/0004-6361/202140366.
- [2] F. Dannert et al., “Large Interferometer For Exoplanets (LIFE). II. Signal simulation, signal extraction, and fundamental exoplanet parameters from single-epoch observations,” *Astronomy & Astrophysics*, vol. 664, A22, 2022, doi:10.1051/0004-6361/202141958.
- [3] B. S. Konrad et al., “Large Interferometer For Exoplanets (LIFE). III. Spectral resolution, wavelength range, and sensitivity requirements based on atmospheric retrieval analyses of an exo-Earth,” *Astronomy & Astrophysics*, vol. 664, A23, 2022, doi:10.1051/0004-6361/202141964.
- [4] J. Kammerer, S. P. Quanz, and F. Dannert, “Large Interferometer For Exoplanets (LIFE). VI. Detecting rocky exoplanets in the habitable zones of Sun-like stars,” *Astronomy & Astrophysics*, vol. 668, A52, 2022, doi:10.1051/0004-6361/202243846.
- [5] J. Kammerer and S. P. Quanz, “Simulating the exoplanet yield of a space-based mid-infrared interferometer based on Kepler statistics,” *Astronomy & Astrophysics*, vol. 609, A4, 2018, doi:10.1051/0004-6361/201731254.
- [6] S. Bryson et al., “The Occurrence of Rocky Habitable-zone Planets around Solar-like Stars from Kepler Data,” *The Astronomical Journal*, vol. 161, no. 1, 36, 2021, doi:10.3847/1538-3881/abc418.
- [7] F. A. Dannert, *How Quiet Is LIFE? Performance of Nulling Interferometers under Instrumental and Fundamental Noise*, doctoral thesis, ETH Zurich, Diss. ETH No. 31341, 2025, doi:10.3929/ethz-c-000788668.
- [8] G. M. Kennedy et al., “Exo-zodi Modeling for the Large Binocular Telescope Interferometer,” *The Astrophysical Journal Supplement Series*, vol. 216, no. 2, 23, 2015, doi:10.1088/0067-0049/216/2/23.
- [9] O. P. Lay, “Systematic errors in nulling interferometers,” *Applied Optics*, vol. 43, no. 33, pp. 6100–6123, 2004, doi:10.1364/AO.43.006100.
- [10] O. P. Lay, “Imaging properties of rotating nulling interferometers,” *Applied Optics*, vol. 44, no. 28, pp. 5859–5871, 2005, doi:10.1364/AO.44.005859.
- [11] R. N. Bracewell, “Detecting nonsolar planets by spinning infrared interferometer,” *Nature*, vol. 274, no. 5673, pp. 780–781, 1978, doi:10.1038/274780a0.
- [12] J. R. P. Angel and N. J. Woolf, “An Imaging Nulling Interferometer to Study Extrasolar Planets,” *The Astrophysical Journal*, vol. 475, no. 1, pp. 373–379, 1997, doi:10.1086/303529.

- [13] O. Guyon, B. Mennesson, E. Serabyn, and S. Martin, “Optimal Beam Combiner Design for Nulling Interferometers,” *Publications of the Astronomical Society of the Pacific*, vol. 125, no. 930, pp. 951–965, 2013, doi:10.1086/671826.
- [14] R. Laugier, N. Cvetojevic, and F. Martinache, “Kernel nullers for an arbitrary number of apertures,” *Astronomy & Astrophysics*, vol. 642, A202, 2020, arXiv:2008.07920.
- [15] R. Norbruis, *Detecting exoplanets with a smallsat-based multi-aperture nulling interferometry telescope*, master’s thesis, Delft University of Technology, 2023.
- [16] P. Binkert, *Signal Extraction, Constraints on Exoplanet Parameters and Molecular Feature Detection During the Search Phase of the Large Interferometer for Exoplanets Mission*, master’s thesis, ETH Zurich, Exoplanets and Habitability Group, 2023.
- [17] F. W. J. Olver et al., eds., *NIST Digital Library of Mathematical Functions*, Release 1.2.7, 15 June 2026. <https://dlmf.nist.gov/>; specific identities: Eq. 10.9.1, Eq. 10.6.6a, and Eq. 10.2.2; accessed 24 July 2026.
- [18] F. Martinache and M. J. Ireland, “Kernel-nulling for a robust direct interferometric detection of extrasolar planets,” *Astronomy & Astrophysics*, vol. 619, A87, 2018, doi:10.1051/0004-6361/201832847.
- [19] O. P. Lay, “Removing instability noise in nulling interferometers,” in *Advances in Stellar Interferometry*, Proc. SPIE, vol. 6268, 62681A, 2006, doi:10.1117/12.670603.
- [20] B. Chazelas et al., “Instrumental stability requirements for exoplanet detection with a nulling interferometer: variability noise as a central issue,” *Applied Optics*, vol. 45, no. 5, pp. 984–992, 2006, doi:10.1364/AO.45.000984.
- [21] L. Kaltenegger, M. Fridlund, and A. Karlsson, “Interferometric space missions for the search for terrestrial exoplanets: requirements on the rejection ratio,” *Astrophysics and Space Science*, vol. 306, no. 3, pp. 147–158, 2006, doi:10.1007/s10509-006-9183-z.
- [22] P. R. Lawson and J. A. Dooley, eds., *Technology Plan for the Terrestrial Planet Finder Interferometer*, JPL Publication 05-5, Jet Propulsion Laboratory, California Institute of Technology, 2005.
- [23] C. S. Cockell et al., “Darwin—a mission to detect and search for life on extrasolar planets,” *Astrobiology*, vol. 9, no. 1, pp. 1–22, 2009, doi:10.1089/ast.2007.0227.
- [24] B. Efron and R. J. Tibshirani, *An Introduction to the Bootstrap*, Monographs on Statistics and Applied Probability, vol. 57, Chapman & Hall/CRC, 1993.
- [25] T. J. DiCiccio and B. Efron, “Bootstrap confidence intervals,” *Statistical Science*, vol. 11, no. 3, pp. 189–228, 1996, doi:10.1214/ss/1032280214.
- [26] M. Falk and E. Kaufmann, “Coverage probabilities of bootstrap-confidence intervals for quantiles,” *The Annals of Statistics*, vol. 19, no. 1, pp. 485–495, 1991, doi:10.1214/aos/1176347995.

- [27] P. J. Bickel and D. A. Freedman, "Some asymptotic theory for the bootstrap," *The Annals of Statistics*, vol. 9, no. 6, pp. 1196–1217, 1981, doi:10.1214/aos/1176345637.
- [28] D. W. K. Andrews and M. Buchinsky, "A three-step method for choosing the number of bootstrap repetitions," *Econometrica*, vol. 68, no. 1, pp. 23–51, 2000, doi:10.1111/1468-0262.00092.
- [29] K. Krishnamoorthy and T. Mathew, *Statistical Tolerance Regions: Theory, Applications, and Computation*, John Wiley & Sons, 2009, doi:10.1002/9780470473900.
- [30] National Aeronautics and Space Administration, *NASA Systems Engineering Handbook*, NASA/SP-2016-6105 Rev2, 2017.
- [31] D. P. Thunnissen, "Method for determining margins in conceptual design," *Journal of Spacecraft and Rockets*, vol. 41, no. 1, pp. 85–92, 2004, doi:10.2514/1.9211.
- [32] Q. C. Curran, D. Allaire, and K. E. Willcox, "Sensitivity analysis methods for mitigating uncertainty in engineering system design," *Systems Engineering*, vol. 21, no. 3, pp. 191–209, 2018, doi:10.1002/sys.21422.
- [33] S. B. Shaklan, L. Marchen, J. J. Green, and O. P. Lay, "The Terrestrial Planet Finder Coronagraph dynamics error budget," in *Techniques and Instrumentation for Detection of Exoplanets II*, Proc. SPIE, vol. 5905, pp. 110–121, 2005, doi:10.1117/12.617890.
- [34] N. W. Tuchow, C. C. Stark, and E. Mamajek, "HPIC: The Habitable Worlds Observatory Preliminary Input Catalog," *The Astronomical Journal*, vol. 167, no. 3, 139, 2024, doi:10.3847/1538-3881/ad25ec.
- [35] F. Menti et al., "Database of Candidate Targets for the LIFE Mission," *Research Notes of the AAS*, vol. 8, no. 10, 267, 2024, doi:10.3847/2515-5172/ad887e.
- [36] P. A. Huber, F. A. Dannert, R. Laugier, T. Birbacher, A. M. Glauser, J. T. Hansen, and S. P. Quanz, "Analytical and numerical instrumental noise simulations for the Large Interferometer For Exoplanets (LIFE)," in *Optical and Infrared Interferometry and Imaging IX*, Proc. SPIE, vol. 13095, 2024, doi:10.1117/12.3018644.